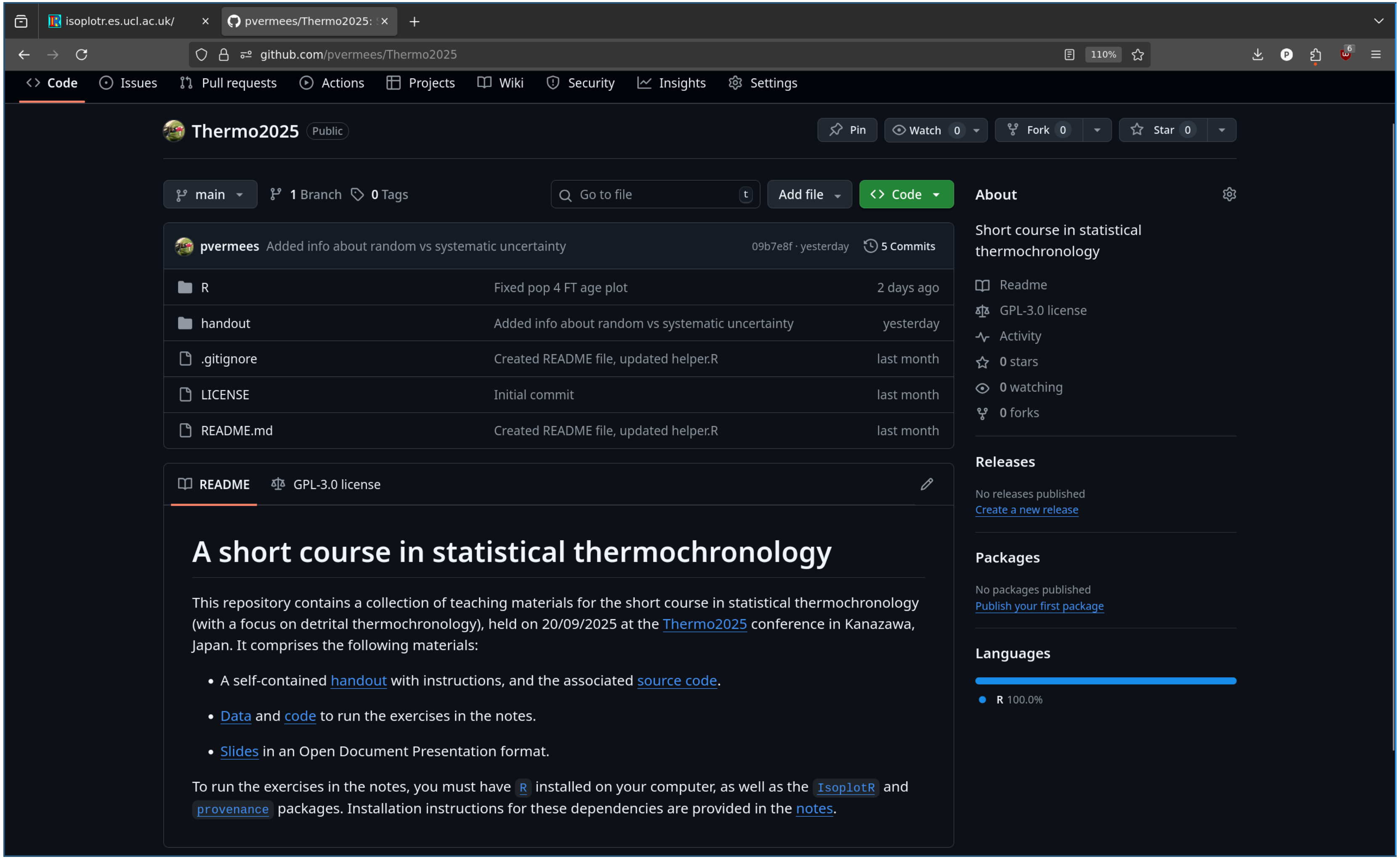
https://github.com/pvermees/Thermo2025



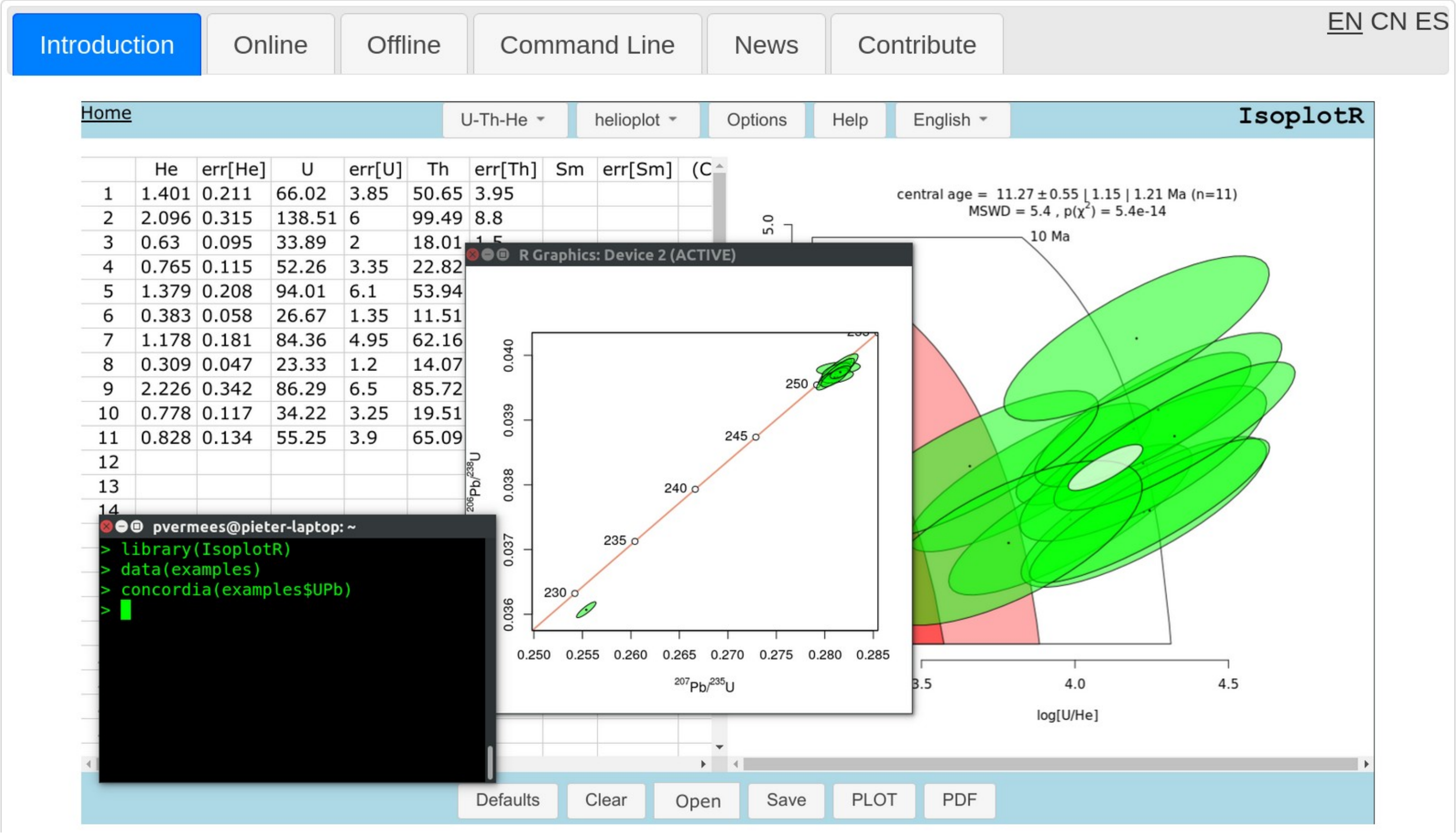
½ hr

1 hr

1 hr

Contents

https://isoplotr.es.ucl.ac.uk



Install R from:

https://cran.r-project.org/bin/

OR

https://posit.co/download/rstudio-desktop/

Parent Directory		-
linux/	2020-07-07 11:45	-
macos/	2005-04-19 09:45	-
macosx/	2025-06-18 22:17	-
windows/	2024-03-05 14:36	-

Introduction | Online | Offline | Command Line | News | Contribute

EN CN ES

To run the offline version of the program, install *R* from http://r-project.org and enter the following command at the prompt:

```
install.packages('IsoplotRgui')
```

Once the package has been installed, you can run the browser-based graphical user interface by typing:

```
library(IsoplotRgui)
IsoplotR()
```

This will open a new tab in your internet browser. To stop the GUI, enter the following command at the prompt:

```
stopIsoplotR()
```

Ages vs. dates

true values	measured data
$\rho_s = 1 \times 10^6 \text{ cm}^{-2}$	$N_s = 15$
$\rho_i = 2 \times 10^6 \text{ cm}^{-2}$	$N_i = 42$
$\rho_d = 2.5 \times 10^6 \text{ cm}^{-2}$	$\hat{\rho}_d = 2.55 \pm 0.05 \times 10^6 \text{ cm}^{-2}$
$\zeta = 350 \text{ yr}^{-1}\text{cm}^2$	$\hat{\zeta} = 355 \pm 10 \text{ yr}^{-1}\text{cm}^2$

$$t = \frac{1}{\lambda_{238}} \ln\left(1 + \frac{1}{2}\lambda_{238}\zeta\rho_d\frac{\rho_s}{\rho_i}\right) = 215 \text{ Ma}$$

$$\hat{t} = \frac{1}{\lambda_{238}} \ln\left(1 + \frac{1}{2}\lambda_{238}\hat{\zeta}\hat{\rho}_d\frac{N_s}{N_i}\right) = 160 \text{ Ma}$$

$$s[\hat{t}] = \hat{t}\sqrt{\underbrace{\frac{1}{N_s} + \frac{1}{N_i}}_{\text{random}} + \underbrace{\left(\frac{s[\hat{\zeta}]}{\hat{\zeta}}\right)^2 + \left(\frac{s[\hat{\rho}_d]}{\hat{\rho}_d}\right)^2}_{\text{systematic}}} = 48 \text{ Ma}$$

$$s[\hat{t}] = \hat{t}\sqrt{\underbrace{\frac{1}{N_s} + \frac{1}{N_i}}_{\text{random}} + \underbrace{\left(\frac{s[\hat{\zeta}]}{\hat{\zeta}}\right)^2 + \left(\frac{s[\hat{\rho}_d]}{\hat{\rho}_d}\right)^2}_{\text{systematic}}}$$

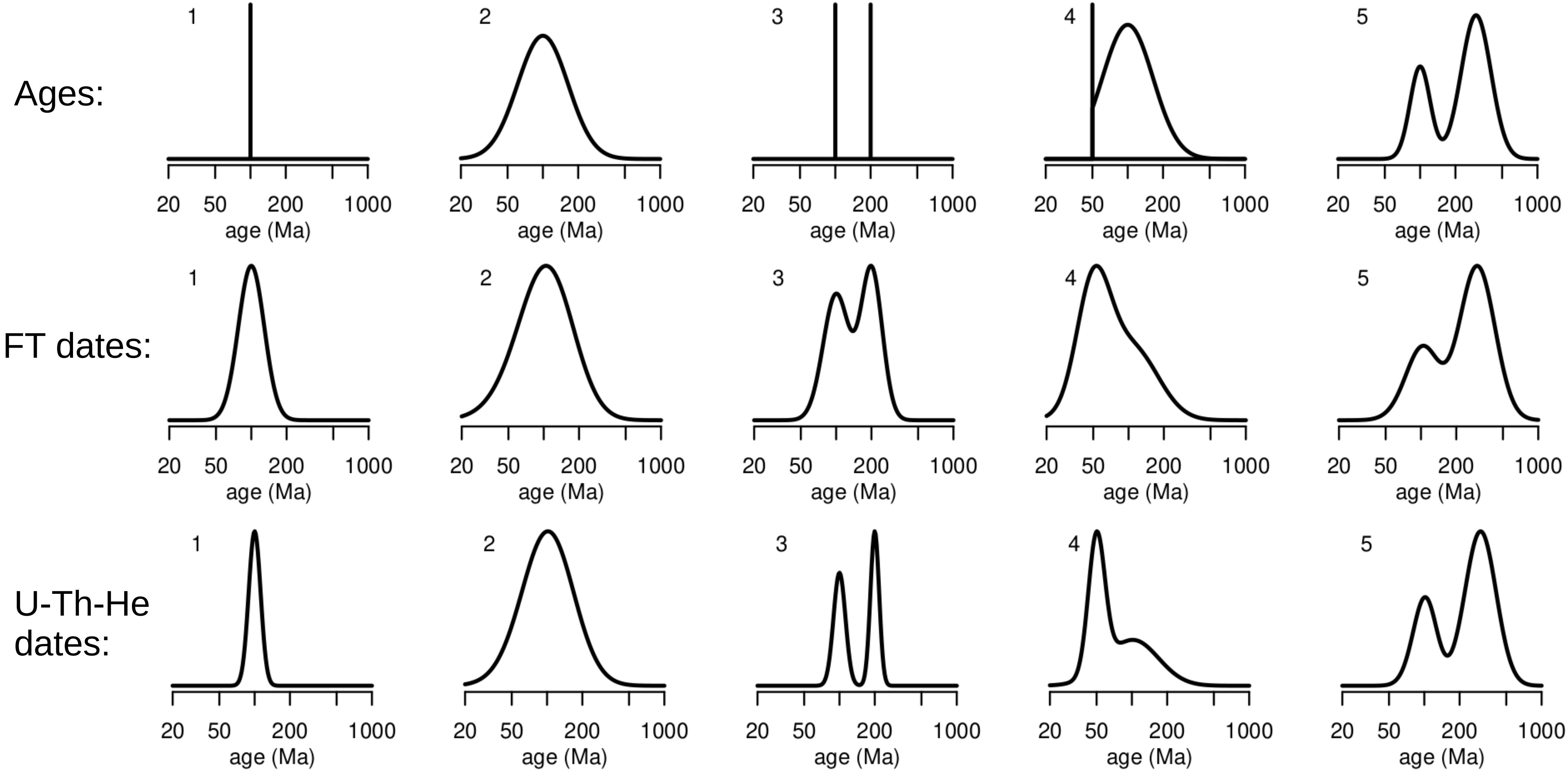
#	1	2	3	4	5	6	7	8	9	10	$\sum$
N_s	38	48	114	13	25	49	39	18	26	24	394
N_i	132	187	565	51	122	249	159	50	80	79	1674
$\hat{t}$	125	111	88	111	89	86	106	156	141	132	$\boxed{102.1}$
$s[\hat{t}] = \sqrt{1/N_s + 1/N_i}$	23	18	9	34	20	13	19	43	32	31	5.7
$s[\bar{t}] = \sqrt{1/\sum N_s + 1/\sum N_i + (s[\hat{\zeta}]/\hat{\zeta})^2 + (s[\hat{\rho}_d]/\hat{\rho}_d)^2}$											6.4

Pooled date

$$
\begin{aligned}
[\mathrm{He}] &= a(t)\,[\mathrm{U}] + b(t)\,[\mathrm{Th}] \\
\text{where } a(t) &= 8\frac{137.818}{138.818}(e^{\lambda_{238}t} - 1) + 7\frac{1}{138.818}(e^{\lambda_{235}t} - 1) \\
\text{and } b(t) &= 6(e^{\lambda_{232}t} - 1)
\end{aligned}
$$

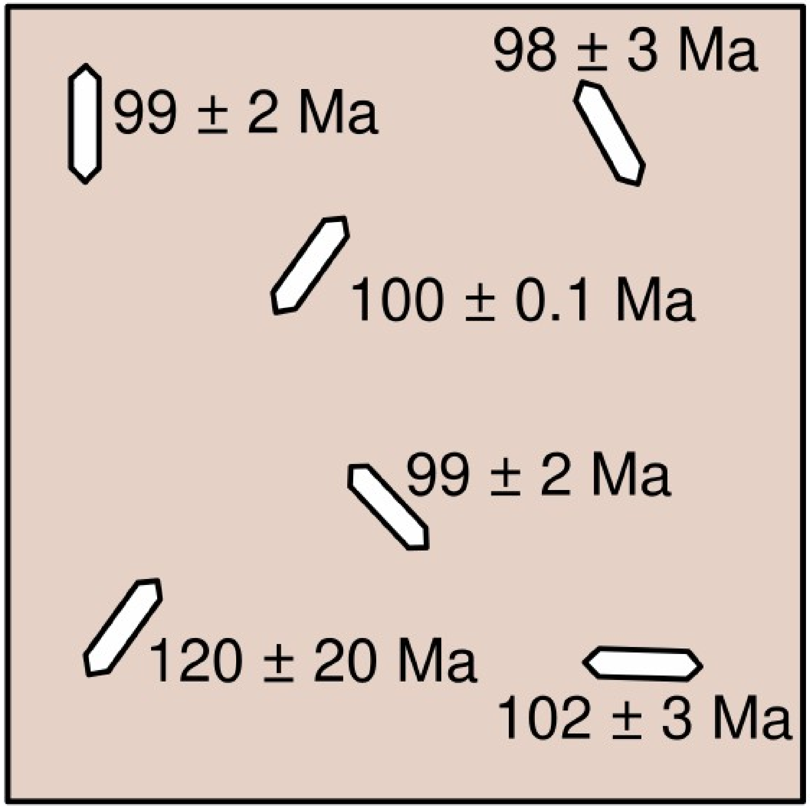
$$
s[\hat{t}] = \frac{\sqrt{a(\hat{t})^2\, s[\mathrm{U}]^2 + 2\,a(\hat{t})\,b(\hat{t})\,s[\mathrm{U{,}Th}] + b(\hat{t})^2\, s[\mathrm{Th}]^2 + s[\mathrm{He}]^2}}{8\,\frac{137.818}{138.818}\,\lambda_{238}\,e^{\lambda_{238}\hat{t}} + 7\,\frac{\lambda_{235}}{138.818}\,e^{\lambda_{235}\hat{t}}}
$$

Dates are always more scattered than ages



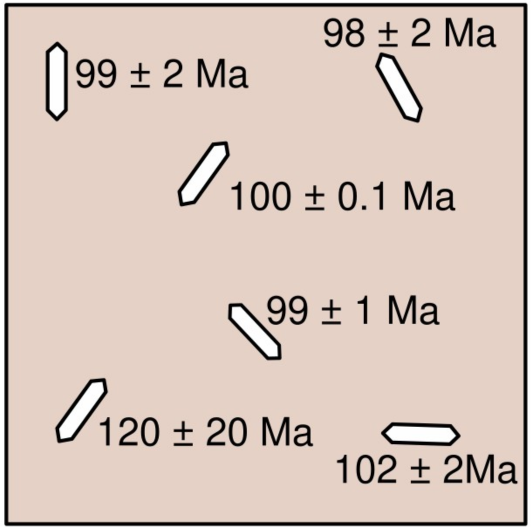
true age=100 Ma

measured dates:



true age=100 Ma

measured dates:



mean: $$\bar{t} = \frac{99 + 98 + 100 + 99 + 120 + 102}{6} = 103$$

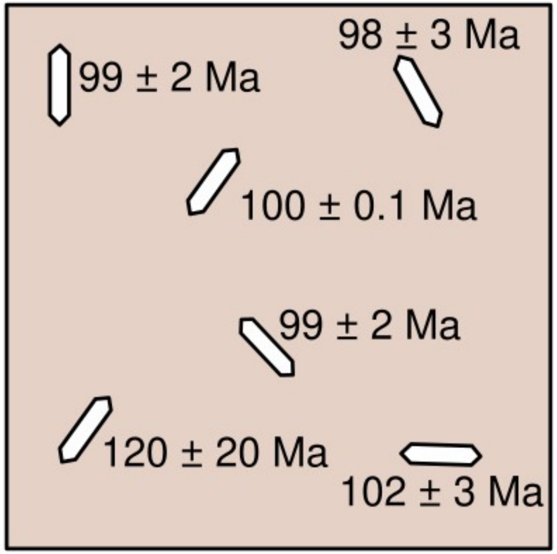
standard deviation: $$s[t] = \sqrt{\frac{(99-103)^2 + (98-103)^2 + (100-103)^2 + (99-103)^2 + (120-103)^2 + (102-103)^2}{5}} = 8.4$$

standard error: $$s[\bar{t}] = \sqrt{\frac{(99-103)^2 + (98-103)^2 + (100-103)^2 + (99-103)^2 + (120-103)^2 + (102-103)^2}{5 \cdot 6}} = 3.4$$

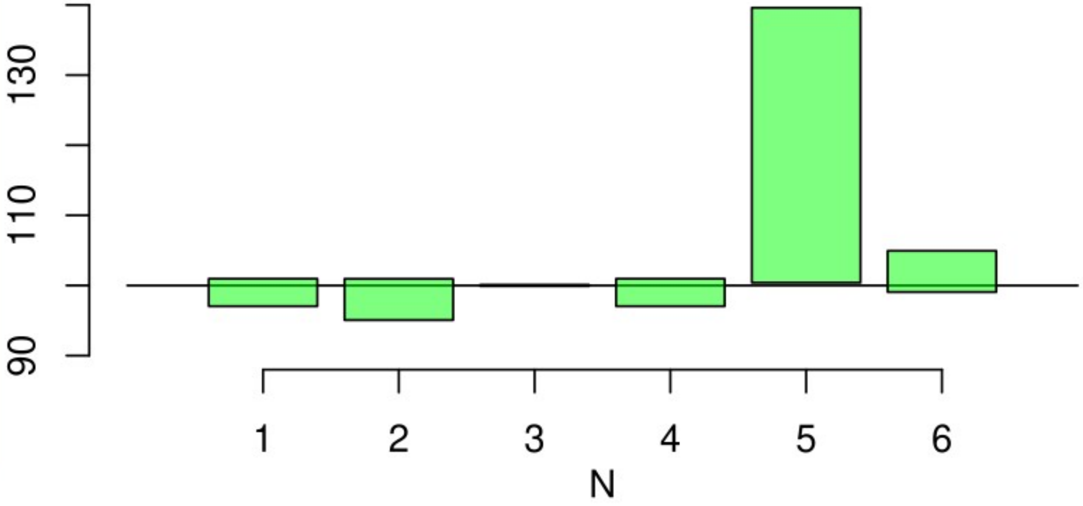
$$\bar{t} = \frac{\sum_{i=1}^{n} t_i/s[t_i]^2}{\sum_{i=1}^{n} 1/s[t_i]^2} \qquad s[\bar{t}] = \frac{1}{\sqrt{\sum_{i=1}^{n} 1/s[t_i]^2}}$$

true age=100 Ma

measured dates:



weighted mean = 99.996 ± 0.098

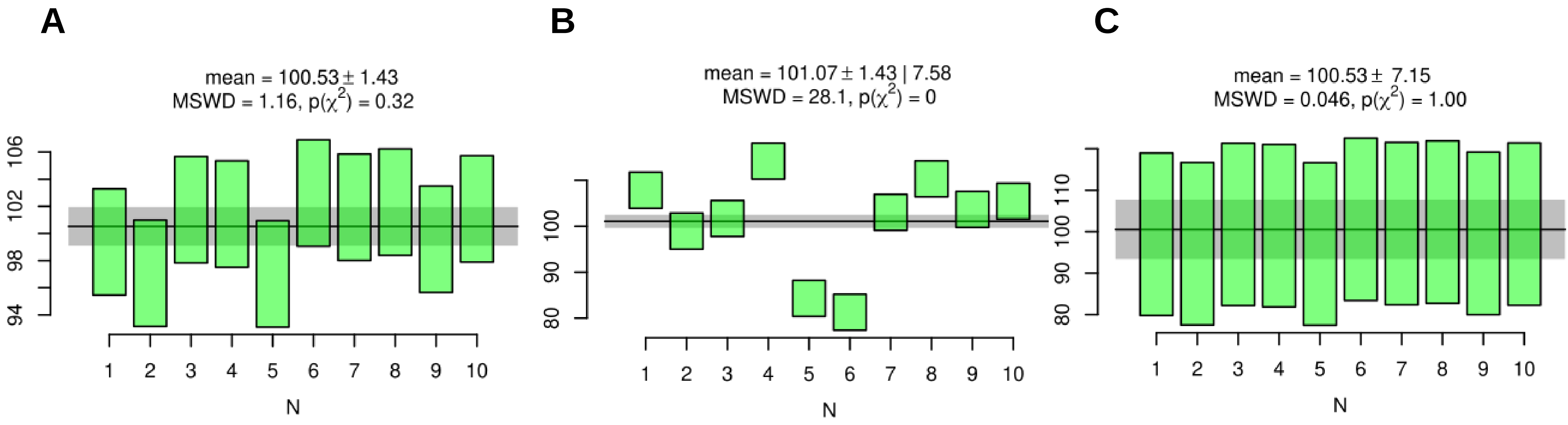


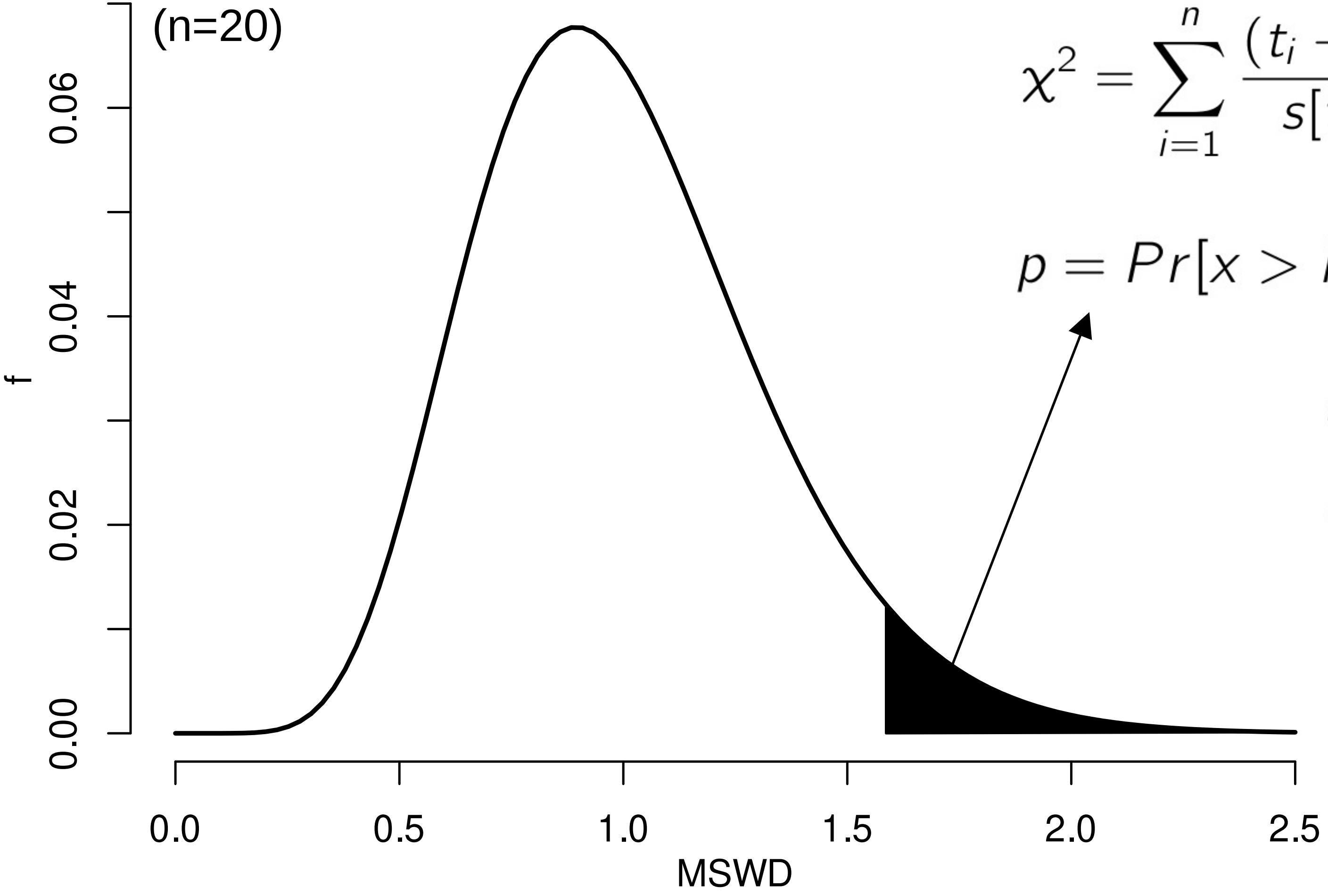
$$\chi^2 = \sum_{i=1}^{n} \frac{(t_i - \bar{t})^2}{s[t_i]^2} \qquad MSWD = \frac{\chi^2}{n-1}$$

MSWD ≈ 1

MSWD>>1 (overdispersion)

MSWD ≈ 0 (underdispersion)





$$\chi^2 = \sum_{i=1}^{n} \frac{(t_i - \bar{t})^2}{s[t_i]^2} \quad MSWD = \frac{\chi^2}{n-1}$$

$$p = Pr[x > MSWD]$$

$$MSWD \gg 1 \Leftrightarrow p \approx 0$$

$$MSWD \approx 1 \Leftrightarrow 0 < p < 1$$

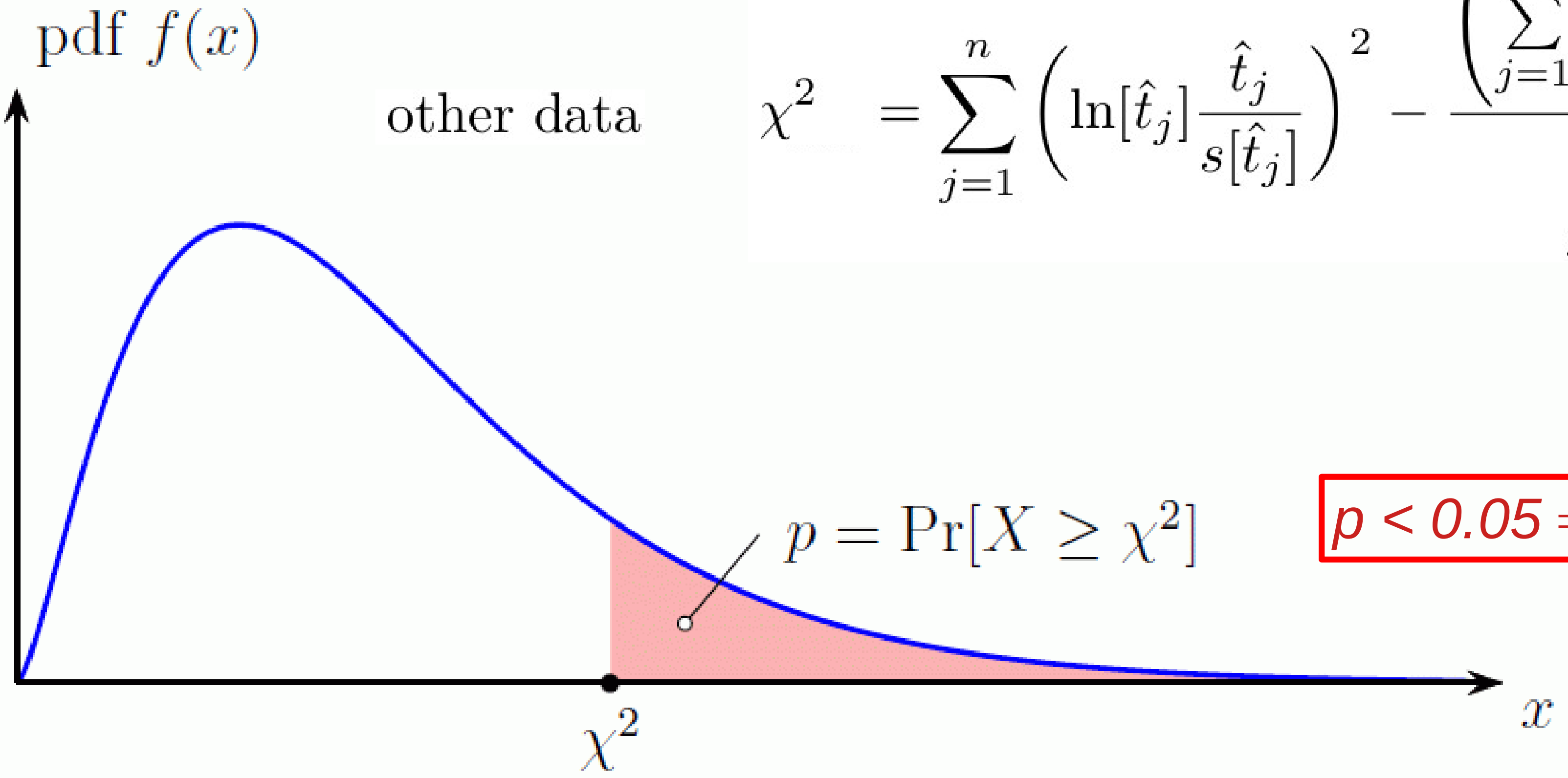
$$MSWD \approx 0 \Leftrightarrow p \approx 1$$

$\boxed{p < 0.05 \Rightarrow \text{overdispersed}}$

(MSWD > 1.6)

EDM $$\chi^2 = \frac{1}{N_s N_i} \sum_{j=1}^{n} \frac{(N_{sj} N_i - N_{ij} N_s)^2}{N_{sj} + N_{ij}}$$

other data $$\chi^2 = \sum_{j=1}^{n} \left(\ln[\hat{t}_j] \frac{\hat{t}_j}{s[\hat{t}_j]} \right)^2 - \frac{\left(\sum_{j=1}^{n} \ln[\hat{t}_j] \left[\frac{s[\hat{t}_j]}{\hat{t}_j} \right]^2 \right)^2}{\sum_{j=1}^{n} \left(\frac{\hat{t}_j}{s[\hat{t}_j]} \right)^2}$$



$p < 0.05 \Rightarrow$ *overdispersed*

Research paper
Geoscience Frontiers 7 (2016) 581–589

Strategies towards statistically robust interpretations of *in situ* U–Pb zircon geochronology

Christopher J. Spencer [a,*], Christopher L. Kirkland [b,c], Richard J.M. Taylor [a]

*"… with an n of 20 the acceptable [MSWD] is 0.35-1.65 […] Weighted averages with [MSWD] values that fall outside of this range **should not be used"***

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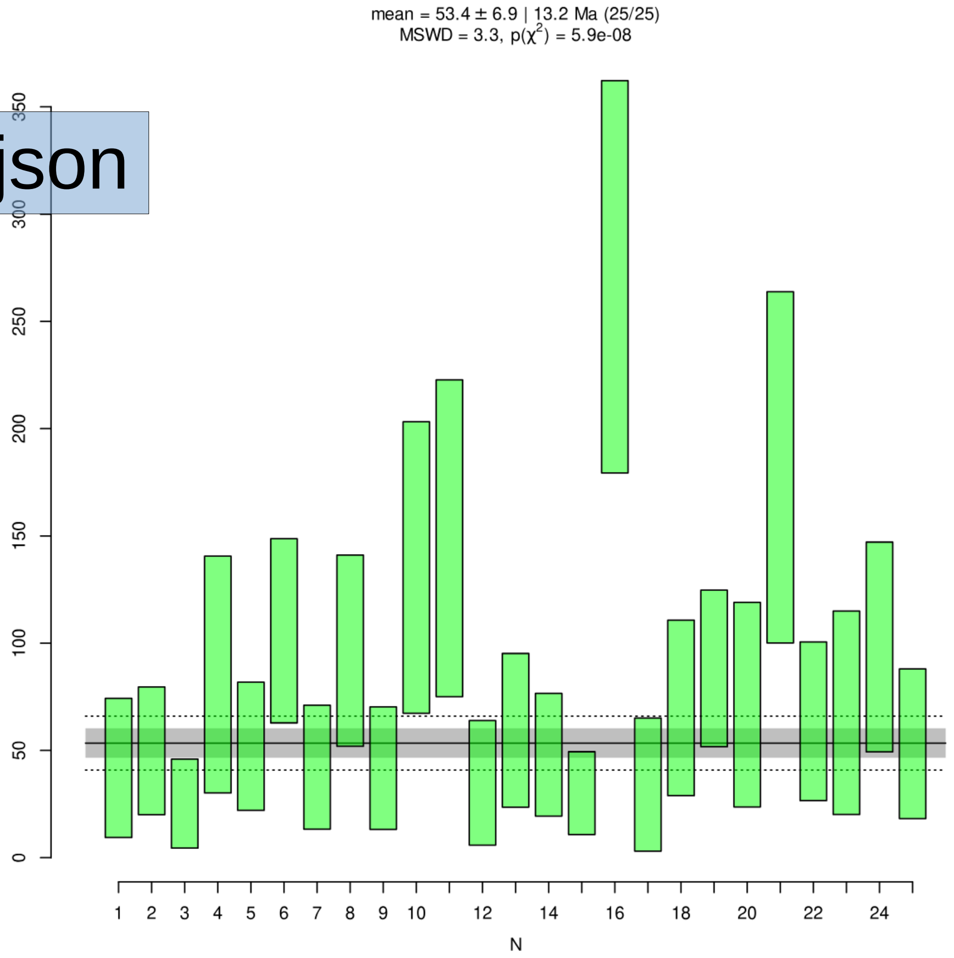
IsoplotR

ζ: 350 ± 2 yr cm^2

ρ_D: 2500000 ± 50000 1/cm^2

	Ns	Ni	(C)	(omit)	(comment)	F	G	H	I	J	K	L
1	7	73										
2	12	105										
3	6	104										
4	11	56										
5	13	109										
6	29	119										
7	9	93										
8	22	99										
9	9	94										
10	20	64										
11	21	61										
12	6	75										
13	12	88										
14	12	109										
15	10	145										
16	55	87										
17	5	64										
18	13	81										
19	27	133										
20	10	61										
21	27	64										
22	13	89										
23	9	58										
24	19	84										
25	10	82										
26												
27												
28												

FT3.json



Defaults | Clear | Open | Save | PLOT | PDF

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ζ: 350 ± 2 yr cm^2

ρ_D: 2500000 ± 50000 1/cm^2

	Ns	Ni	(C)	(omit)	(comment)	F	G	H	I	J	K	L
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4	11	56										
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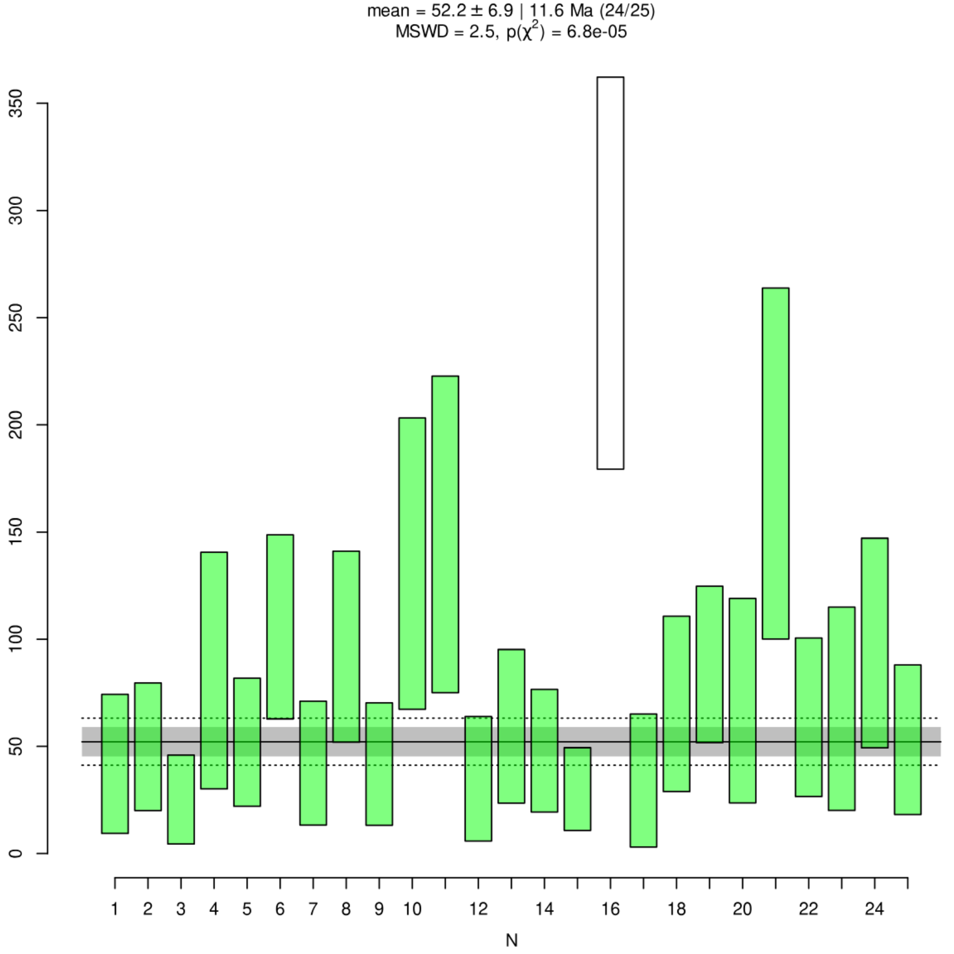
6	29	119										
7	9	93										
8	22	99										
9	9	94										
10	20	64										
11	21	61										
12	6	75										
13	12	88										
14	12	109										
15	10	145										
16	55	87		x								
17	5	64										
18	13	81										
19	27	133										
20	10	61										
21	27	64										
22	13	89										
23	9	58										
24	19	84										
25	10	82										
26												
27												
28												



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IsoplotR

ζ: 350 ± 2 yr cm^2

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	Ns	Ni	(C)	(omit)	(comment)	F	G	H	I	J	K	L
1	7	73										
2	12	105										
3	6	104		x								
4	11	56										
5	13	109										
6	29	119		x								
7	9	93										
8	22	99										
9	9	94										
10	20	64		x								
11	21	61		x								
12	6	75										
13	12	88										
14	12	109										
15	10	145		x								
16	55	87		x								
17	5	64										
18	13	81										
19	27	133										
20	10	61										
21	27	64		x								
22	13	89										
23	9	58										
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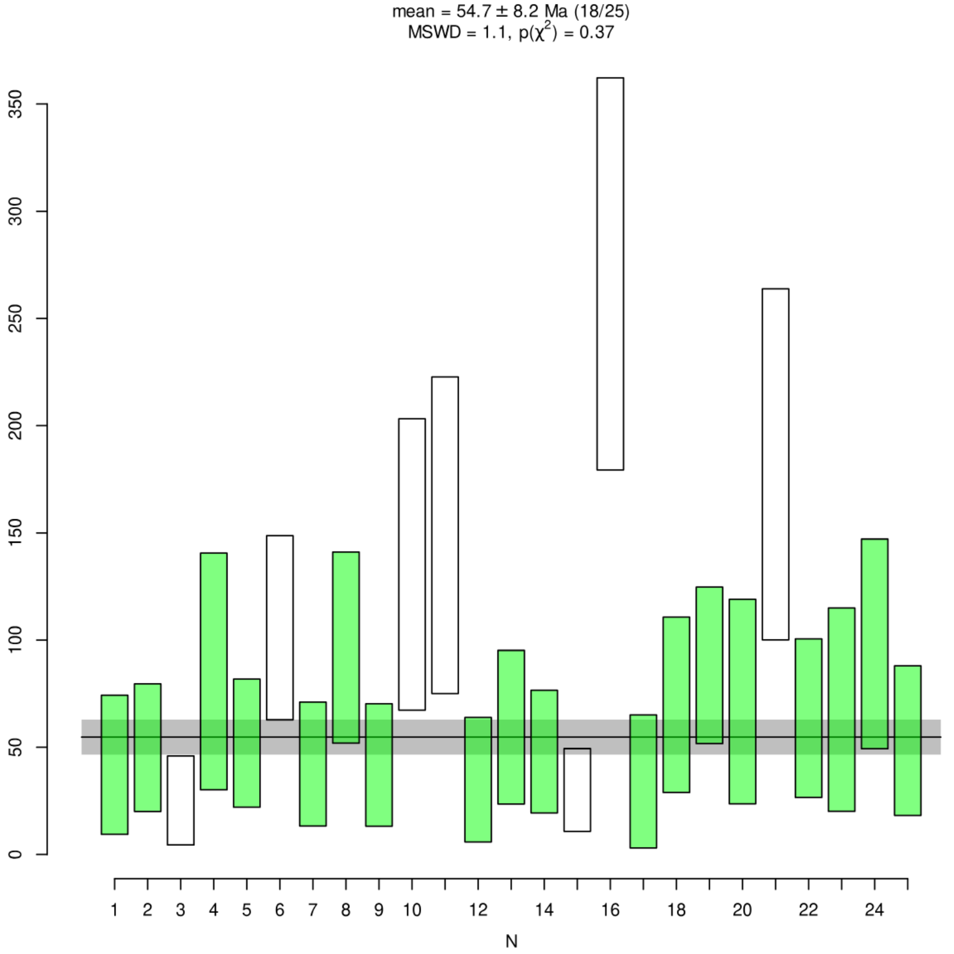
25	10	82										
26												
27												
28												



Defaults | Clear | Open | Save | PLOT | PDF

Home | fission tracks | weighted mean | Options | Help | English | **IsoplotR**

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	Ns	Ni	(C)	(omit)	(comment)	F	G	H	I	J	K	L
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2	12	105										
3	6	104		X								
4	11	56										
5	13	109										
6	29	119		X								
7	9	93										
8	22	99										
9	9	94										
10	20	64		X								
11	21	61		X								
12	6	75										
13	12	88										
14	12	109										
15	10	145		X								
16	55	87		X								
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22	13	89										
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26												
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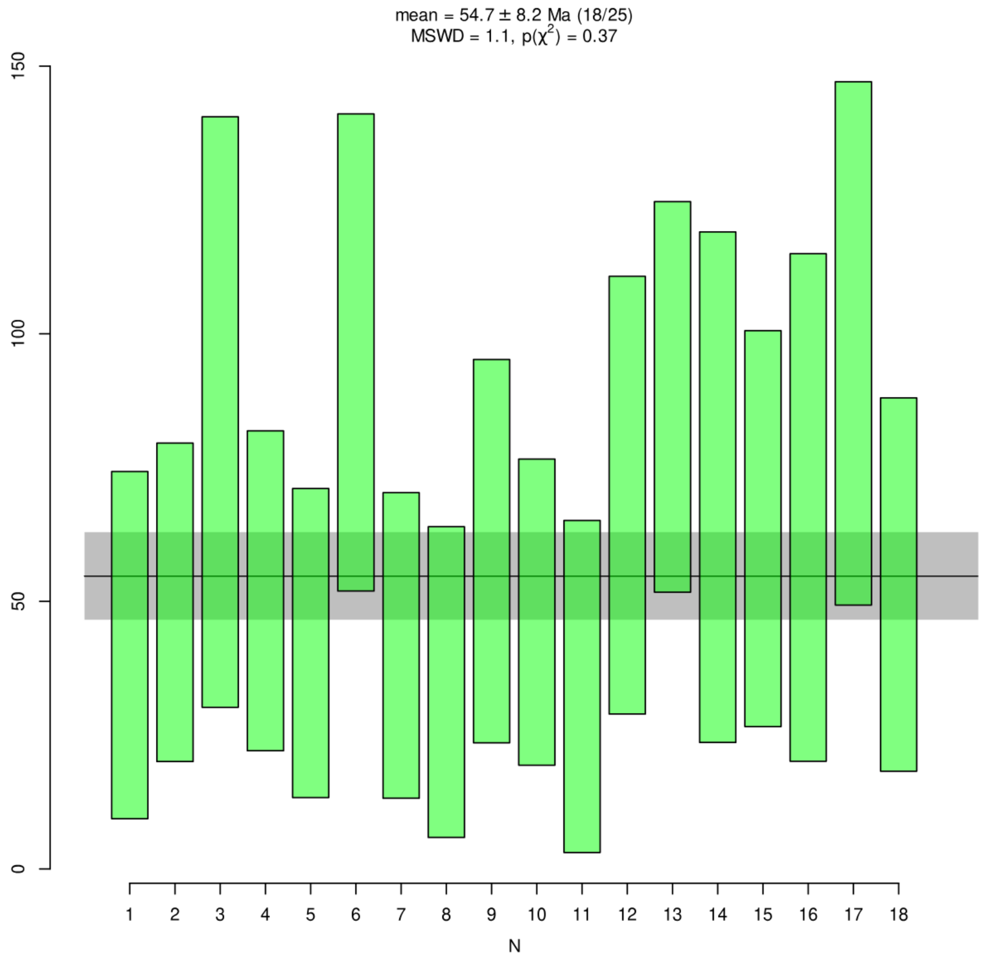
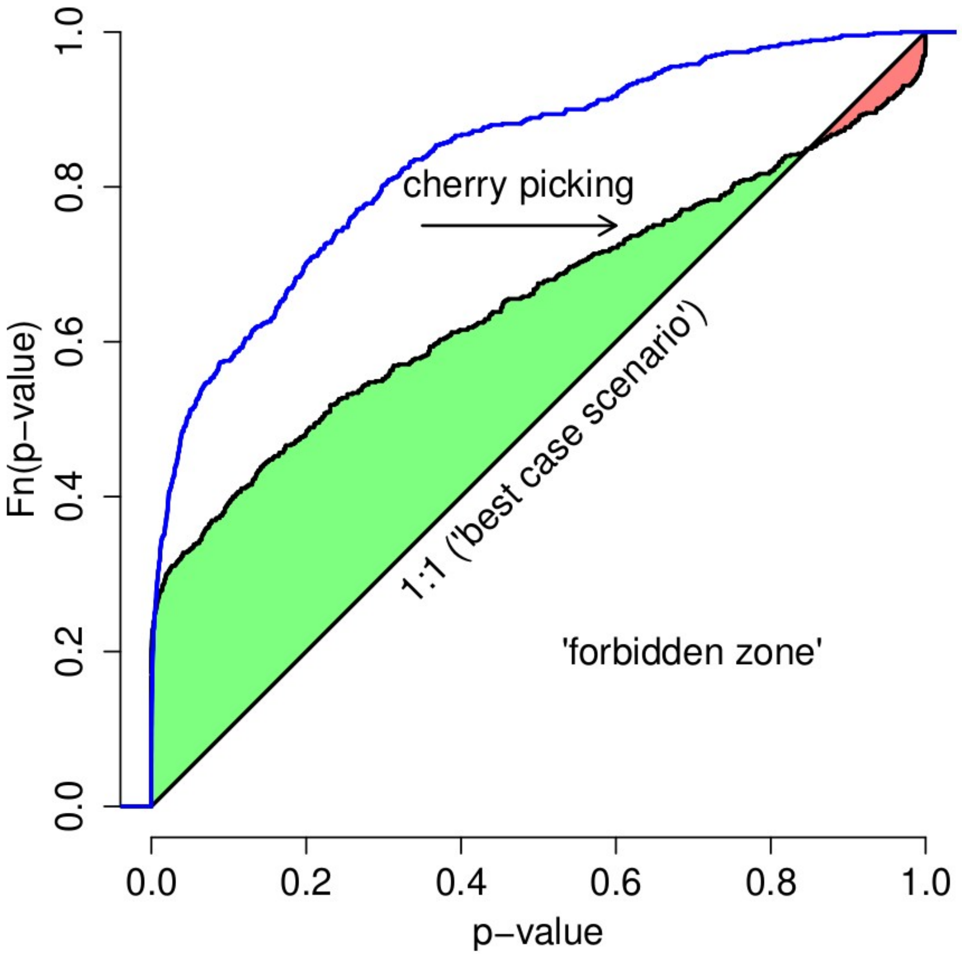


Defaults | Clear | Open | Save | PLOT | PDF



cherry picking
1:1 ('best case scenario')
'forbidden zone'
Fn(p−value)
p−value
0.0
0.2
0.4
0.6
0.8
1.0

Research paper

Geoscience Frontiers 7 (2016) 581–589

Strategies towards statistically robust interpretations of *in situ* U–Pb zircon geochronology

Christopher J. Spencer [a,*], Christopher L. Kirkland [b,c], Richard J.M. Taylor [a]

~~*"… with an n of 20 the acceptable [MSWD] is 0.35-1.65 […] Weighted averages with [MSWD] values that fall outside of this range **should not be used"***~~

PERSPECTIVE

National Science Review
12: nwaf036, 2025
https://doi.org/10.1093/nsr/nwaf036
Advance access publication 7 February 2025

EARTH SCIENCES

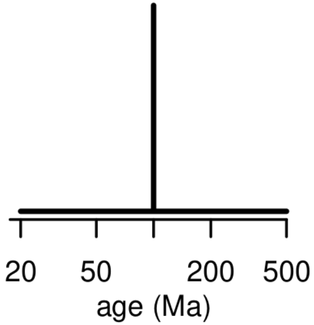
Special Topic: Geochronology

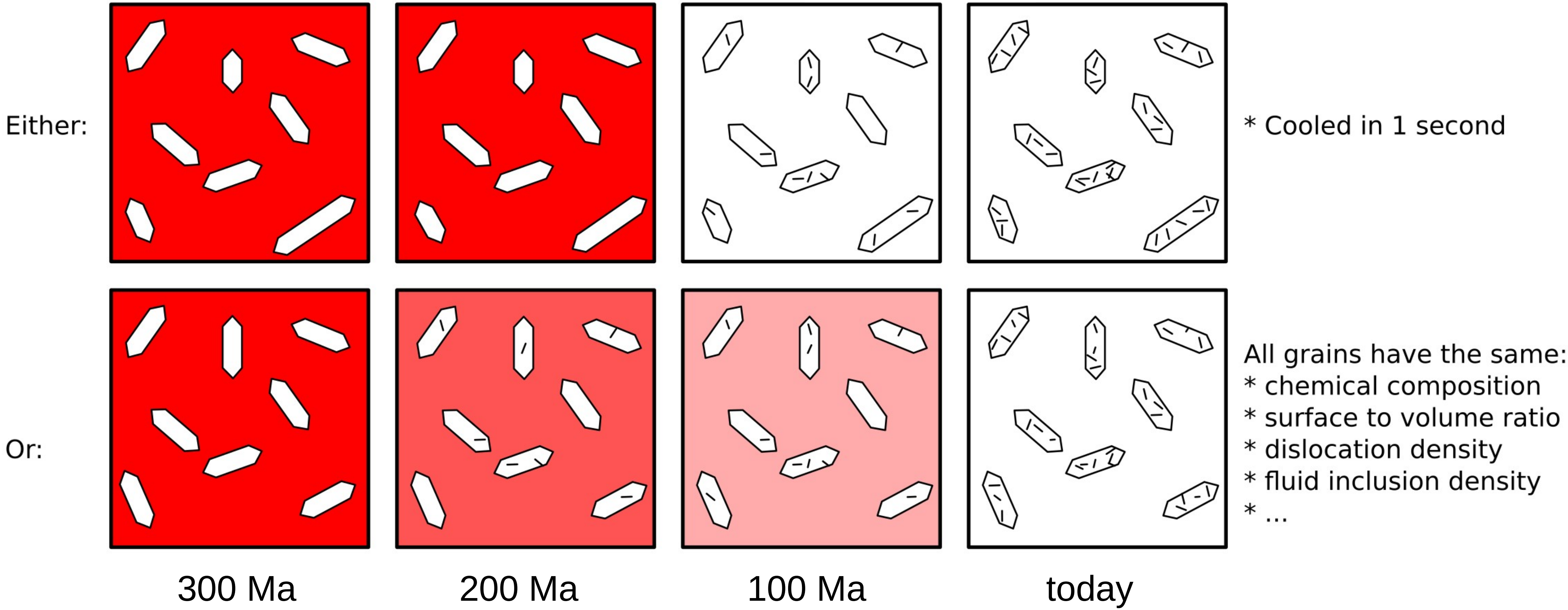
High MSWDs are not the problem, low ones are

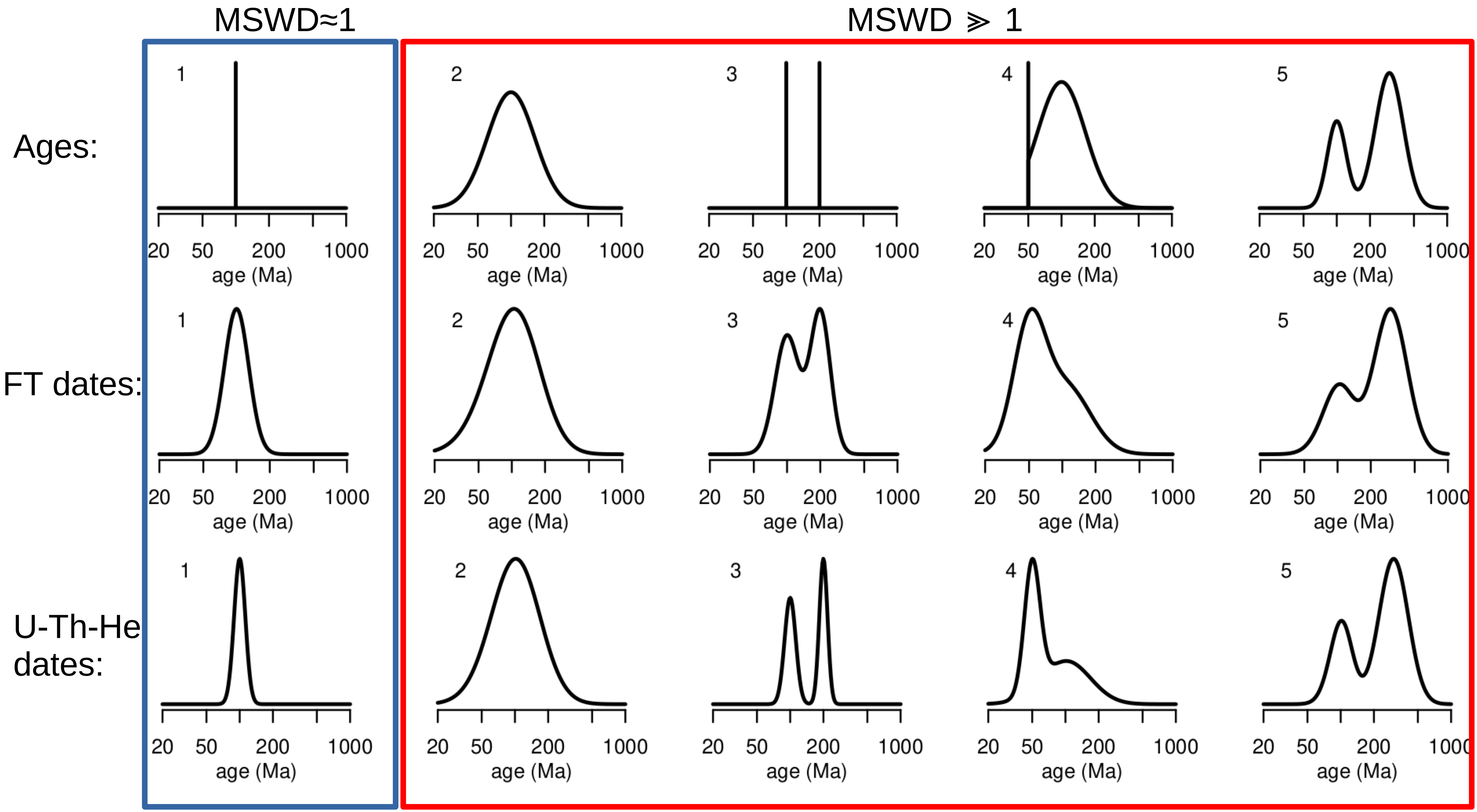
Pieter Vermeesch[1]

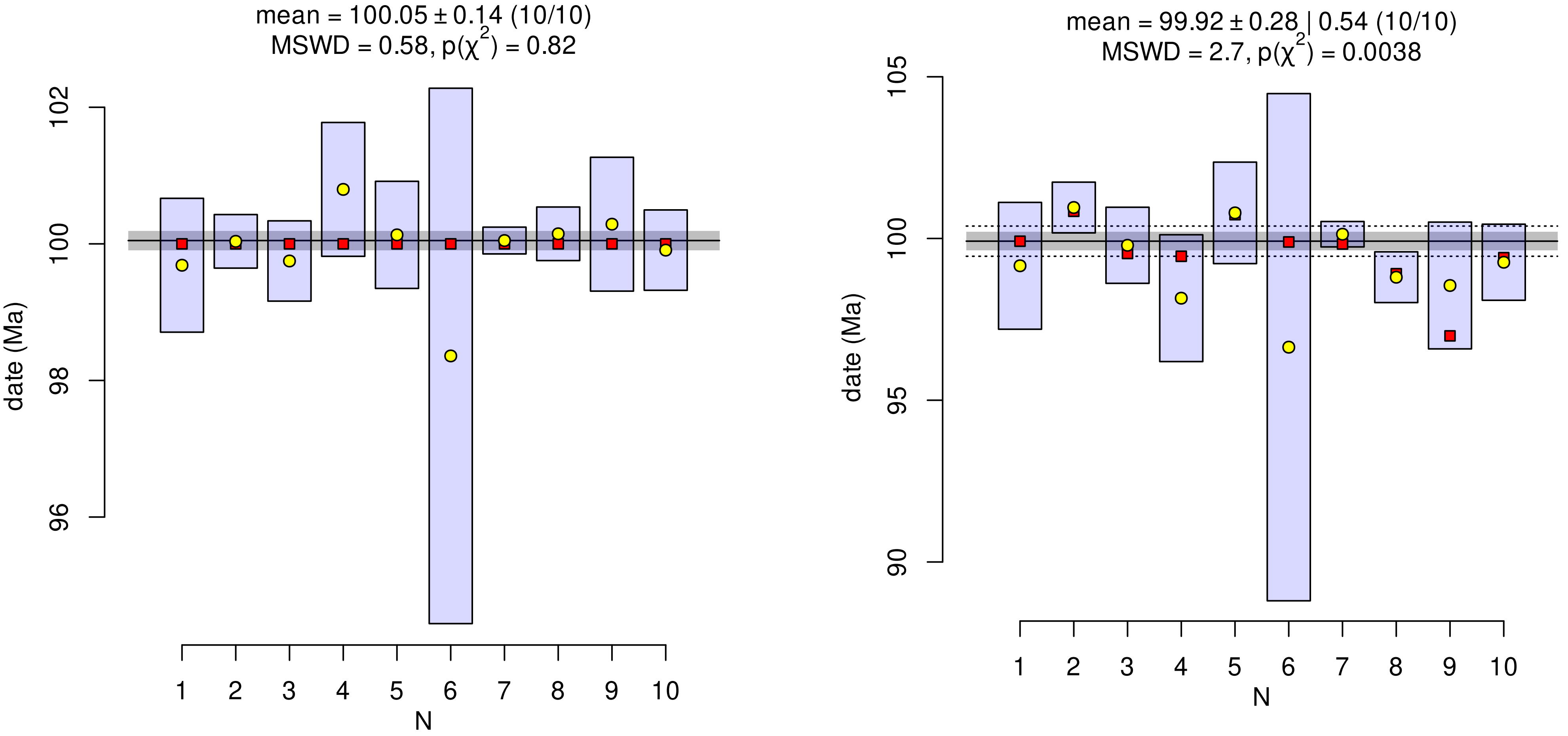
Use this paper to defend yourself against misinformed reviewers

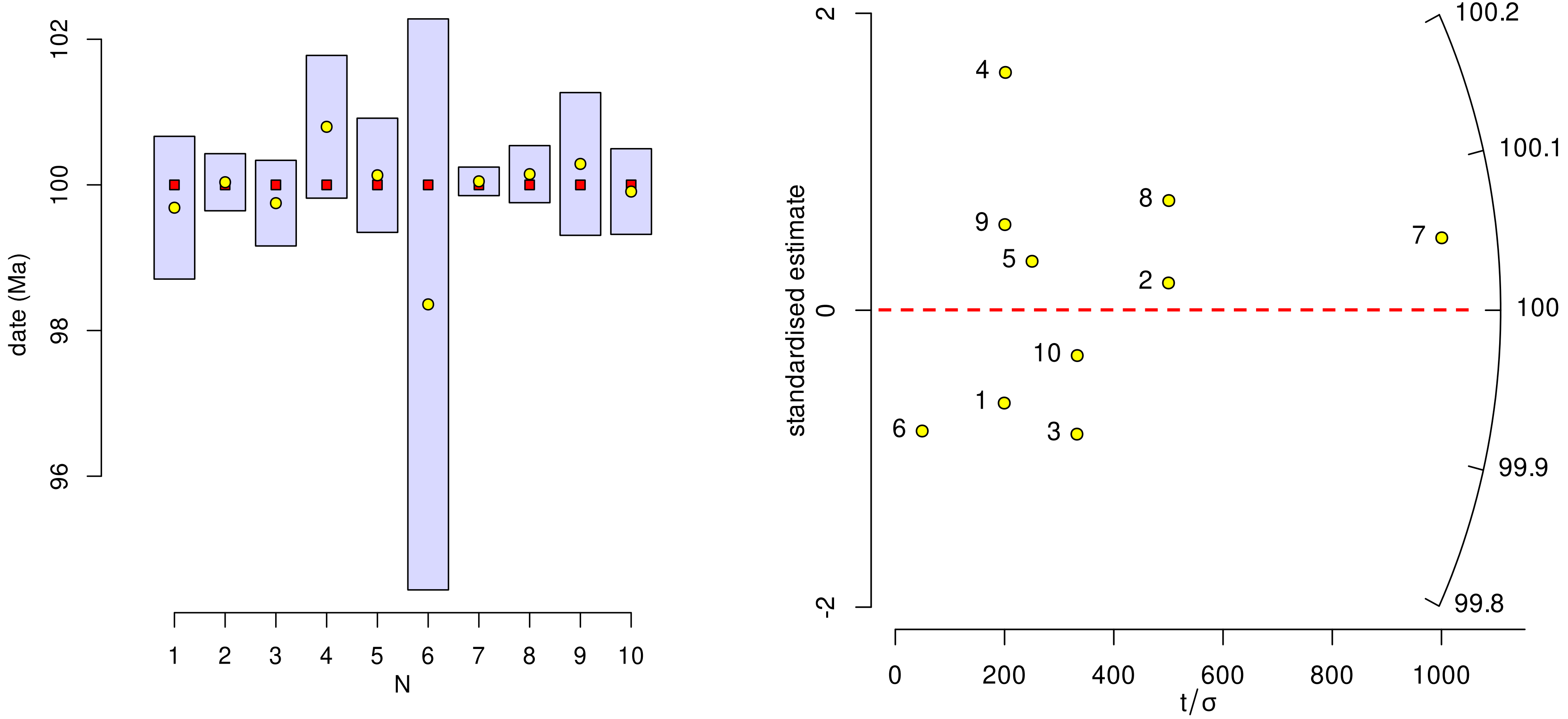
What it means to NOT be overdispersed:





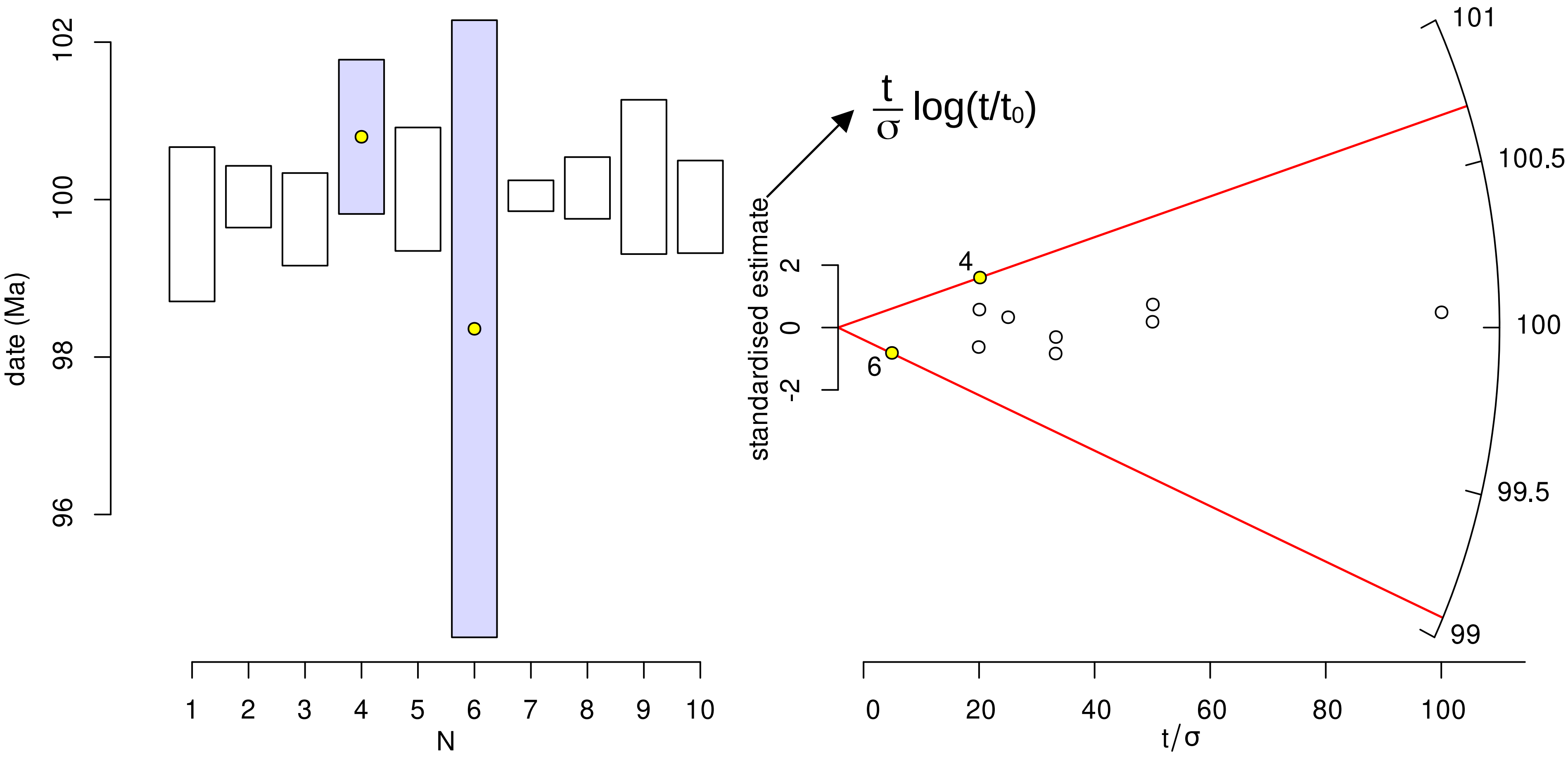

MSWD≈1
MSWD ≫ 1
Ages:
FT dates:
U-Th-He dates:
1
2
3
4
5
20 50 200 1000
age (Ma)


mean = 100.05 ± 0.14 (10/10)
MSWD = 0.58, p(χ2) = 0.82
date (Ma)
96
98
100
102
1
2
3
4
5
6
7
8
9
10
N
mean = 99.92 ± 0.28 | 0.54 (10/10)
MSWD = 2.7, p(χ2) = 0.0038
date (Ma)
90
95
100
105
1
2
3
4
5
6
7
8
9
10
N

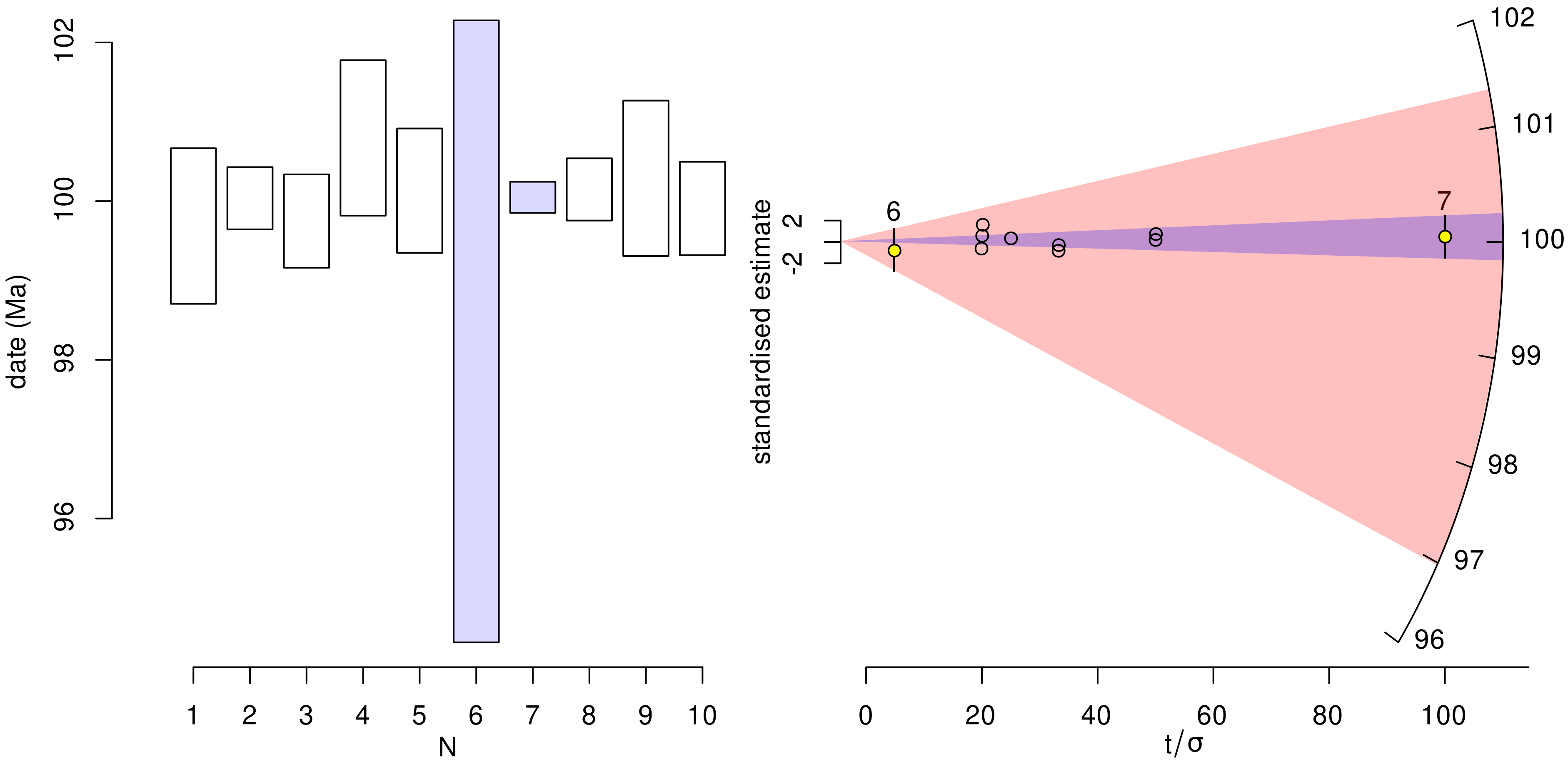


THE RADIAL PLOT: GRAPHICAL ASSESSMENT OF SPREAD IN AGES

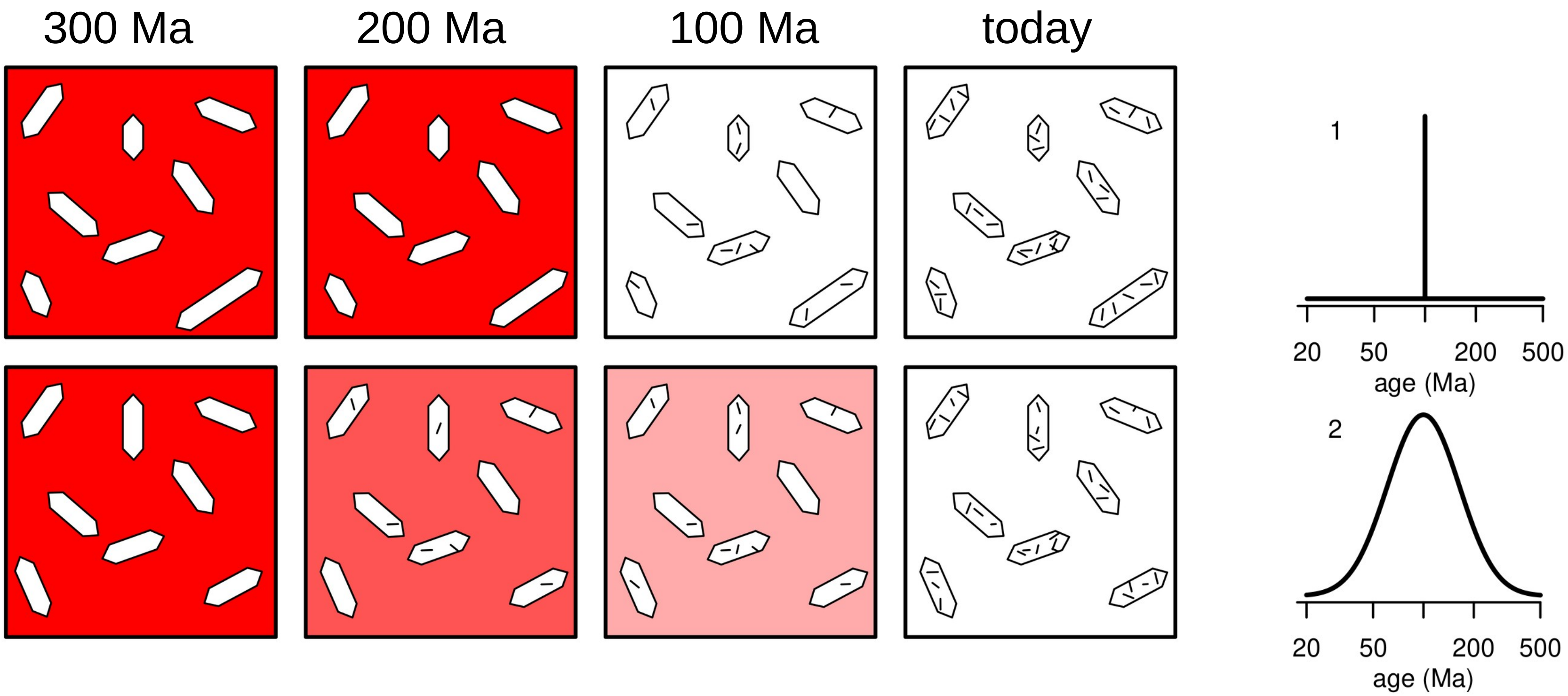
R. F. GALBRAITH



date (Ma)
96
98
100
102
1
2
3
4
5
6
7
8
9
10
N
$\frac{t}{\sigma}\log(t/t_0)$
standardised estimate
2
0
-2
4
6
101
100.5
100
99.5
99
0
20
40
60
80
100
t/σ



date (Ma)
96
98
100
102
1
2
3
4
5
6
7
8
9
10
N
standardised estimate
-2
2
6
7
0
20
40
60
80
100
t/σ
96
97
98
99
100
101
102



EDM $\ln[\rho_s/\rho_i] \sim \mathcal{N}(\mu, \sigma^2)$

other data $\ln[t] \sim \mathcal{N}(\mu, \sigma^2)$

$\hat{\mu}$ and $\hat{\sigma}$ $\rightarrow$ $\exp[\hat{\mu}]$ $\rightarrow$ Central age

$\hat{\mu}$ and $\hat{\sigma}$ $\rightarrow$ dispersion

STATISTICAL MODELS FOR MIXED FISSION TRACK AGES

R. F. GALBRAITH* and G. M. LASLETT†

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fission tracks | radial plot | Options | Help | English

IsoplotR

ζ: 350 ± 10 yr cm^2

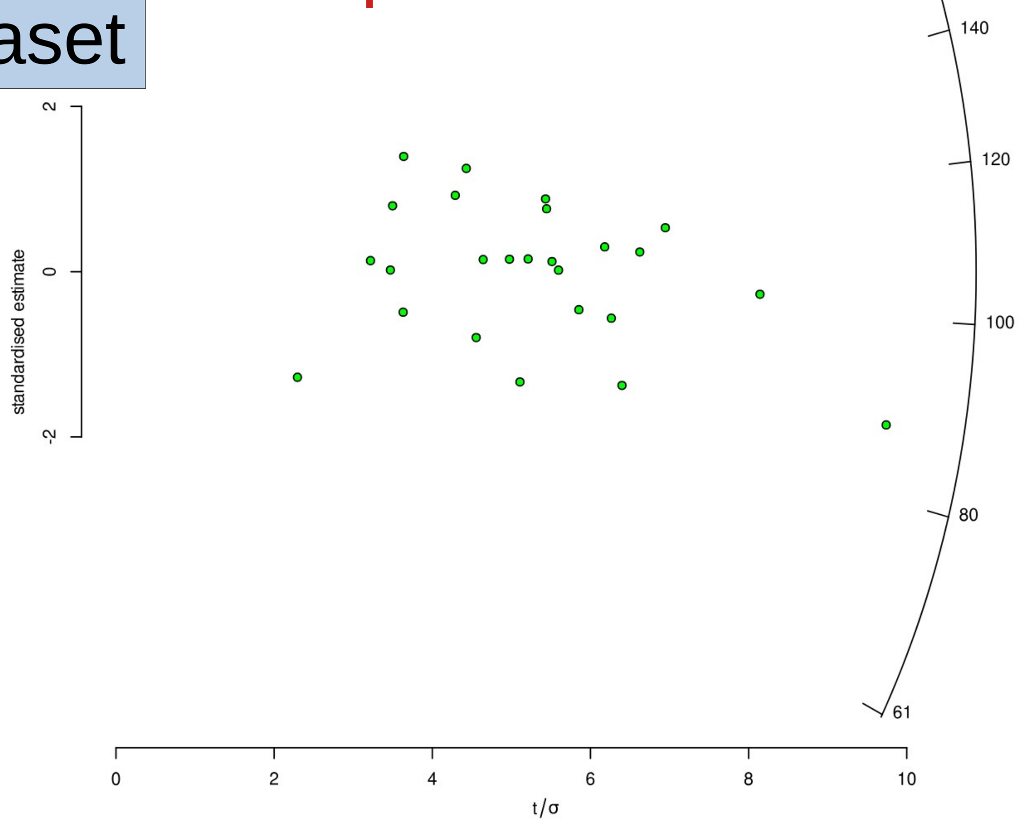
ρ_D: 2500000 ± 10000 1/cm^2

pooled age = 103.5 ± 7.4 Ma (n=25)
central age = 103.5 ± 9.4 Ma
MSWD = 0.72, p(χ^2) = 0.84

pooled = central

default dataset

	Ns	Ni	(C)	(omit)	(comment)	F	G	H	I	J
1	38	132								
2	48	187								
3	114	565								
4	13	51								
5	25	122								
6	49	249								
7	39	159								
8	18	50								
9	26	80								
10	24	79								
11	55	217								
12	38	152								
13	61	231								
14	82	347								
15	16	75								
16	38	135								
17	42	186								
18	31	123								
19	15	61								
20	27	107								
21	34	135								
22	6	43								
23	48	215								
24	16	52								
25	31	165								
26										
27										
28										



Defaults | Clear | Open | Save | PLOT | PDF

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IsoplotR

ζ: 350 ± 2 yr cm^2

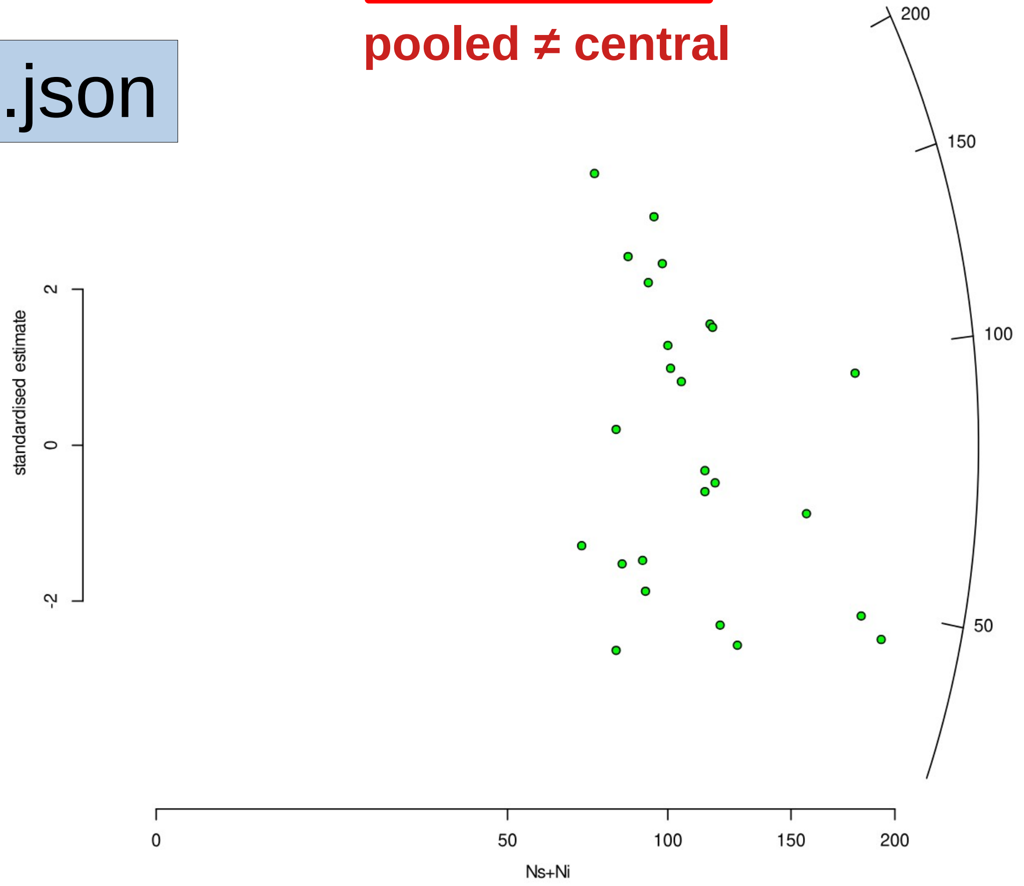
ρ_D: 2500000 ± 50000 $1/cm^2$

	Ns	Ni	(C)	(omit)	(comment)	F	G	H	I	J	K	L
1	22	64										
2	8	84										
3	16	98										
4	22	71										
5	10	110										
6	13	69										
7	5	77										
8	20	80										
9	24	92										
10	15	99										
11	20	137										
12	19	86										
13	24	74										
14	24	51										
15	19	82										
16	8	76										
17	32	148										
18	10	117										
19	7	64										
20	26	69										
21	18	165										
22	9	82										
23	18	175										
24	16	102										
25	24	93										
26												
27												
28												

FT2.json

pooled age = 78.9 ± 8.1 Ma (n=25)
central age = 82 ± 16 Ma
MSWD = 3.7, $p(\chi^2)$ = 1.7e-09
dispersion = 43 ± 17%

pooled ≠ central



Defaults | Clear | Open | Save | PLOT | PDF

Home | fission tracks | radial plot | Options | Help | English | IsoplotR

ζ: 9700 ± 50 Myr μm^2

spot size: 35 μm

	Ns	A	U1	err[U1]	(U2)	(err[U2])	(U3)	(err[U3])	(U4)	(err
1	38	4410	0.2848	0.0053	0.4017	0.0063	0.368	0.0061		
2	48	4410	0.4907	0.007	0.4491	0.0067	0.4506	0.0067		
3	114	4410	1.1073	0.0105	1.7847	0.0134	1.4605	0.0121		
4	13	4410	0.1246	0.0035	0.1575	0.004	0.111	0.0033		
5	25	4410	0.3064	0.0055	0.2439	0.0049	0.2414	0.0049		
6	49	4410	0.7553	0.0087	0.8813	0.0094	0.7768	0.0088		
7	39	4410	0.6477	0.008	0.3851	0.0062	0.5782	0.0076		
8	18	1470	0.4525	0.0067						
9	26	2940	0.6265	0.0079	0.3733	0.0061				
10	24	1470	0.6665	0.0082						
11	55	4410	0.565	0.0075	0.3496	0.0059	0.661	0.0081		
12	38	2940	0.5624	0.0075	0.6332	0.008				
13	61	4410	0.5469	0.0074	0.562	0.0075	0.6882	0.0083		
14	82	4410	1.1055	0.0105	0.6209	0.0079	0.9559	0.0098		
15	16	1470	0.3365	0.0058						
16	38	2940	0.826	0.0091	0.5669	0.0075				
17	42	2940	0.749	0.0087	0.7883	0.0089				
18	31	4410	0.2188	0.0047	0.2904	0.0054	0.2018	0.0045		
19	15	1470	0.563	0.0075						
20	27	1470	0.6939	0.0083						
21	34	2940	0.5107	0.0071	0.4176	0.0065				
22	6	1470	0.3732	0.0061						
23	48	1470	1.5108	0.0123						
24	16	1470	0.4085	0.0064						
25	31	4410	0.4958	0.007	0.3928	0.0063	0.3323	0.0058		
26										
27										
28										

Method: ICP (ζ)

Input errors: 1se (abs)

Output errors: ci (abs)

Probability cutoff: 0.05

Show sample numbers?

Propagate external uncertainties?

Transformation: logarithmic

Finite mixtures: 0 (none)

Minimum value: auto

Central value: auto

Maximum value: auto

Maximum precision: auto

Plot character: 21

Significant digits: 2

Fill colour: Custom colour ramp

from: to:

Plot symbol magnification: 1

Font magnification: 1

Colour label:

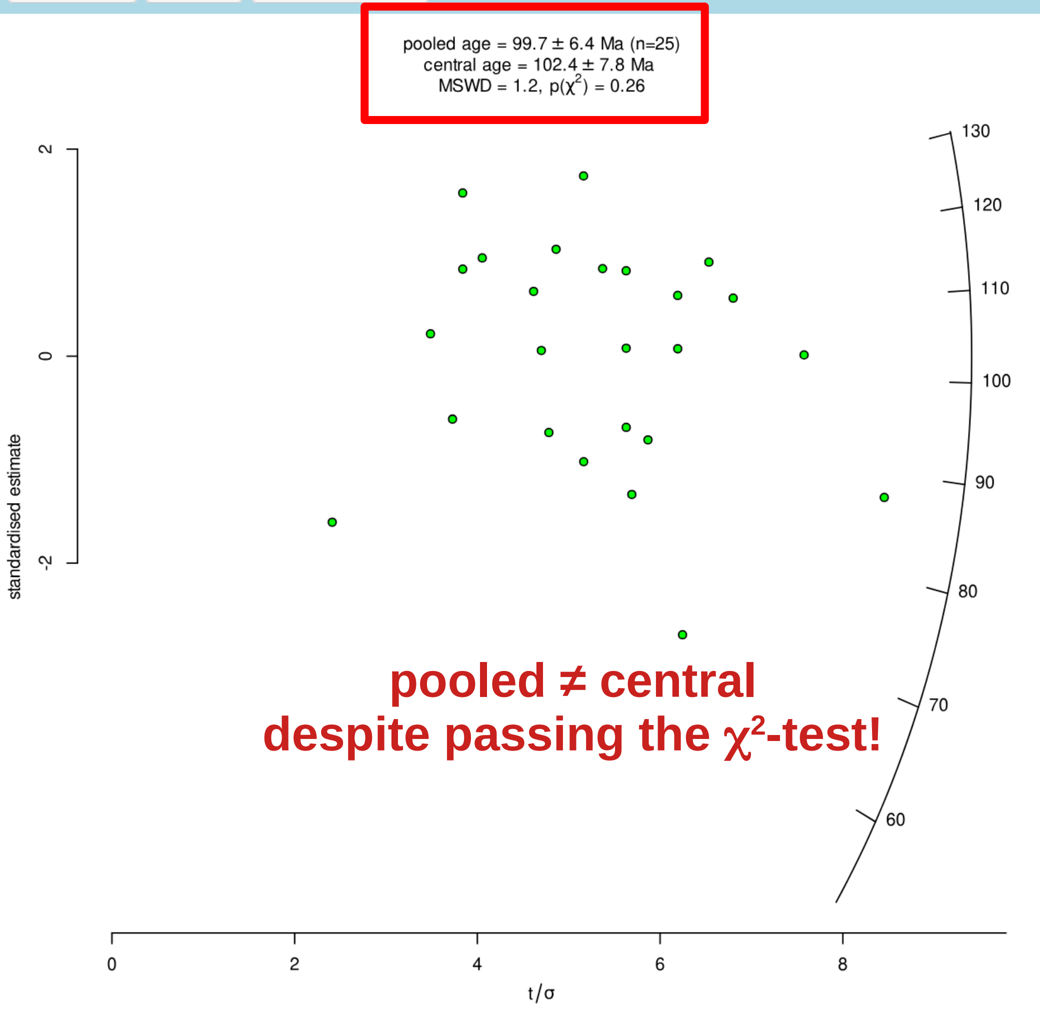
Defaults | Clear | Open | Save | PLOT | PDF

Home | fission tracks | radial plot | Options | Help | English | **IsoplotR**

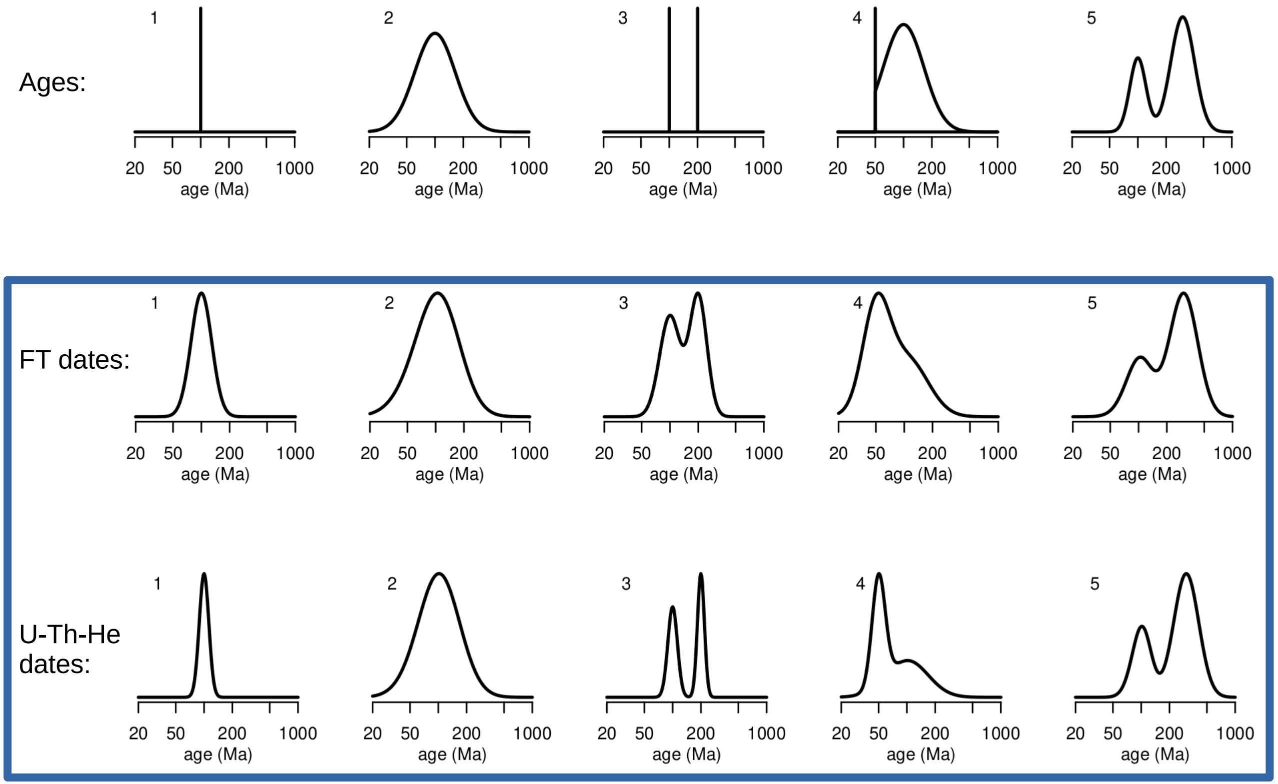
ζ: 9700 ± 50 Myr μm^2

spot size: 35 μm

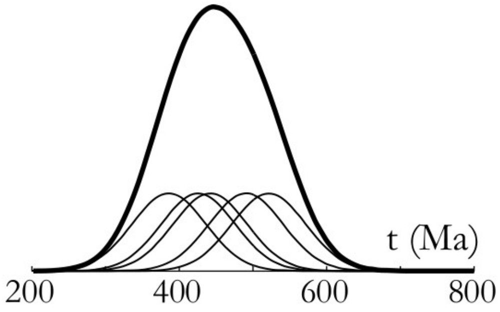
	Ns	A	U1	err[U1]	(U2)	(err[U2])	(U3)	(err[U3])	(U4)	(err
1	38	4410	0.2848	0.0053	0.4017	0.0063	0.368	0.0061		
2	48	4410	0.4907	0.007	0.4491	0.0067	0.4506	0.0067		
3	114	4410	1.1073	0.0105	1.7847	0.0134	1.4605	0.0121		
4	13	4410	0.1246	0.0035	0.1575	0.004	0.111	0.0033		
5	25	4410	0.3064	0.0055	0.2439	0.0049	0.2414	0.0049		
6	49	4410	0.7553	0.0087	0.8813	0.0094	0.7768	0.0088		
7	39	4410	0.6477	0.008	0.3851	0.0062	0.5782	0.0076		
8	18	1470	0.4525	0.0067						
9	26	2940	0.6265	0.0079	0.3733	0.0061				
10	24	1470	0.6665	0.0082						
11	55	4410	0.565	0.0075	0.3496	0.0059	0.661	0.0081		
12	38	2940	0.5624	0.0075	0.6332	0.008				
13	61	4410	0.5469	0.0074	0.562	0.0075	0.6882	0.0083		
14	82	4410	1.1055	0.0105	0.6209	0.0079	0.9559	0.0098		
15	16	1470	0.3365	0.0058						
16	38	2940	0.826	0.0091	0.5669	0.0075				
17	42	2940	0.749	0.0087	0.7883	0.0089				
18	31	4410	0.2188	0.0047	0.2904	0.0054	0.2018	0.0045		
19	15	1470	0.563	0.0075						
20	27	1470	0.6939	0.0083						
21	34	2940	0.5107	0.0071	0.4176	0.0065				
22	6	1470	0.3732	0.0061						
23	48	1470	1.5108	0.0123						
24	16	1470	0.4085	0.0064						
25	31	4410	0.4958	0.007	0.3928	0.0063	0.3323	0.0058		
26										
27										
28										

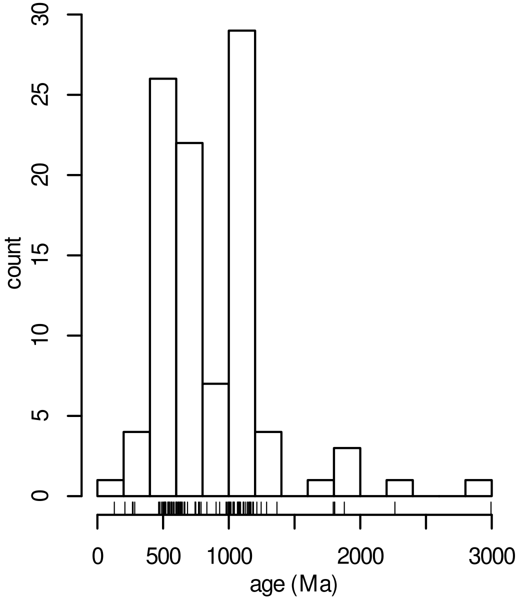


Defaults | Clear | Open | Save | PLOT | PDF

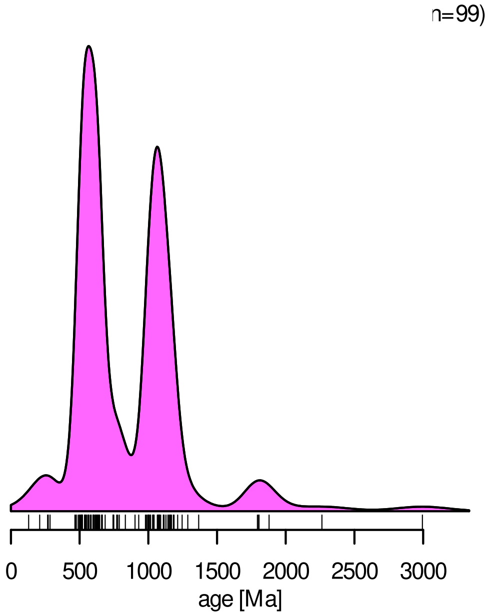


Contents

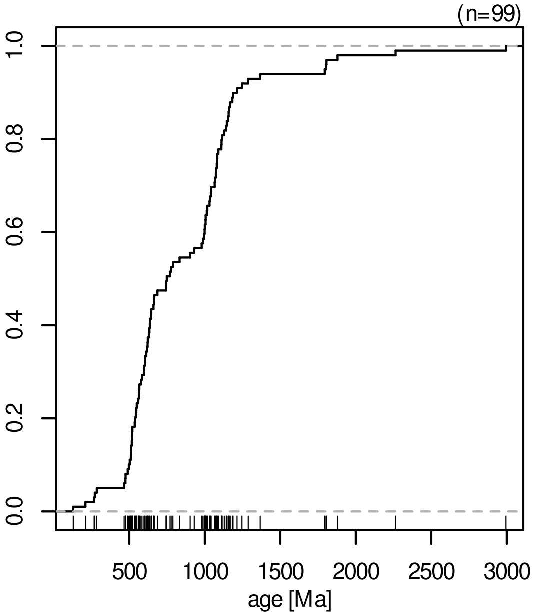




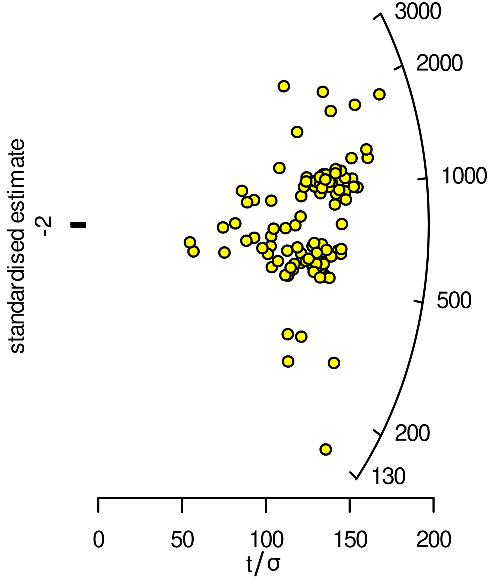
histogram



KDE

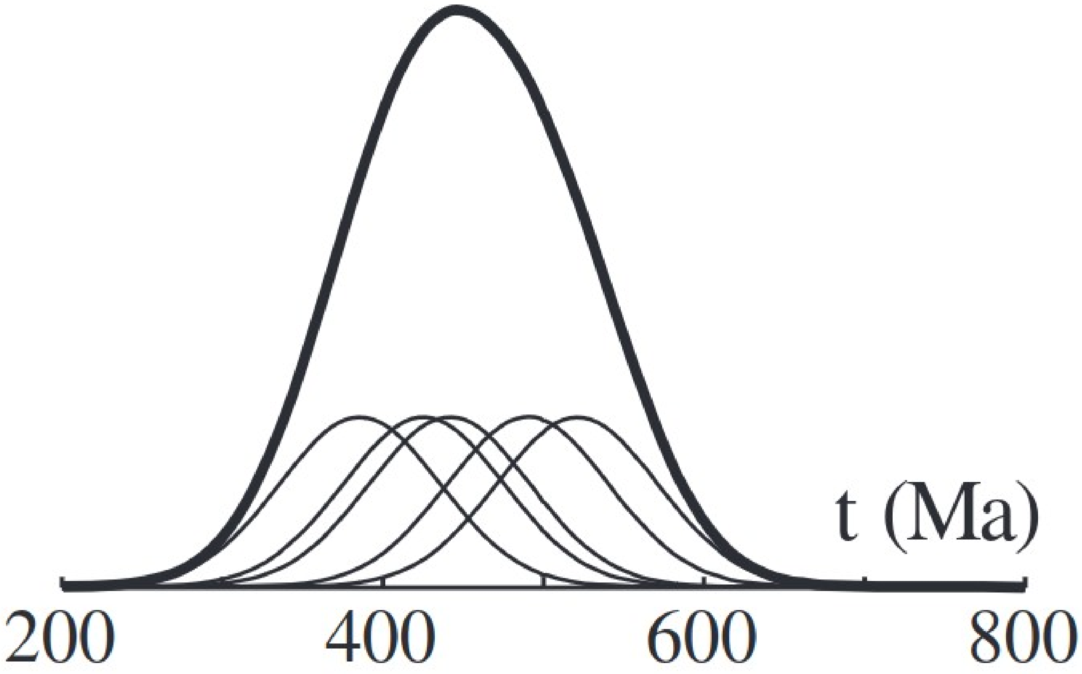


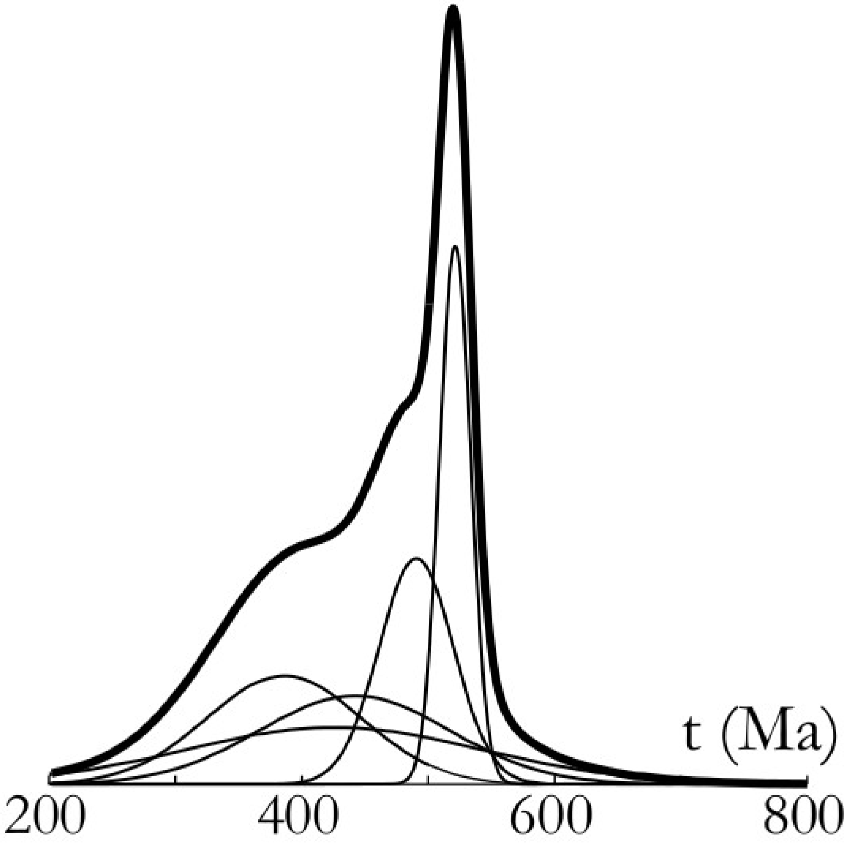
ecdf/CAD

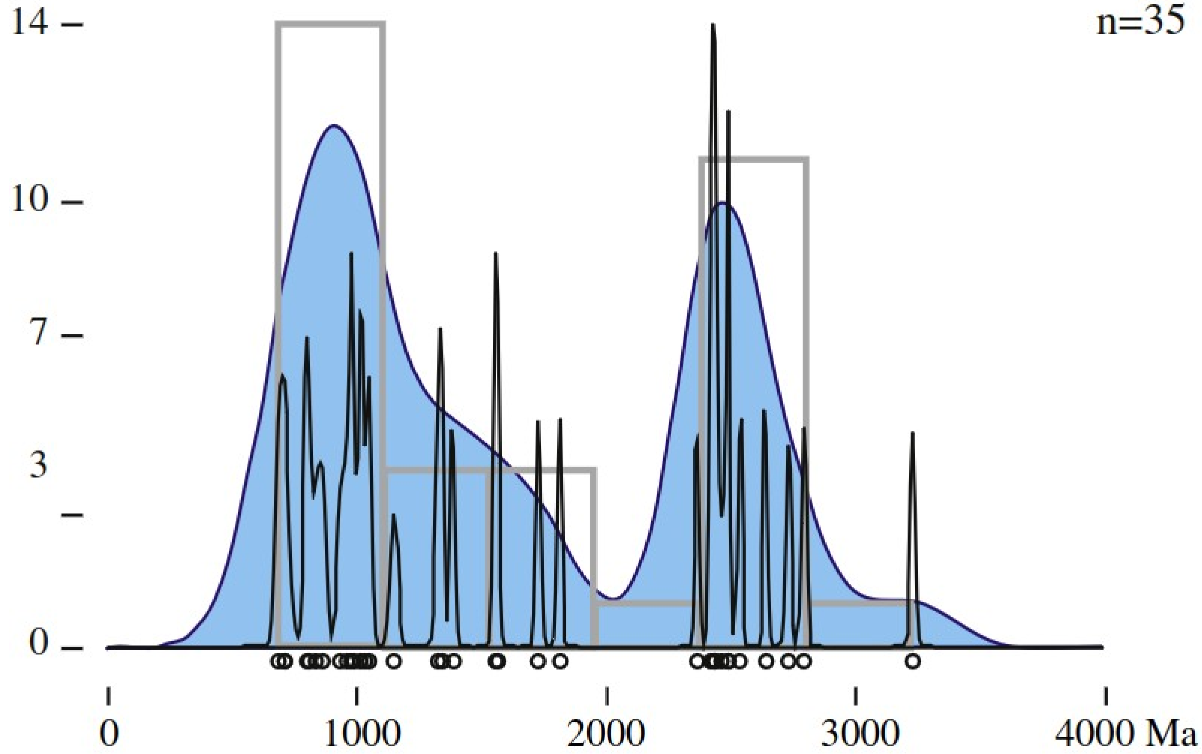


radial plot

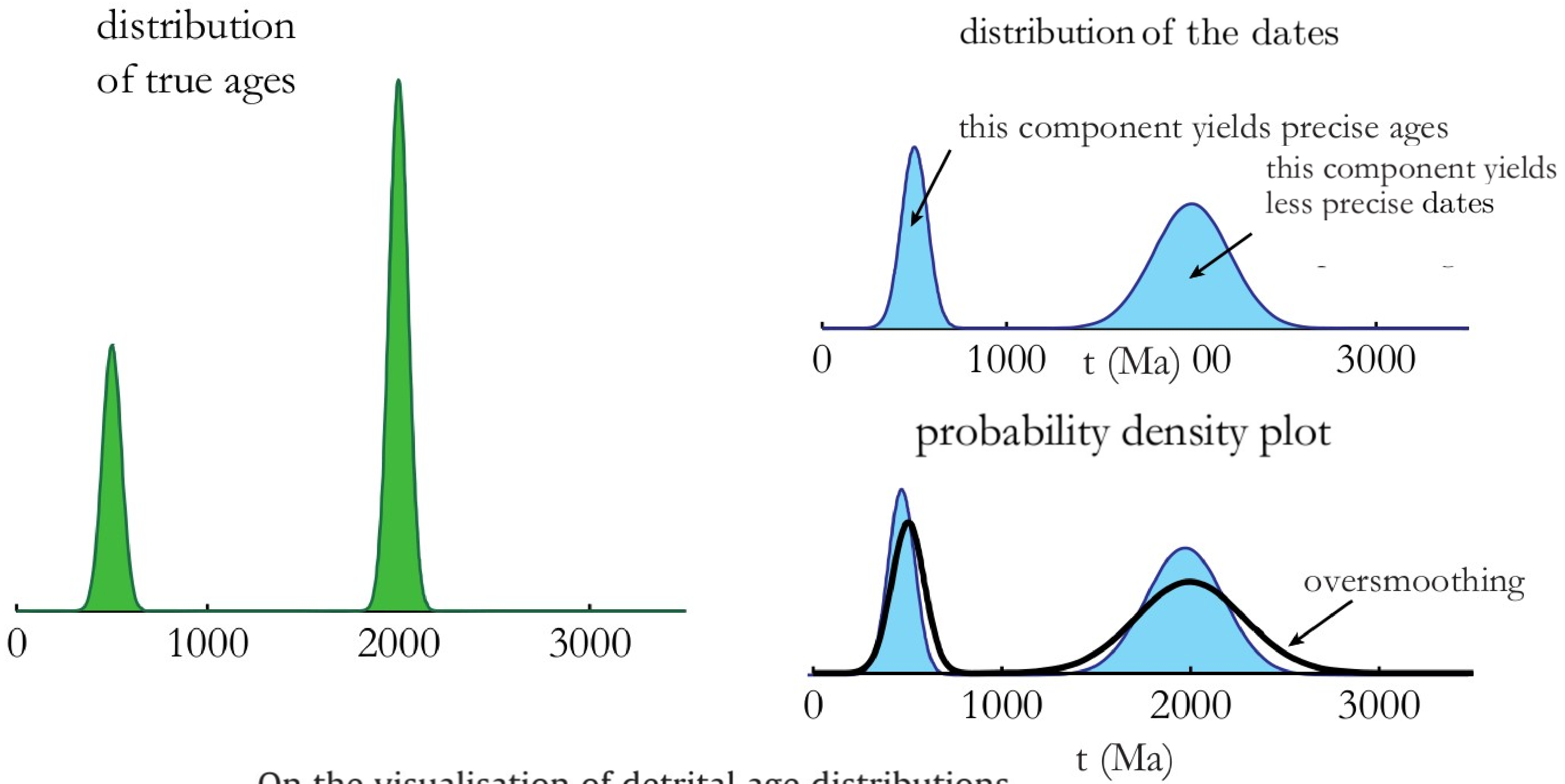
$$\mathrm{KDE}(x) = \frac{1}{nh\sqrt{2\pi}} \sum_{j=1}^{n} \exp\left[-\frac{1}{2}\left(\frac{x-\hat{t}_j}{h}\right)^2\right]$$





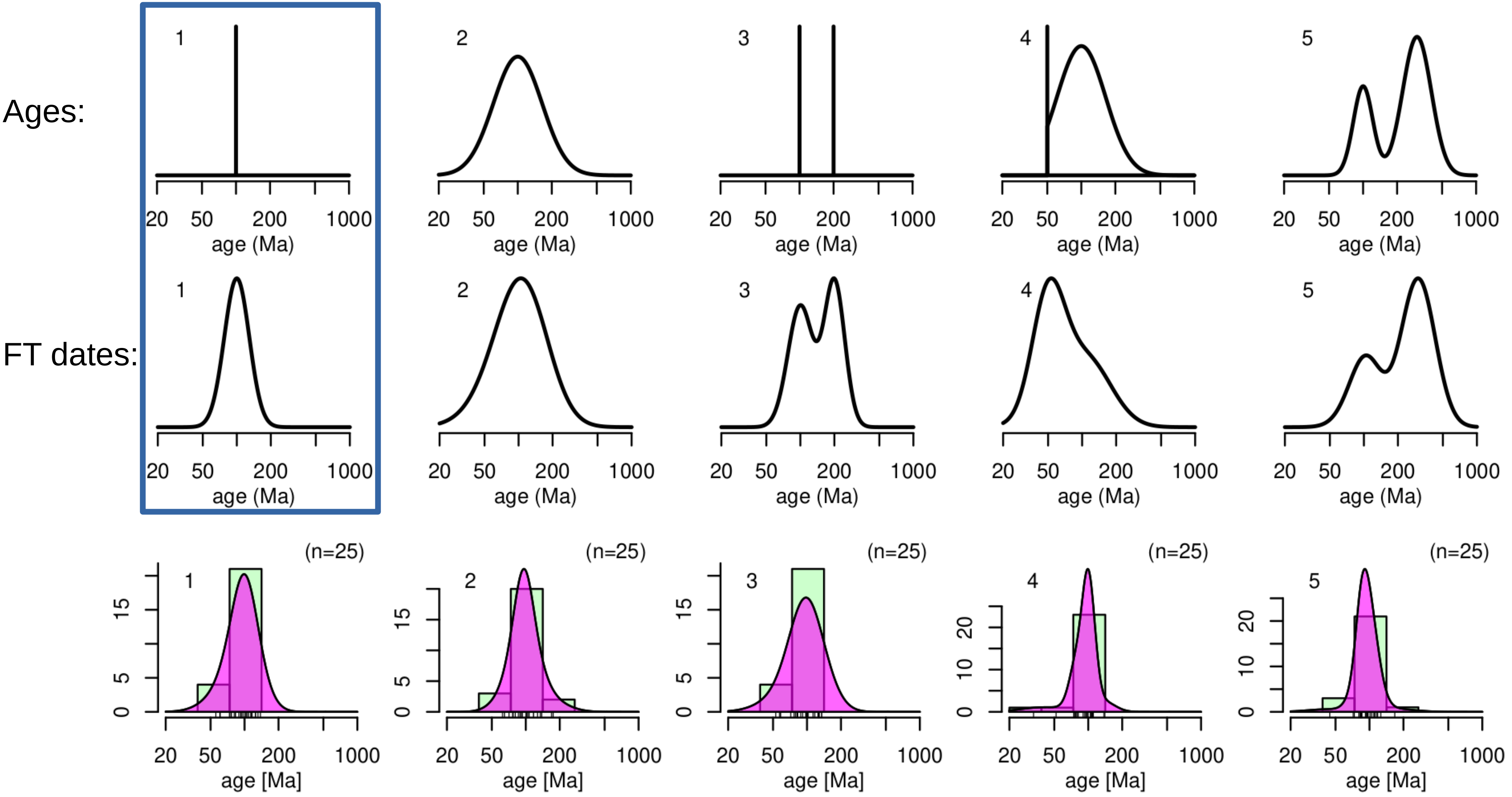


$$PDP(x) = \sum_{i=1}^{N} \frac{1}{e_i\sqrt{2\pi}} \exp^{-(t-x_i)^2/2e_i^2}$$

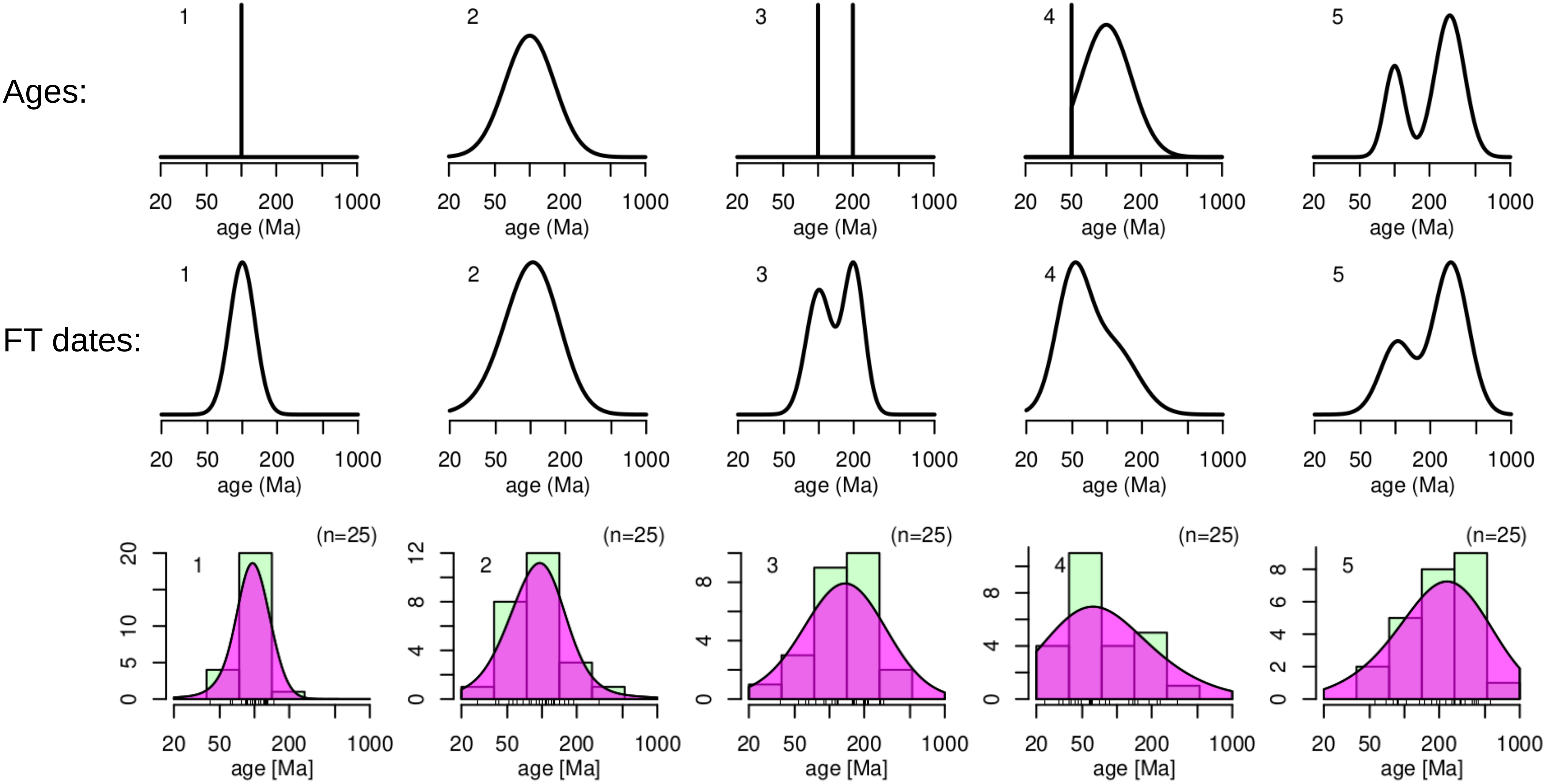


On the visualisation of detrital age distributions

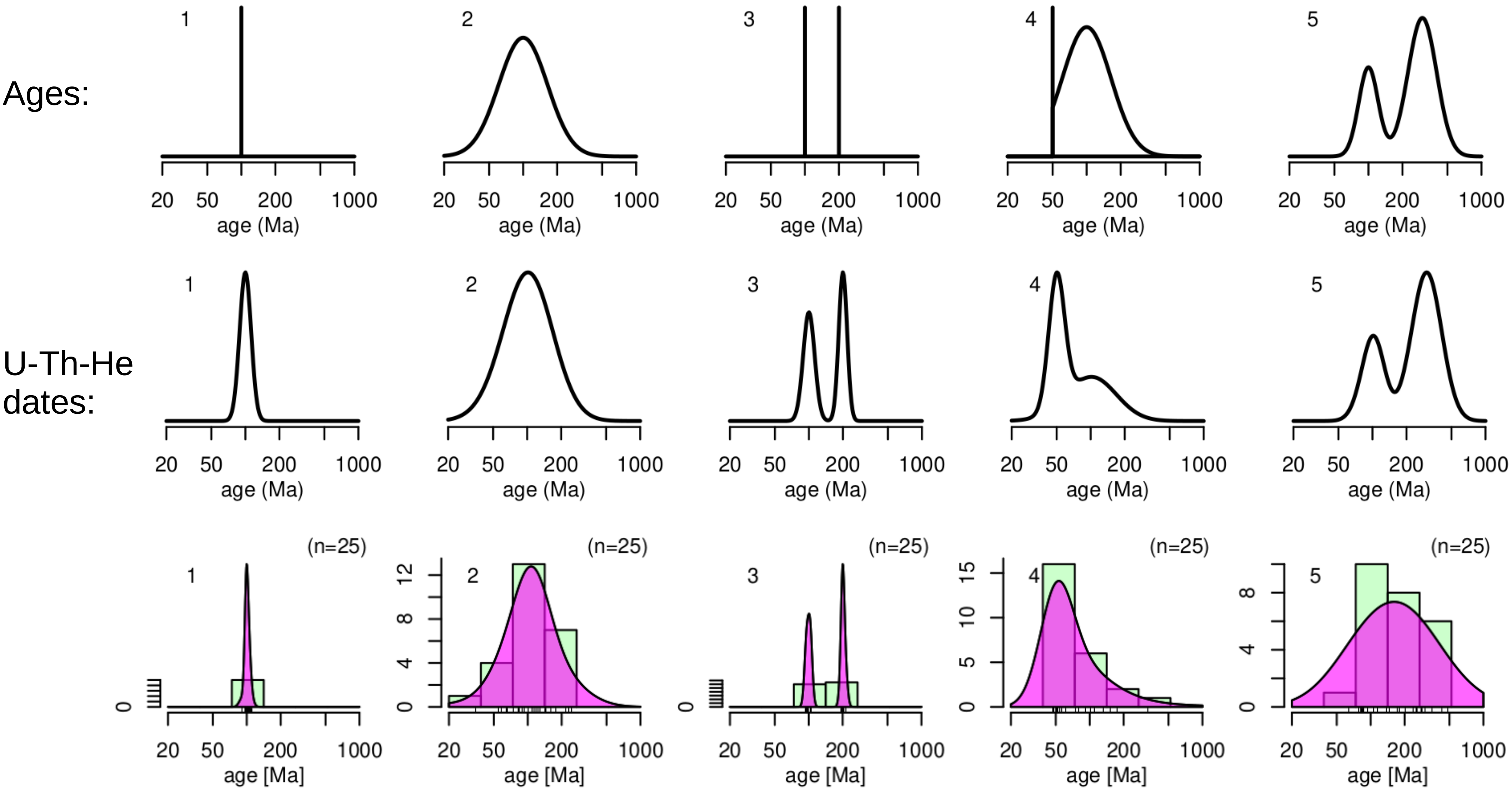
Pieter Vermeesch *



Ages:
FT dates:
1
2
3
4
5
20 50 200 1000
age (Ma)
(n=25)
age [Ma]
0 5 15
0 10 20



Ages:
FT dates:
1
2
3
4
5
20
50
200
1000
age (Ma)
age [Ma]
(n=25)
0
5
10
20
4
8
12
2
6

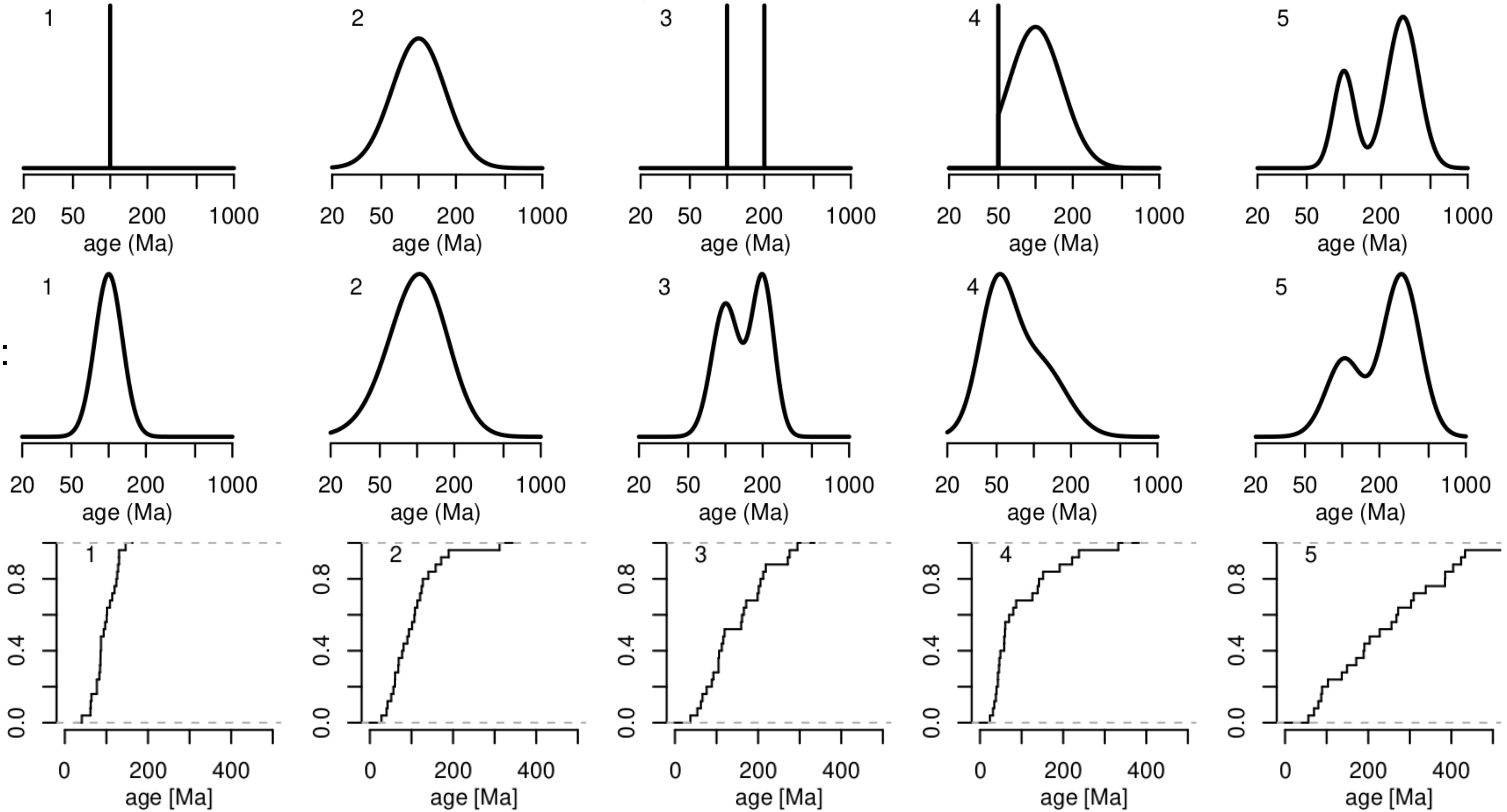


Ages:
U-Th-He
dates:
1
2
3
4
5
20
50
200
1000
age (Ma)
(n=25)
0
4
8
12
5
10
15
age [Ma]

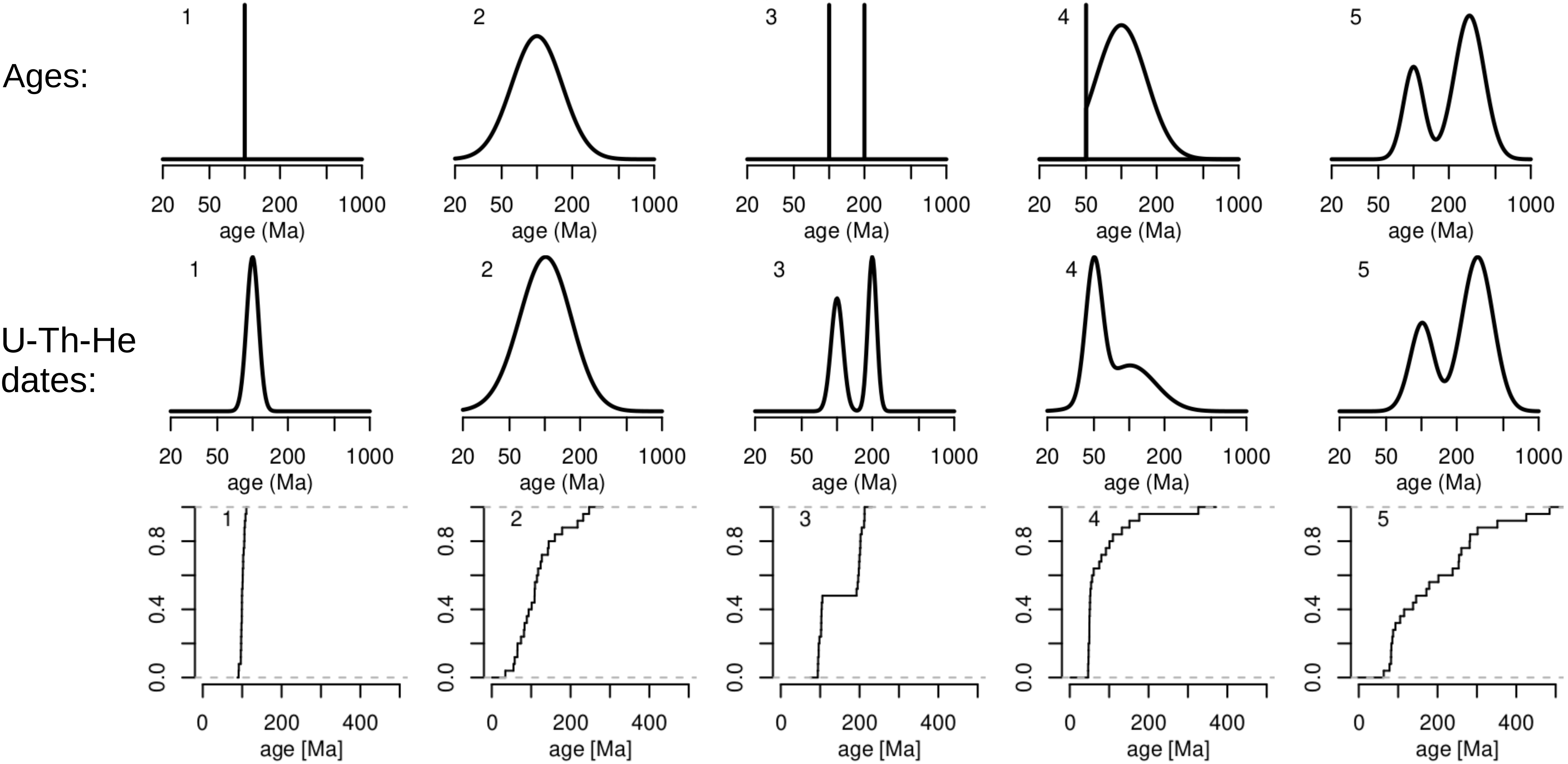
$$\mathrm{CAD}(x) = \frac{1}{n}\sum_{j=1}^{n} I(\hat{t}_j < x)$$

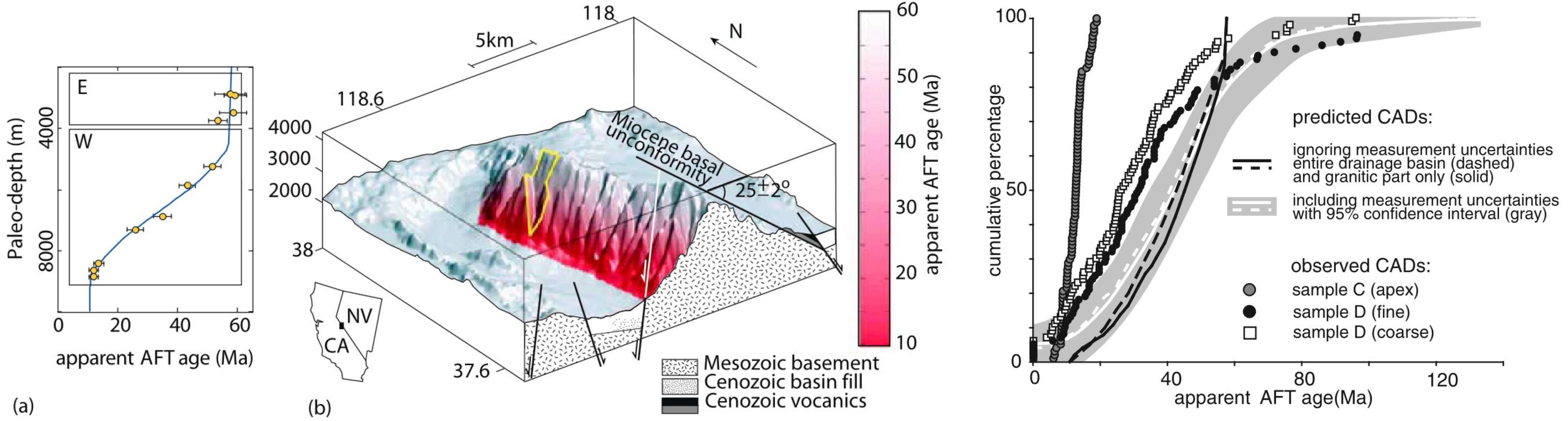
Ages:

FT dates:



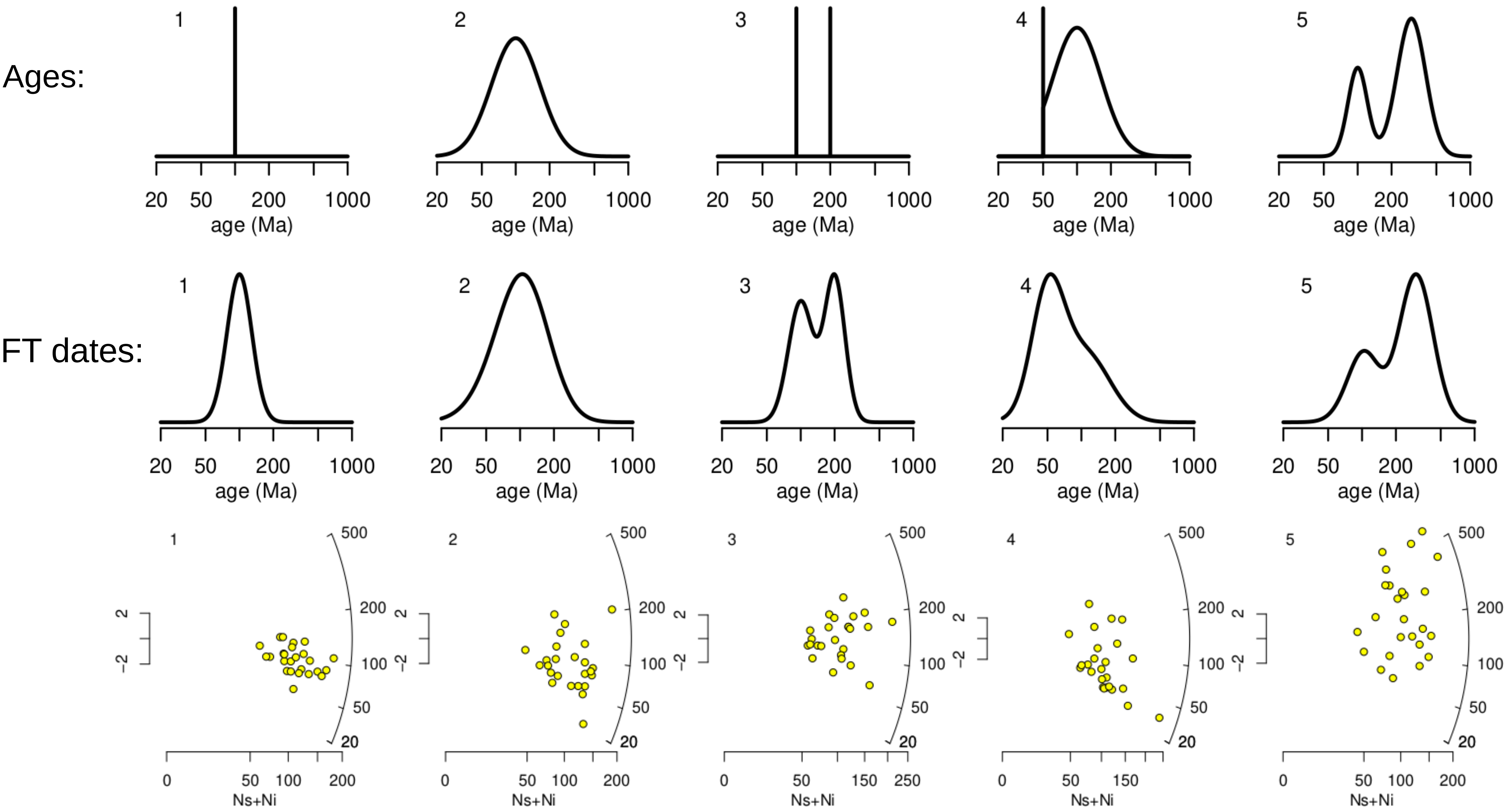
$$\mathrm{CAD}(x) = \frac{1}{n}\sum_{j=1}^{n} I(\hat{t}_j < x)$$



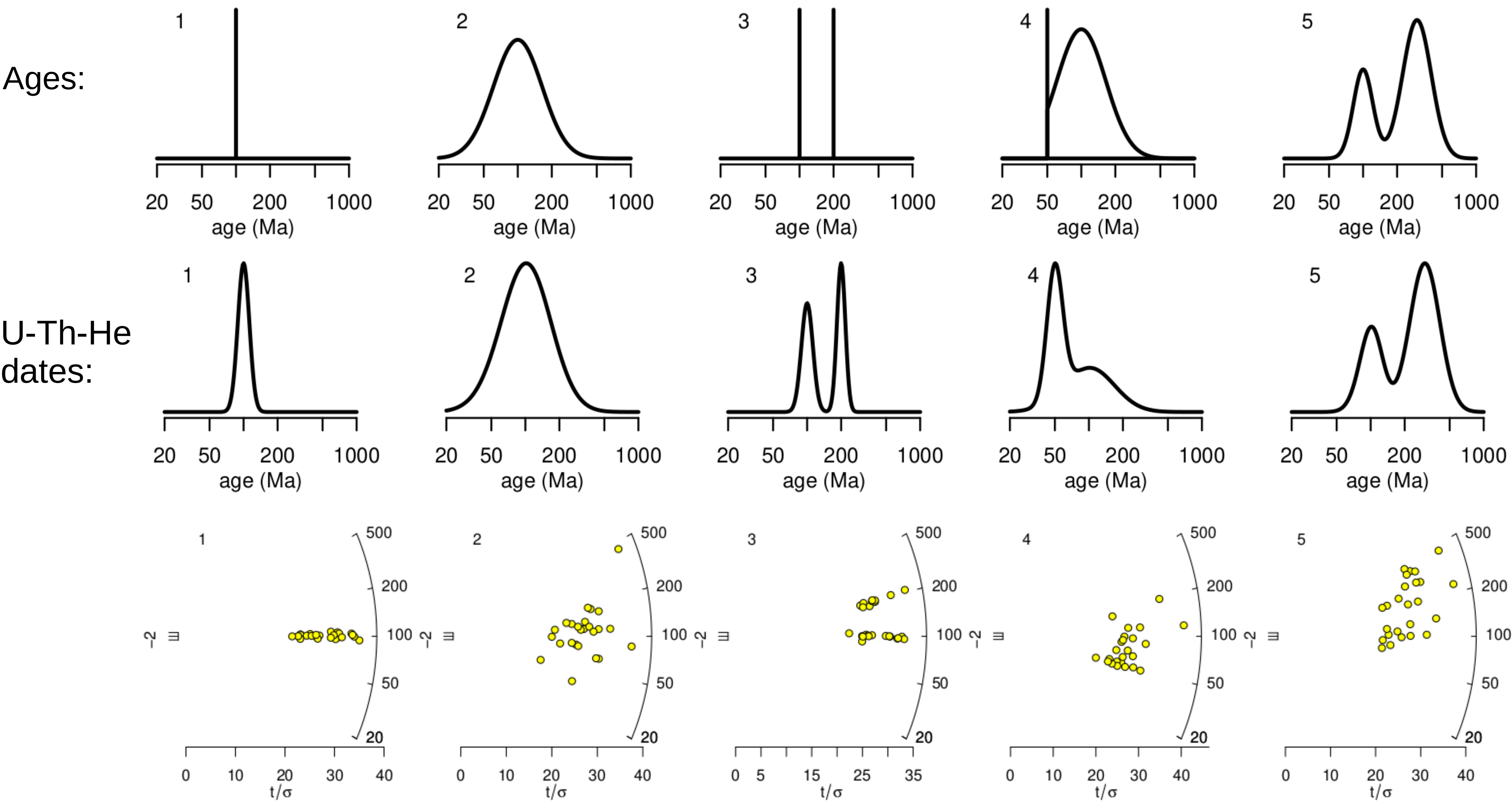


Quantitative geomorphology of the White Mountains (California) using detrital apatite fission track thermochronology

Pieter Vermeesch[1]



Ages:
FT dates:
1
2
3
4
5
20 50 200 1000
age (Ma)
Ns+Ni
500
200
100
50
20
2
−2
0 50 100 200
0 50 150 250
0 50 150


Ages:
U-Th-He dates:
1
2
3
4
5
20 50 200 1000
age (Ma)
500
200
100
50
20
0 10 20 30 40
0 5 15 25 35
t/σ

Home

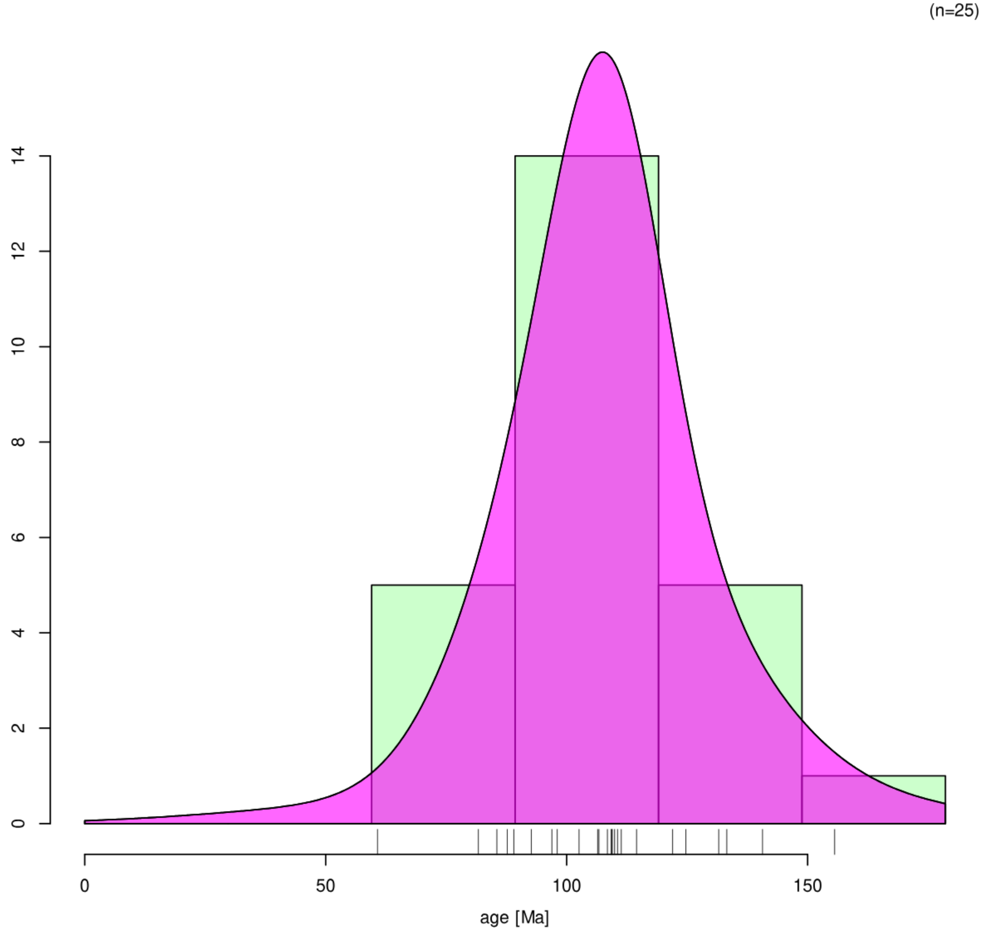
fission tracks | KDE | Options | Help | English

IsoplotR

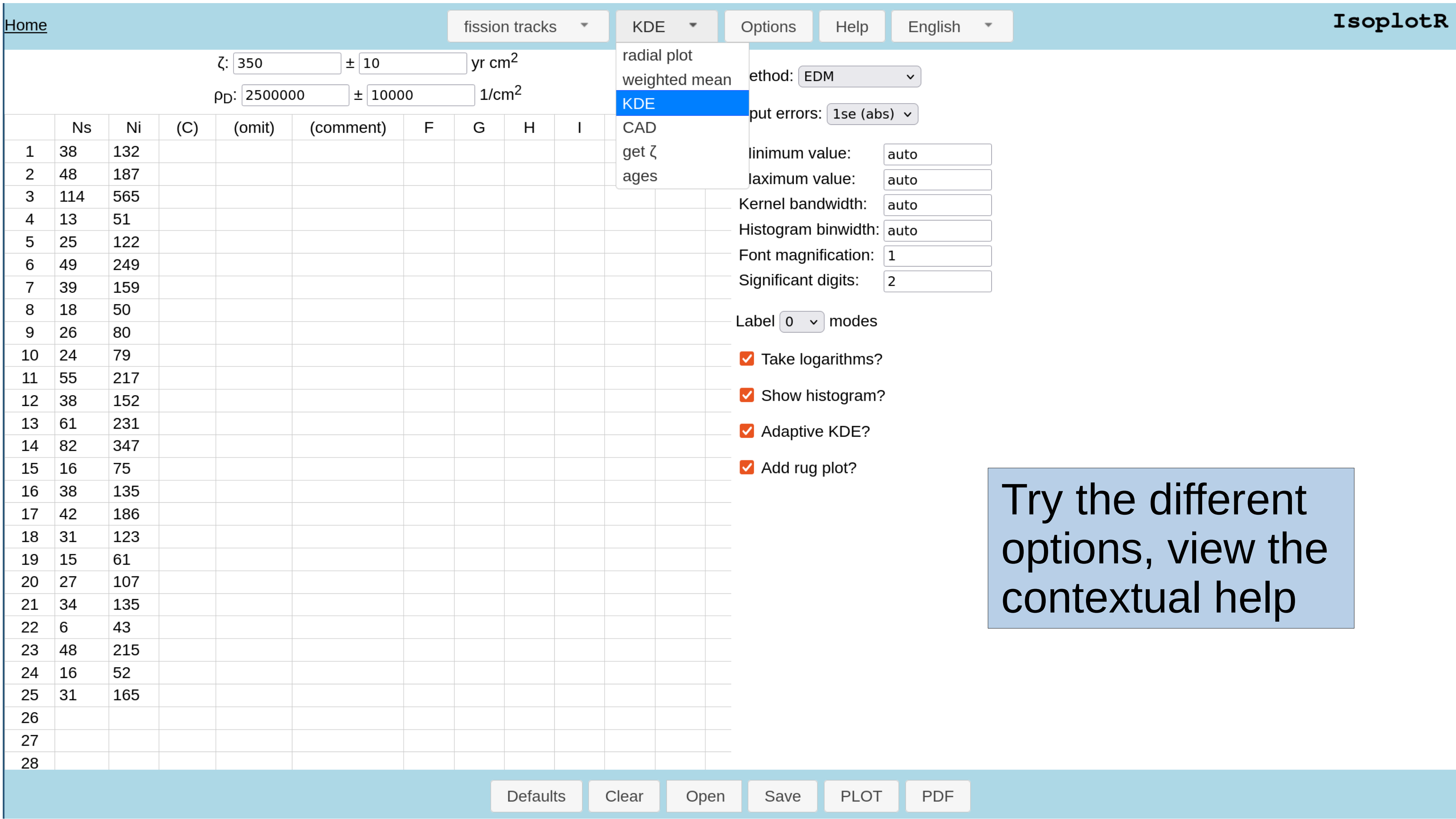
ζ: 350 ± 10 yr cm^2

ρ_D: 2500000 ± 10000 1/cm^2

	Ns	Ni	(C)	(omit)	(comment)	F	G	H	I	J	K	L
1	38	132										
2	48	187										
3	114	565										
4	13	51										
5	25	122										
6	49	249										
7	39	159										
8	18	50										
9	26	80										
10	24	79										
11	55	217										
12	38	152										
13	61	231										
14	82	347										
15	16	75										
16	38	135										
17	42	186										
18	31	123										
19	15	61										
20	27	107										
21	34	135										
22	6	43										
23	48	215										
24	16	52										
25	31	165										
26												
27												
28												



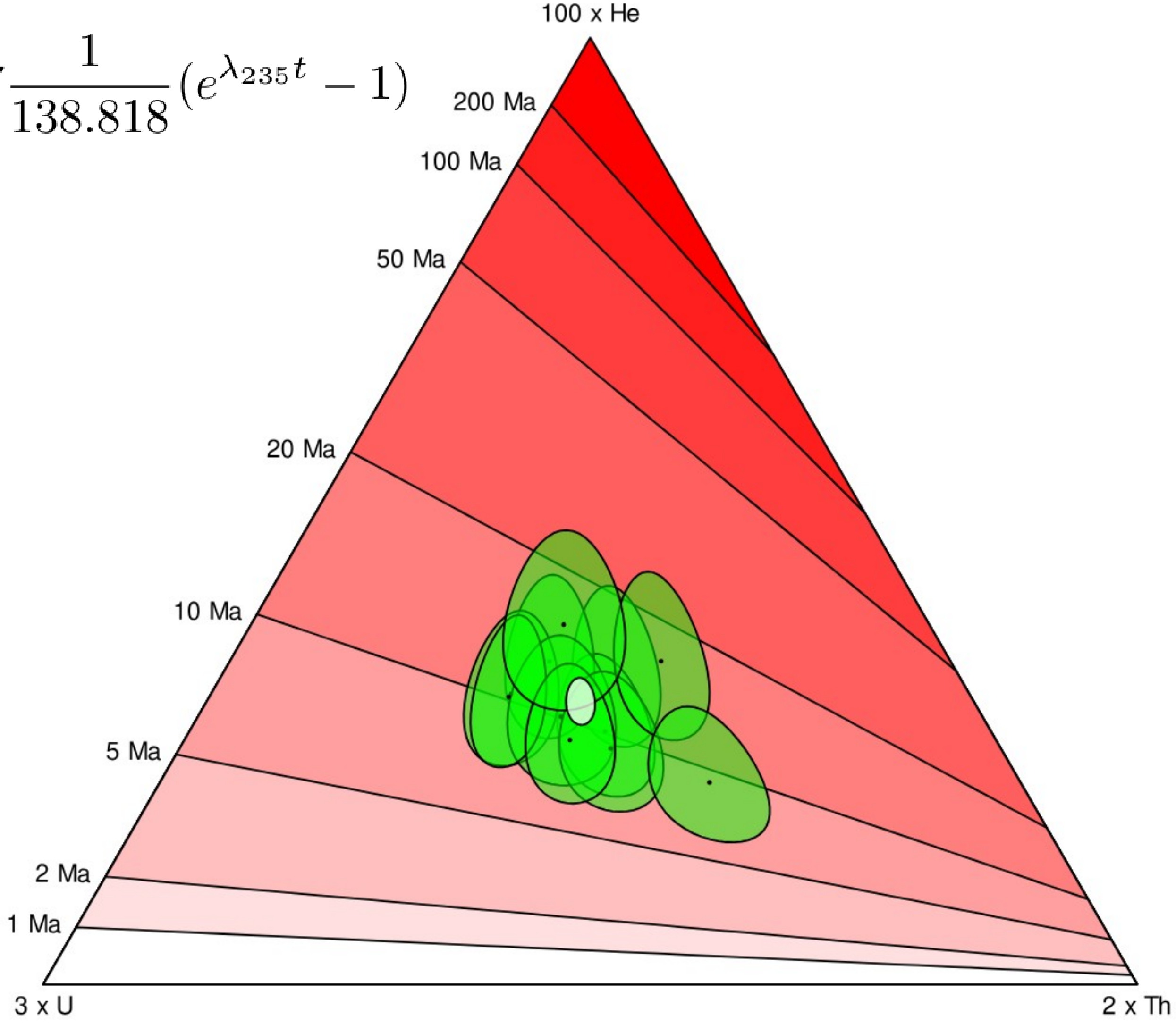
Defaults | Clear | Open | Save | PLOT | PDF

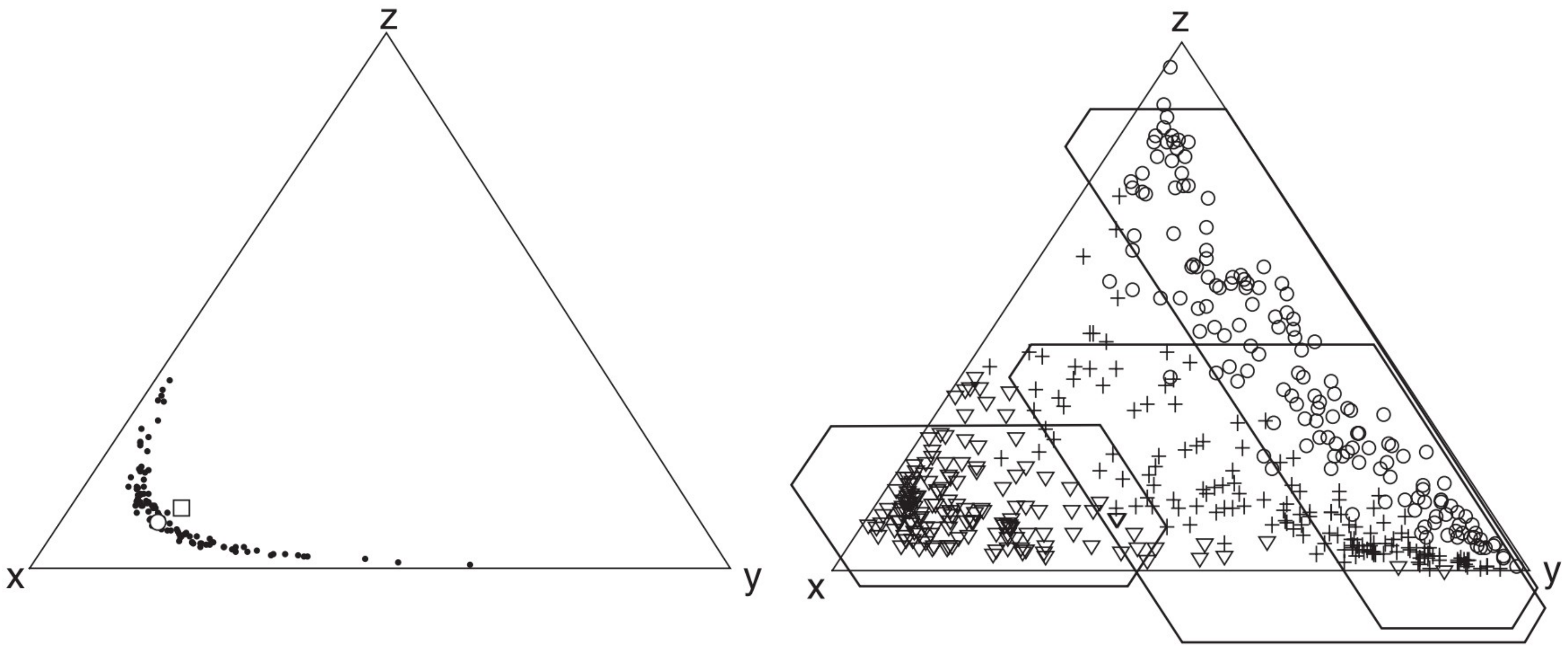


Try the different options, view the contextual help

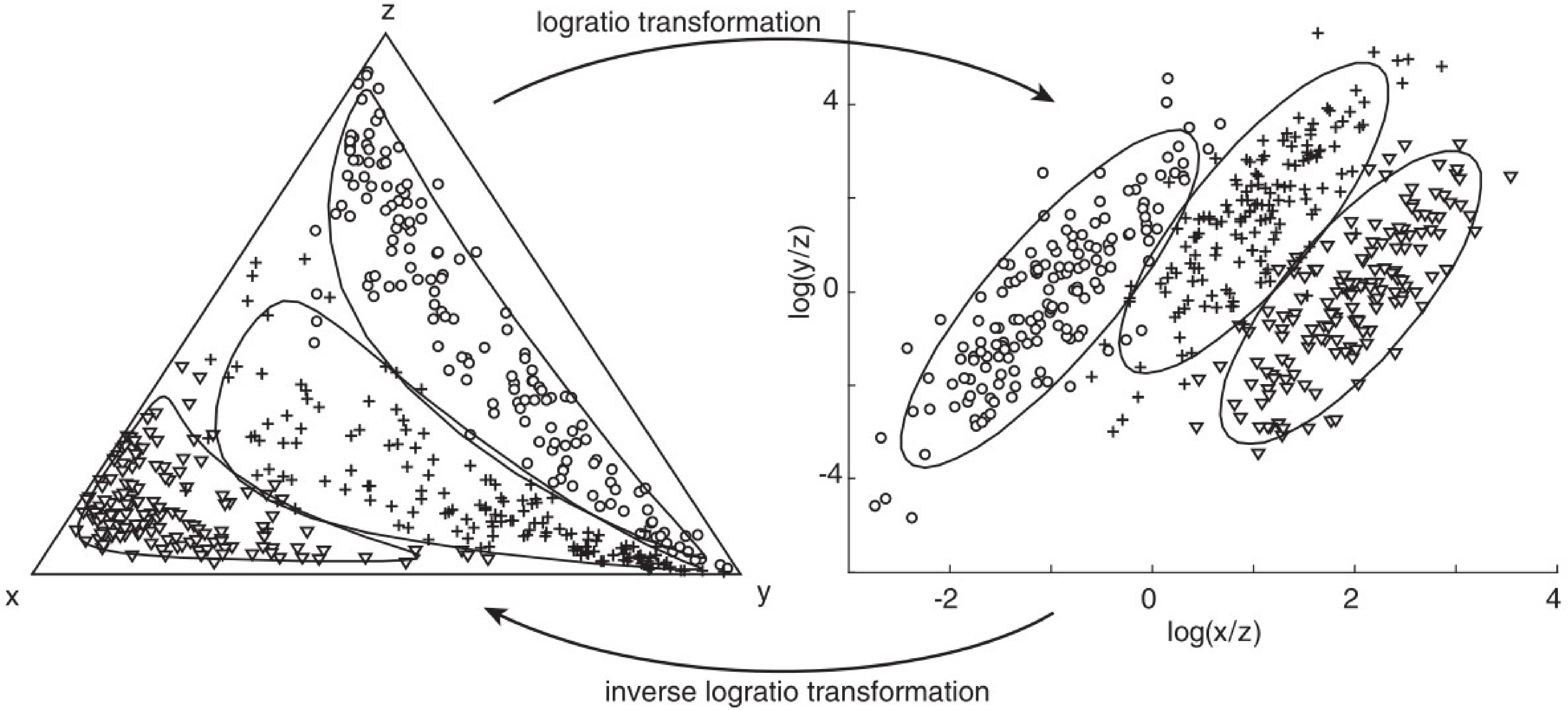
```
library(IsoplotR)

source("helper.R")

FT <- FTsamp(pop=1,ng=50)

kde(FT)

cad(FT)

radialplot(FT)
```

$$[\mathrm{He}] = a(t)[\mathrm{U}] + b(t)[\mathrm{Th}]$$
$$\text{where } a(t) = 8\frac{137.818}{138.818}(e^{\lambda_{238}t} - 1) + 7\frac{1}{138.818}(e^{\lambda_{235}t} - 1)$$
$$\text{and } b(t) = 6(e^{\lambda_{232}t} - 1)$$

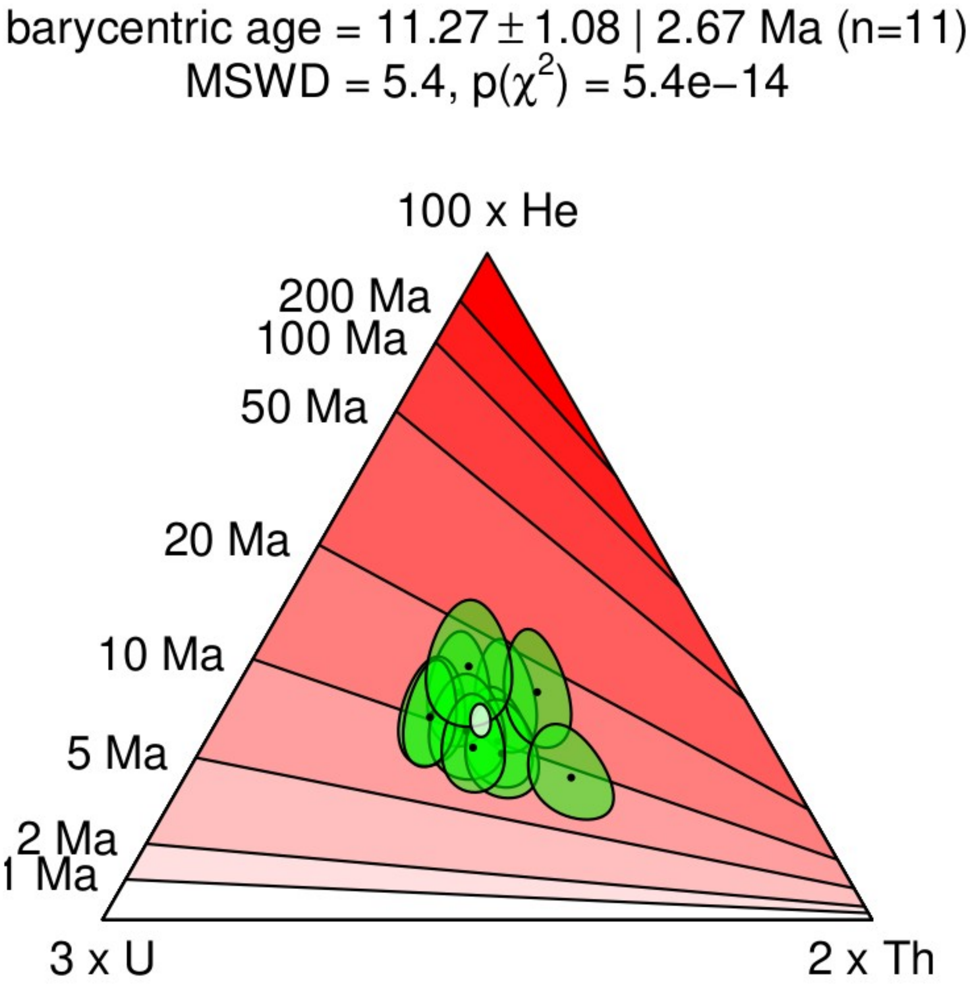


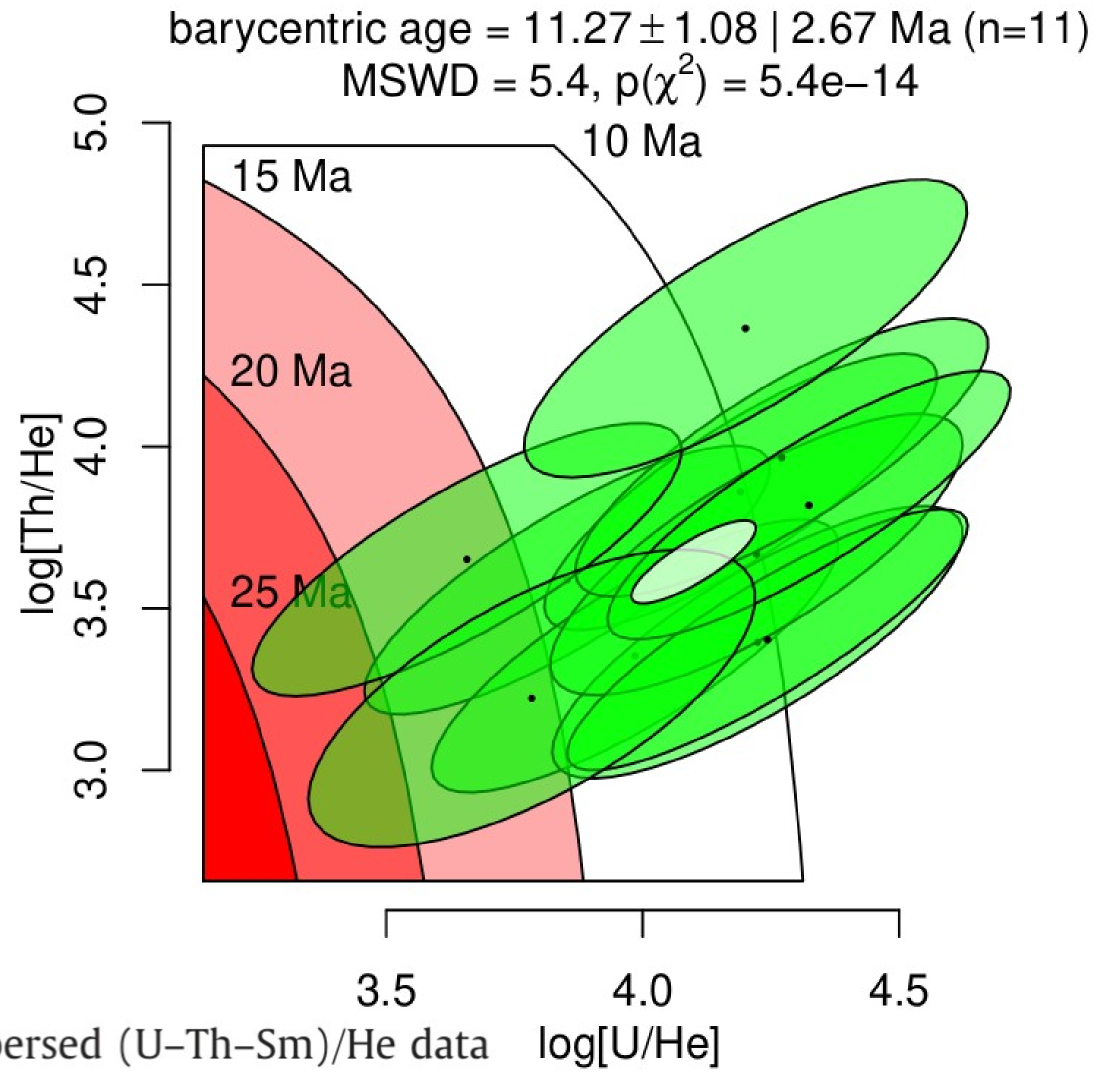


z
x
y
z
x
y



z
x
y
logratio transformation
inverse logratio transformation
log(y/z)
log(x/z)
4
0
-4
-2
0
2
4

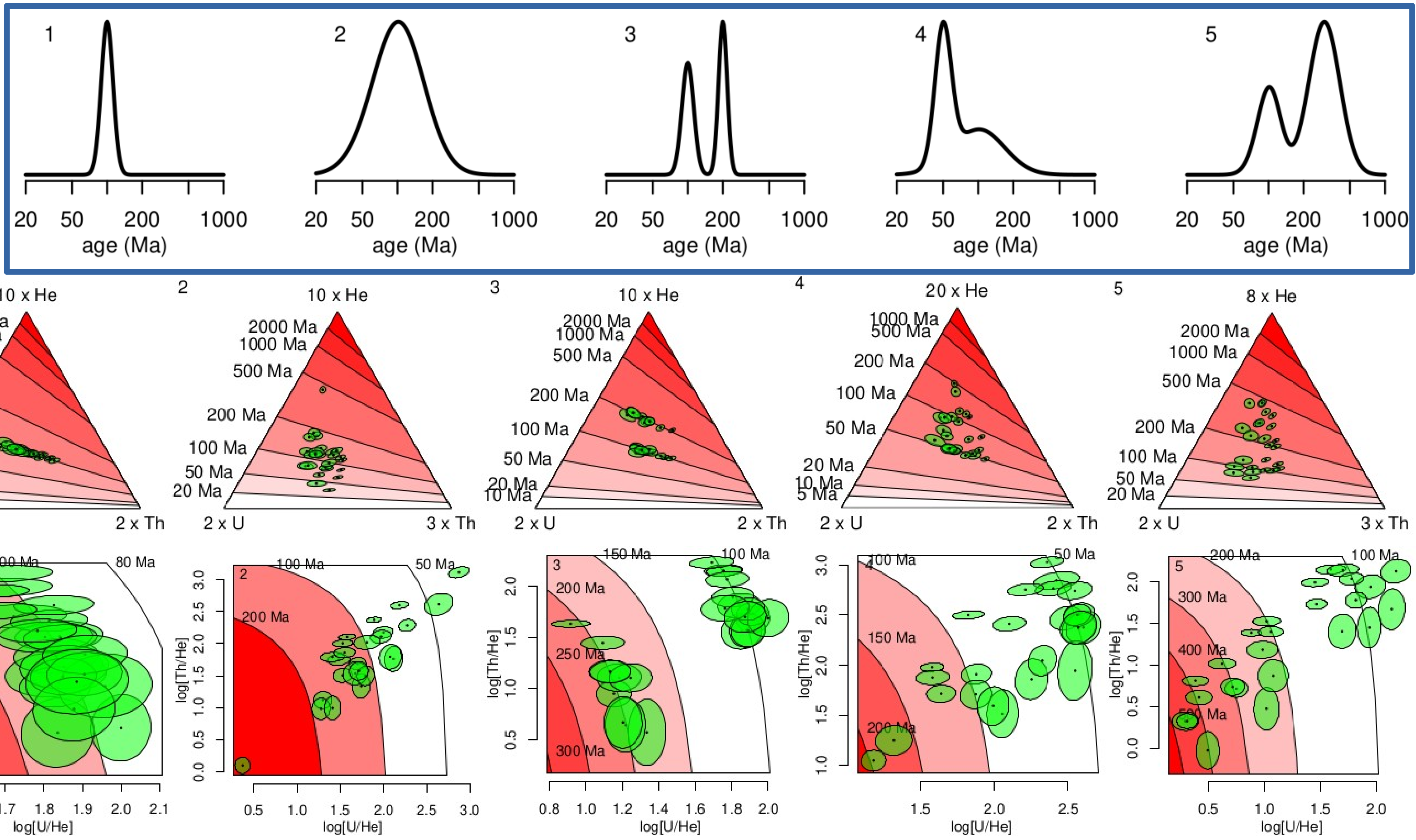




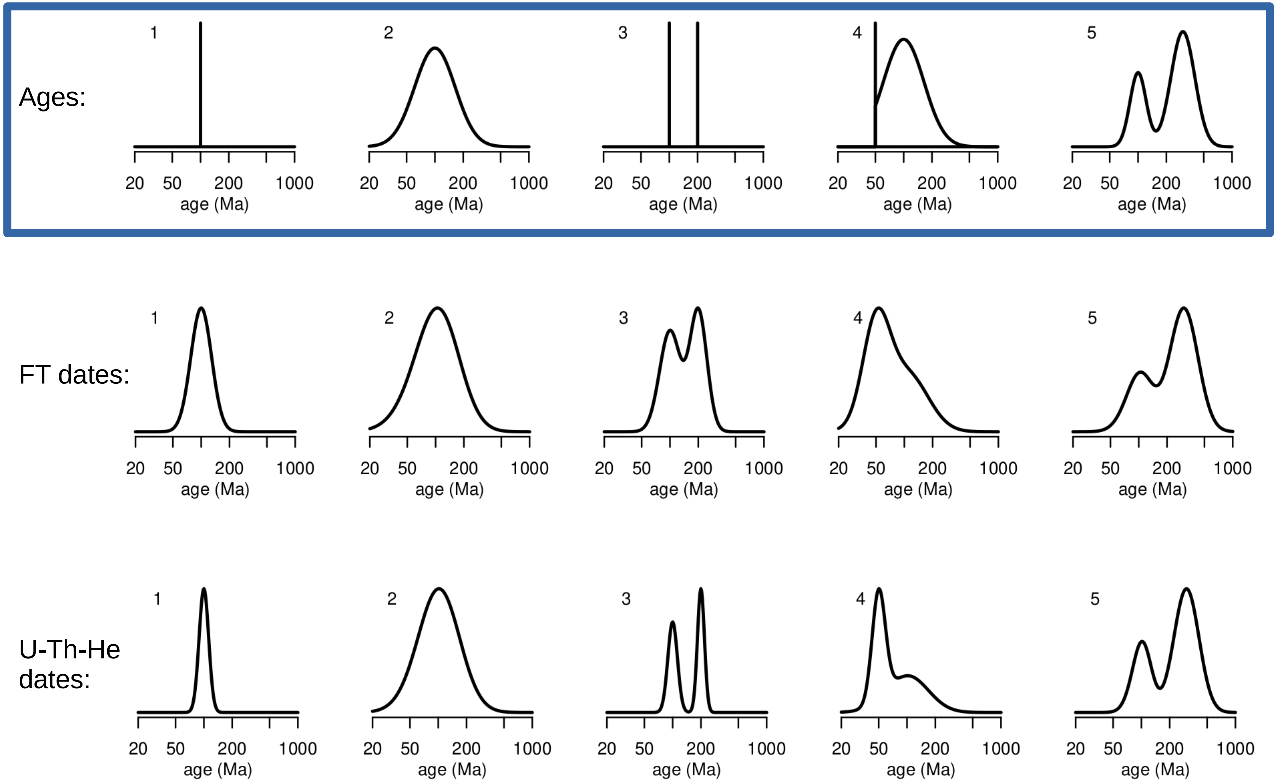
HelioPlot, and the treatment of overdispersed (U–Th–Sm)/He data

Pieter Vermeesch

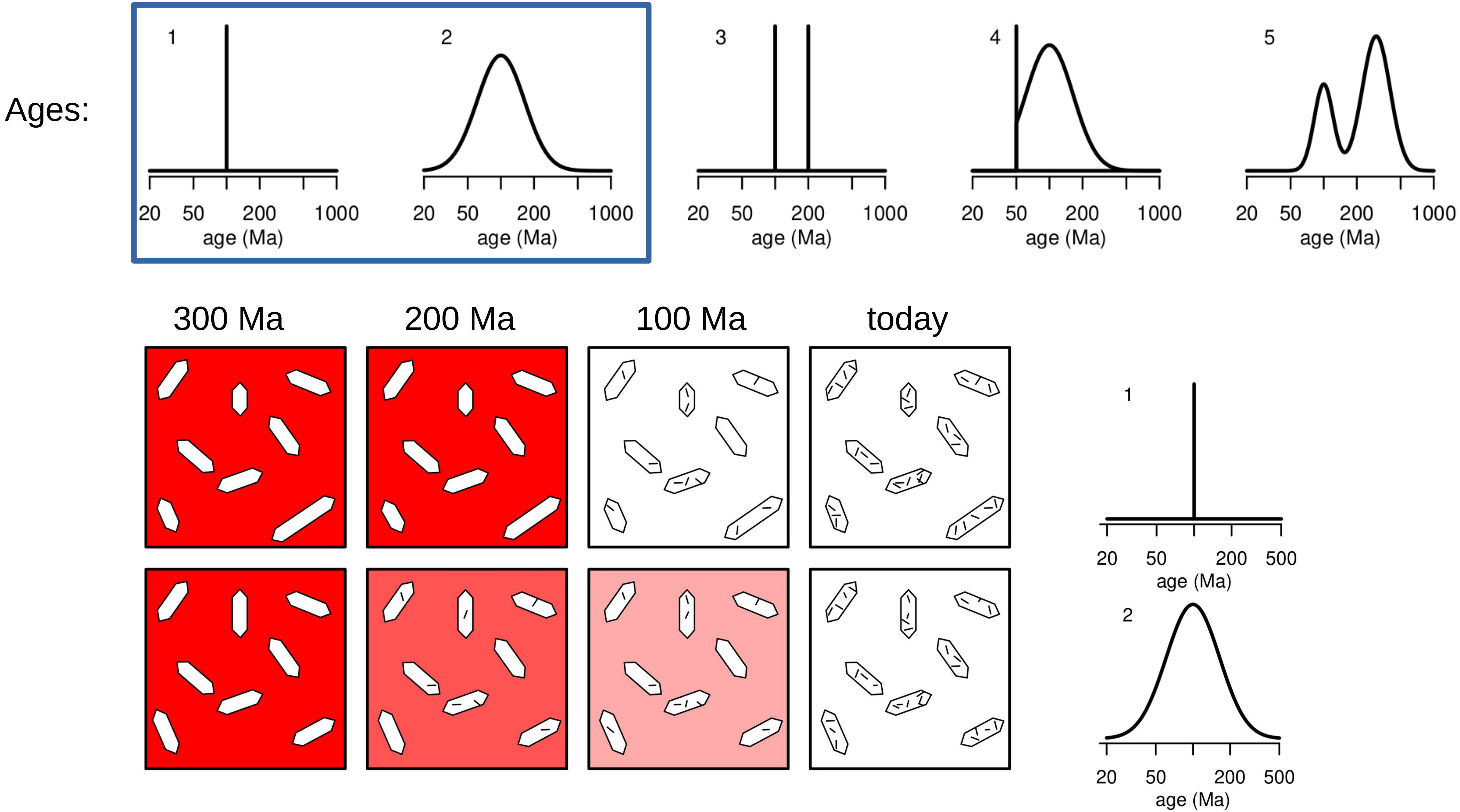
U-Th-He dates:

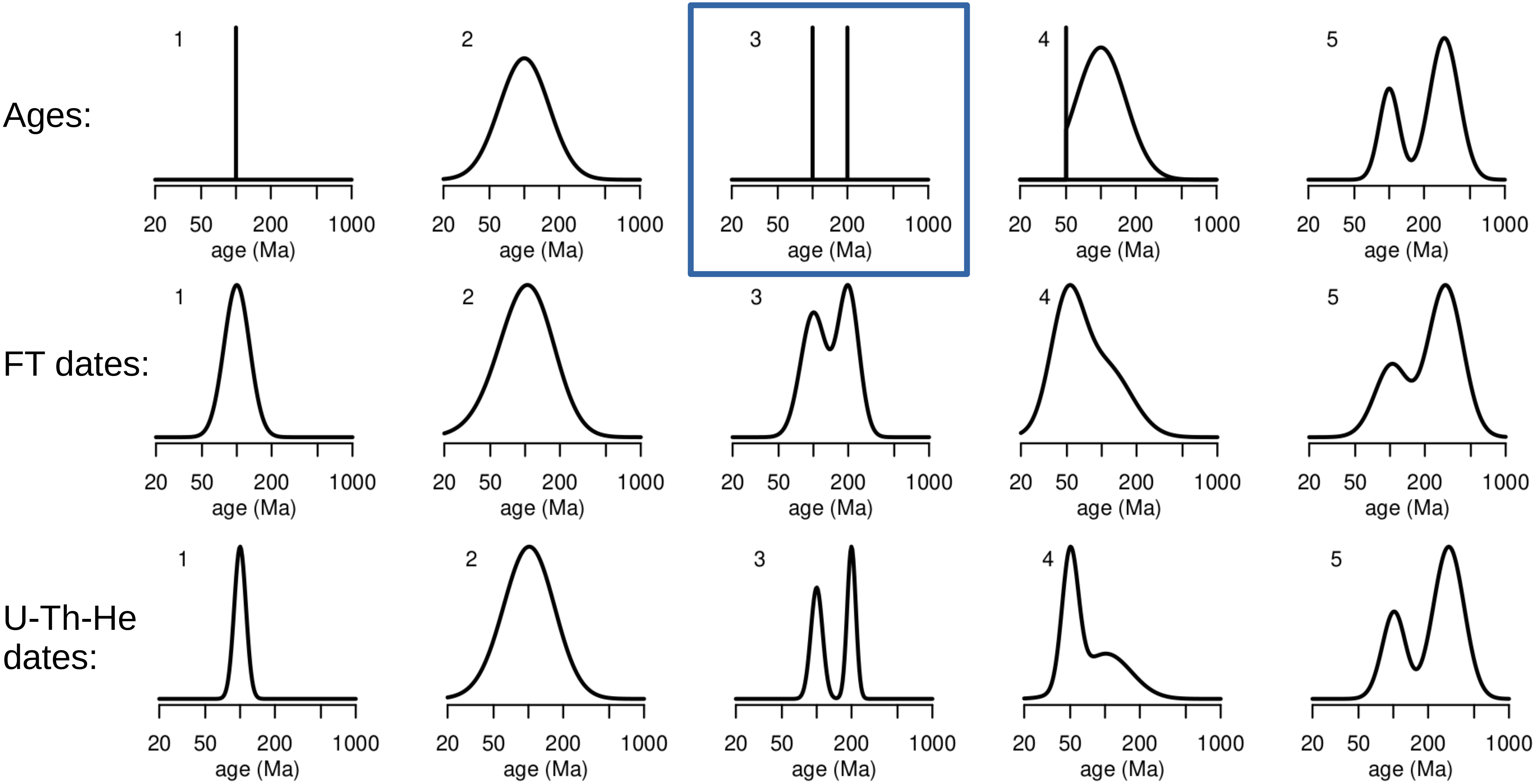

1
2
3
4
5
20 50 200 1000
age (Ma)
10 x He
20 x He
8 x He
2 x U
2 x Th
3 x Th
2000 Ma
1000 Ma
500 Ma
200 Ma
100 Ma
50 Ma
20 Ma
10 Ma
5 Ma
log[Th/He]
log[U/He]
80 Ma
100 Ma
120 Ma
140 Ma
150 Ma
200 Ma
250 Ma
300 Ma
400 Ma
500 Ma

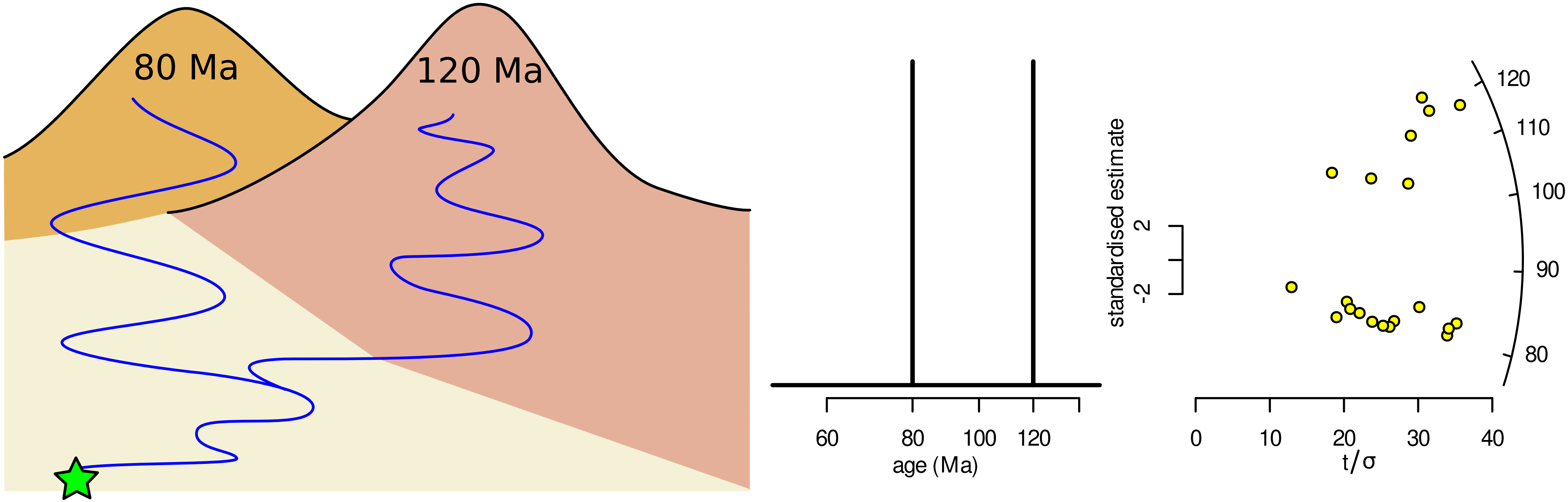
```
He <- UThHesamp(pop=1,ng=20)

helioplot(He)

helioplot(He,logratio=FALSE)
```



Contents


Ages:
1
20 50 200 1000
age (Ma)
2
20 50 200 1000
age (Ma)
3
20 50 200 1000
age (Ma)
4
20 50 200 1000
age (Ma)
5
20 50 200 1000
age (Ma)
300 Ma
200 Ma
100 Ma
today
1
20 50 200 500
age (Ma)
2
20 50 200 500
age (Ma)


Ages:
FT dates:
U-Th-He dates:
1
2
3
4
5
20 50 200 1000
age (Ma)


80 Ma
120 Ma
60
80
100
120
age (Ma)
standardised estimate
2
-2
0
10
20
30
40
t/σ
120
110
100
90
80

Maximise

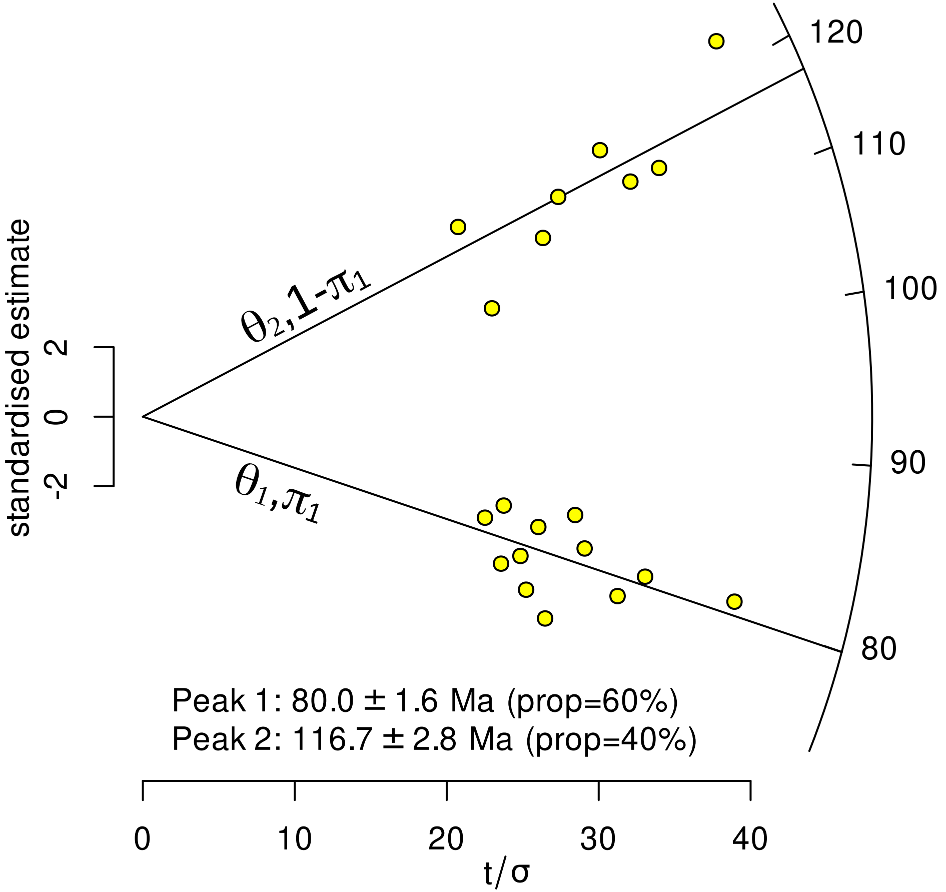
$$\mathcal{L}(\pi_k, \theta_k, k = 1 \ldots K) = \sum_{i=1}^{n} \ln\left[\sum_{k=1}^{K} \pi_k f(\theta_k | z_i, s[z_i])\right]$$

where $\pi_K = 1 - \sum_{k-1}^{K-1} \pi_k$

$z_i = \ln(t_i)$

$s[z_i] = \frac{s[t_i]}{t_i}$, and

$$f(\theta_k | z_i, s[z_i]) = \frac{1}{\sqrt{2\,\pi}\, s[z_i]} \exp\left[-\frac{(z_i - \theta_k)^2}{2\, s[z_i]^2}\right]$$



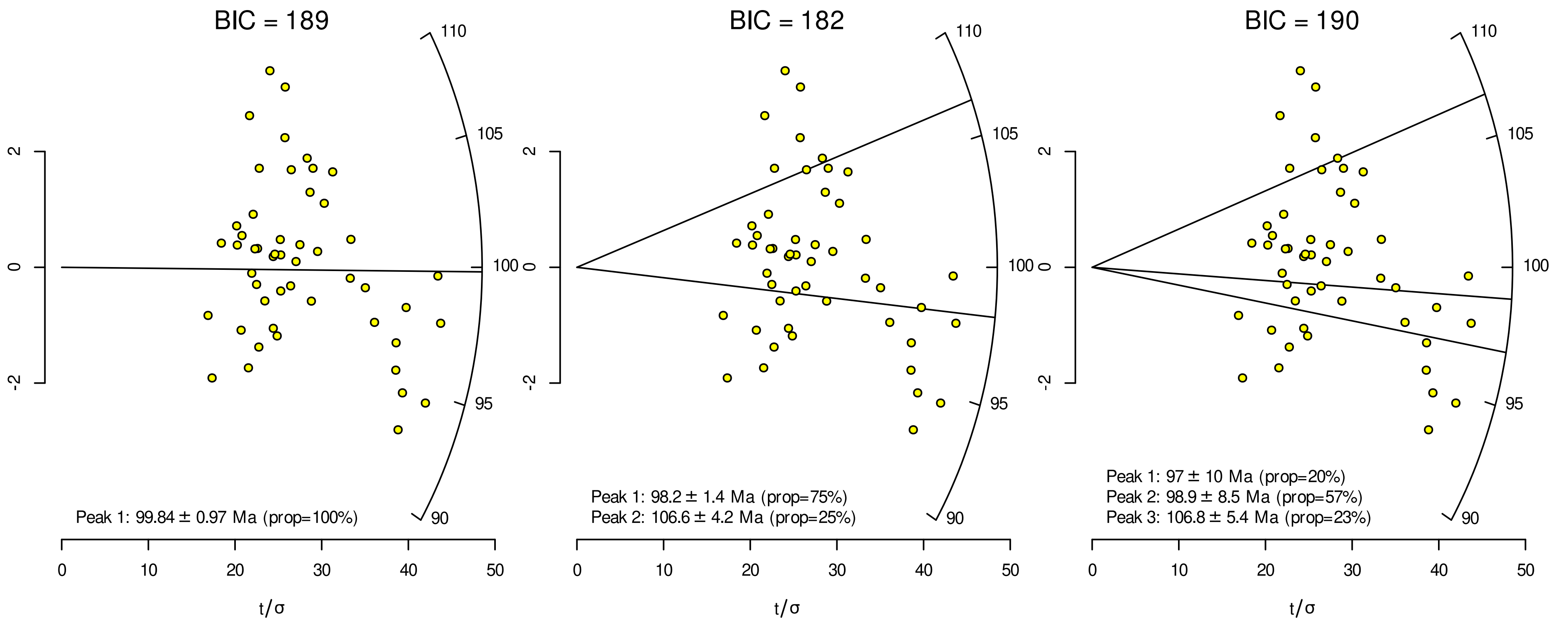
STATISTICAL MODELS FOR MIXED FISSION TRACK AGES

R. F. GALBRAITH* and G. M. LASLETT†

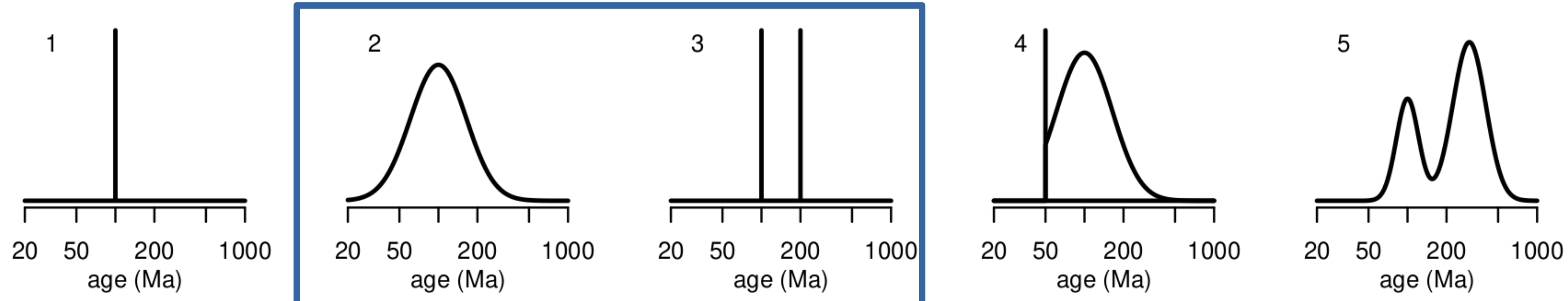
How many peaks to fit?

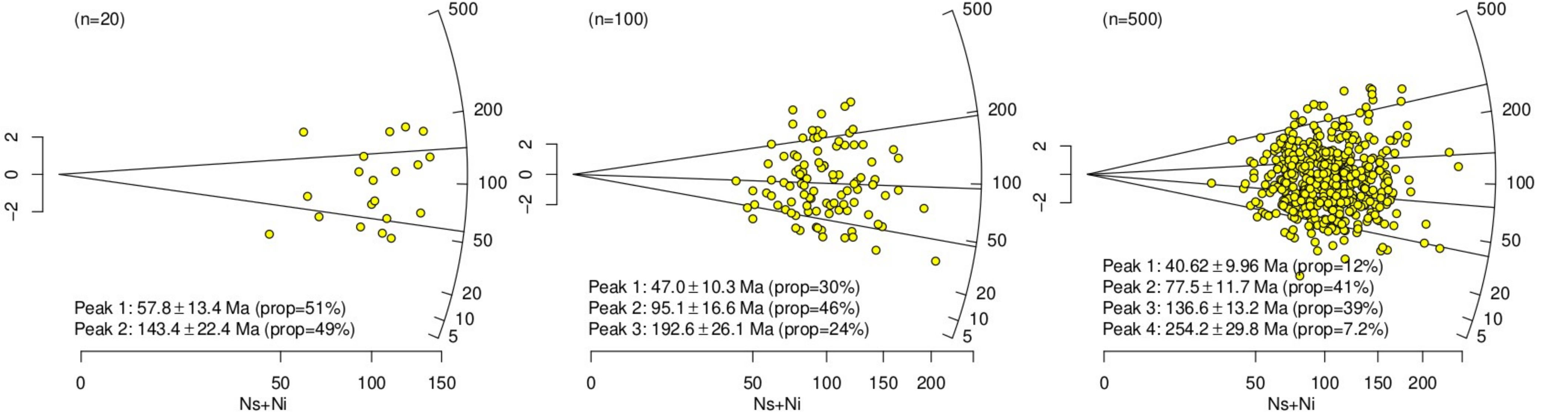
$$\mathcal{L}(\pi_k, \theta_k, k = 1 \ldots K) = \sum_{i=1}^{n} \ln \left[\sum_{k=1}^{K} \pi_k f(\theta_k | z_i, s[z_i]) \right]$$

$$\mathrm{BIC} = -2\mathcal{L}_{\max} + p \ln(n)$$



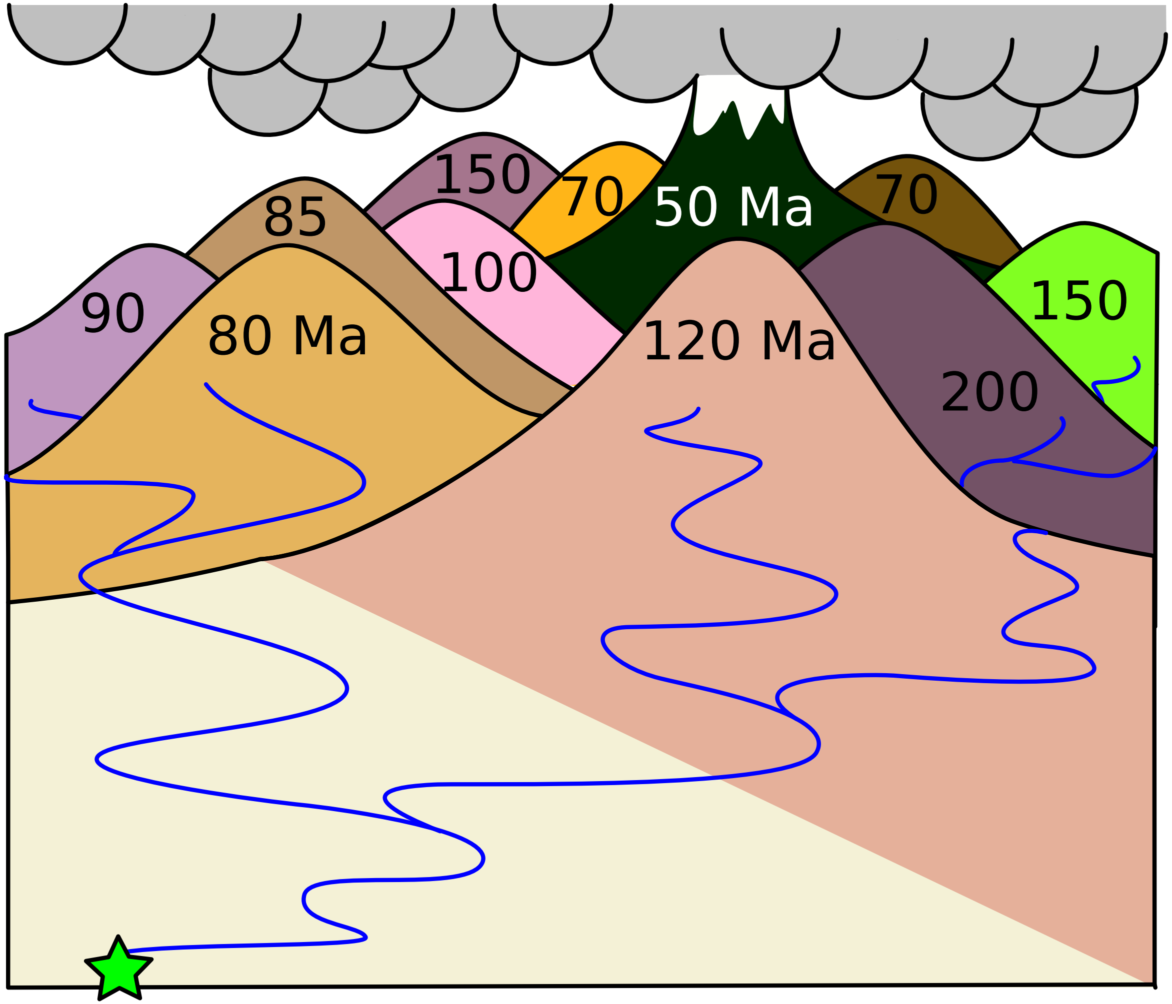
Ages:

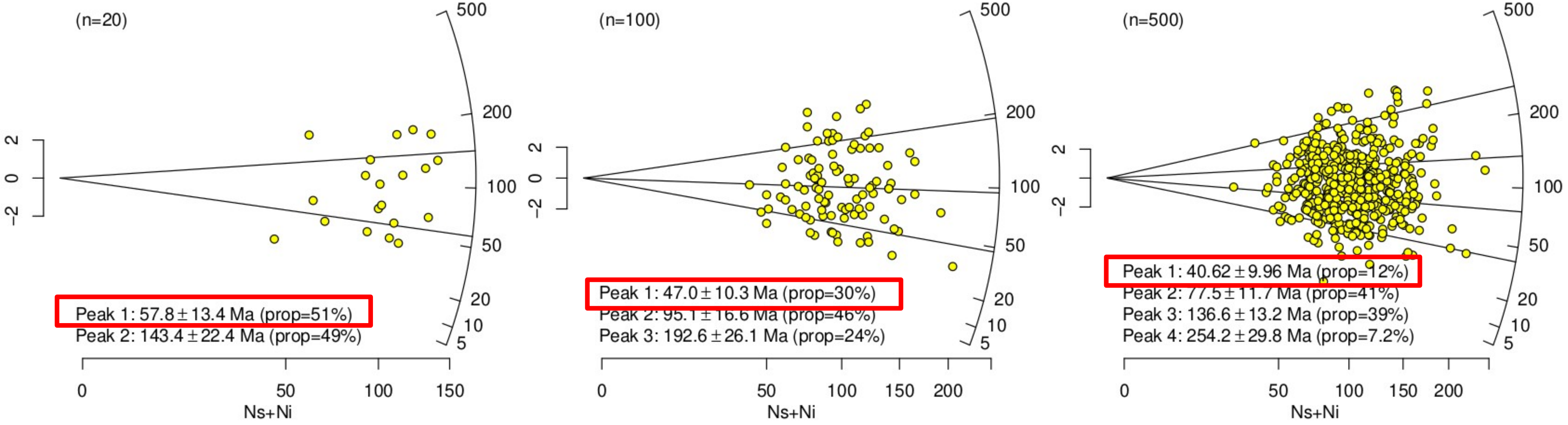




Ghost age components in detrital thermochronology

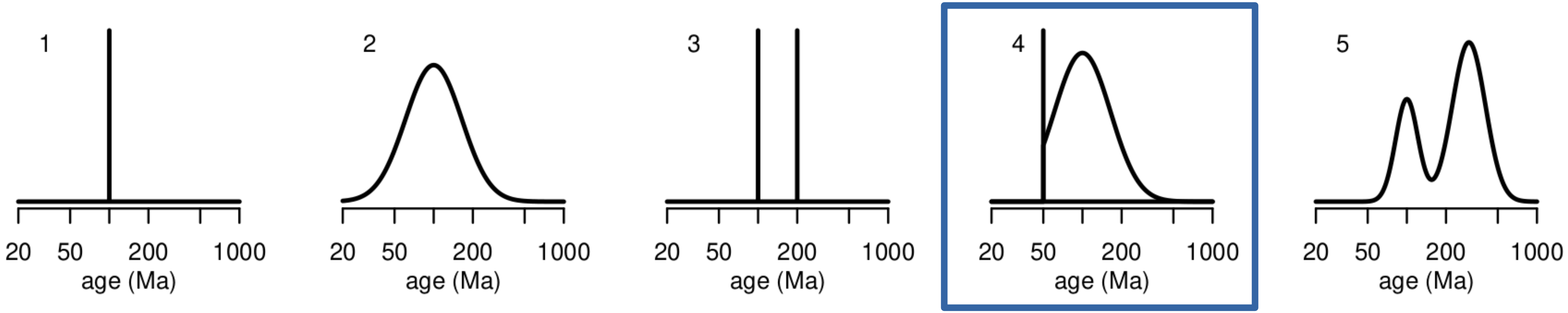
B. Härtel [a,*], Pieter Vermeesch [b], Eva Enkelmann [a], Stijn Glorie [c]

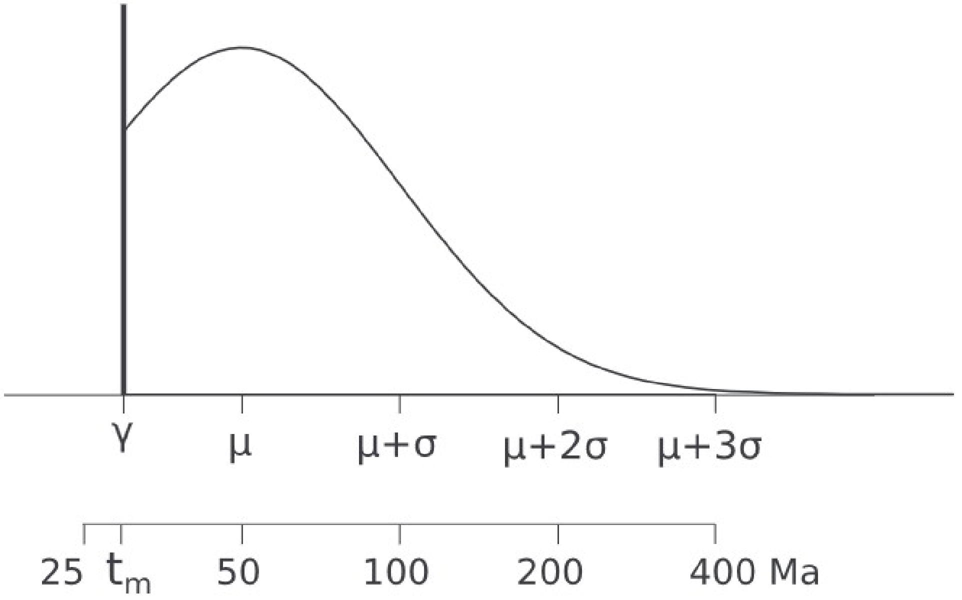

150
70
50 Ma
70
85
90
100
80 Ma
150
120 Ma
200

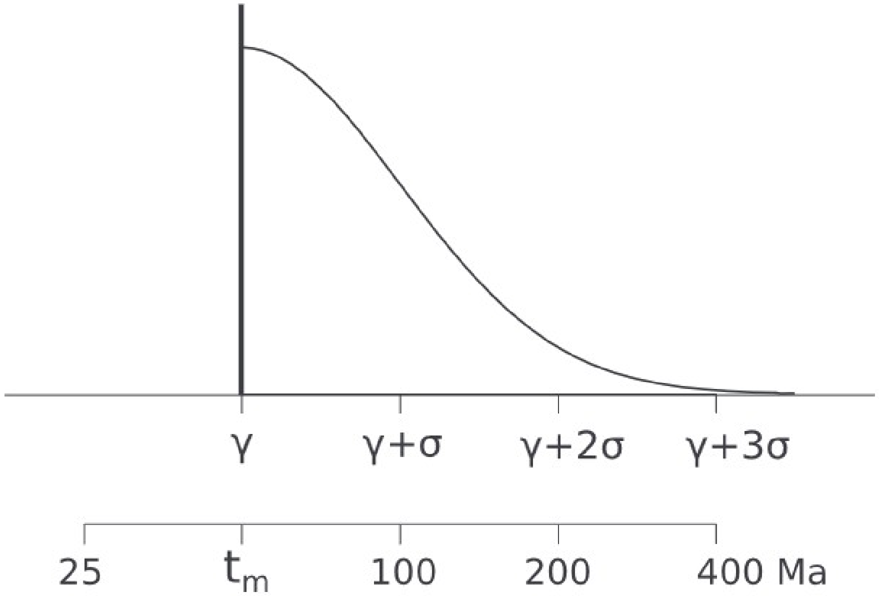


The minimum age drifts to younger ages with increasing sample size

Ages:







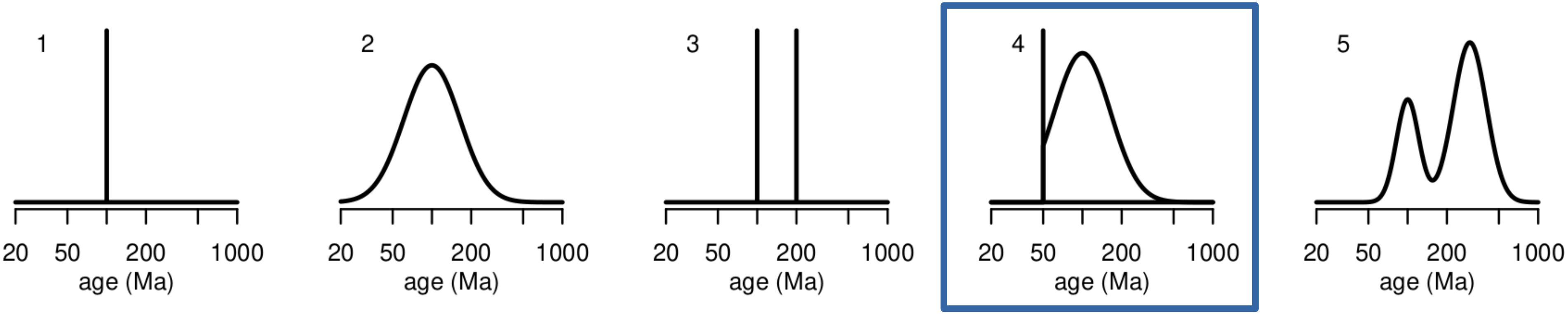
STATISTICAL MODELS FOR MIXED FISSION TRACK AGES

R. F. Galbraith* and G. M. Laslett†

Maximum depositional age estimation revisited

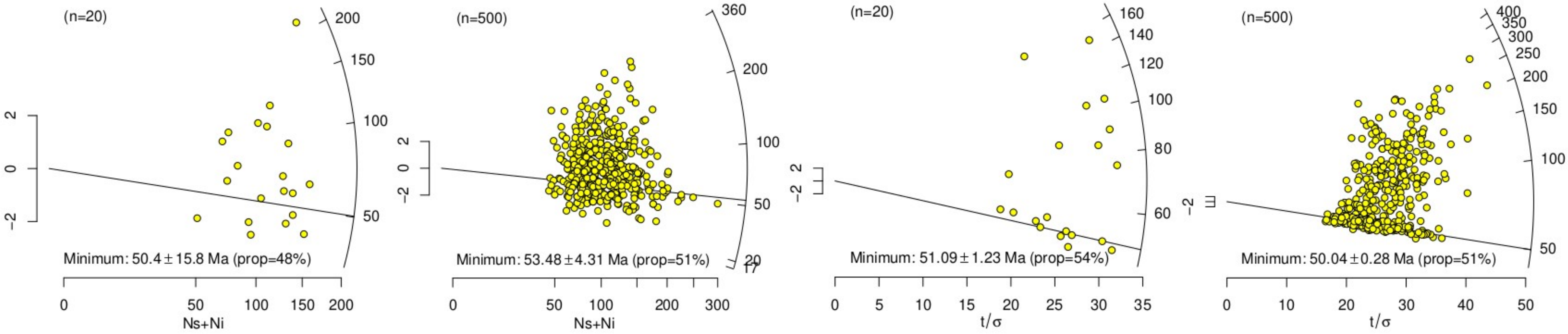
Pieter Vermeesch

Ages:



Fission tracks

U-Th-He



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ζ: 350 ± 2 yr cm^2

ρ_D: 2500000 ± 50000 1/cm^2

	Ns	Ni	(C)	(omit)	(comment)	F	G	H	I	J	K	L
1	22	64										
2	8	84										
3	16	98										
4	22	71										
5	10	110										
6	13	69										
7	5	77										
8	20	80										
9	24	92										
10	15	99										
11	20	137										
12	19	86										
13	24	74										
14	24	51										
15	19	82										
16	8	76										
17	32	148										
18	10	117										
19	7	64										
20	26	69										
21	18	165										
22	9	82										
23	18	175										
24	16	102										
25	24	93										
26												
27												
28												

Method: EDM

Input errors: 1se (abs)

Output errors: ci (abs)

Probability cutoff: 0.05

Show sample numbers?

Propagate external uncertainties?

Transformation: arcsin

Finite mixtures: 2

0 (none)
1
2
3
4
5
auto
minimum

Minimum value: uto

Central value: uto

Maximum value uto

Maximum preci uto

Plot character: 1

Significant digits

Fill colour: Custom colour ramp

from: to:

Plot symbol magnification: 1

Font magnification: 1

Colour label:

FT3.json

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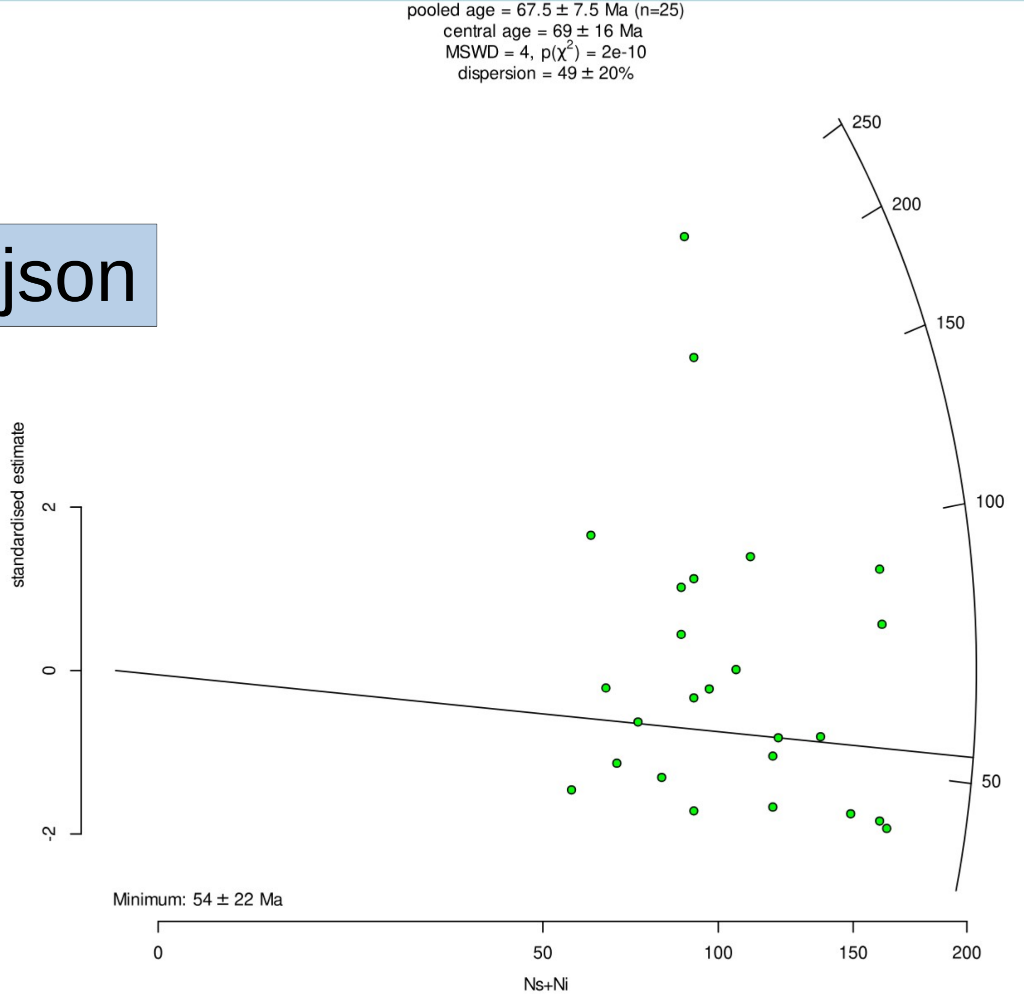
IsoplotR

ζ: 350 ± 2 yr cm^2

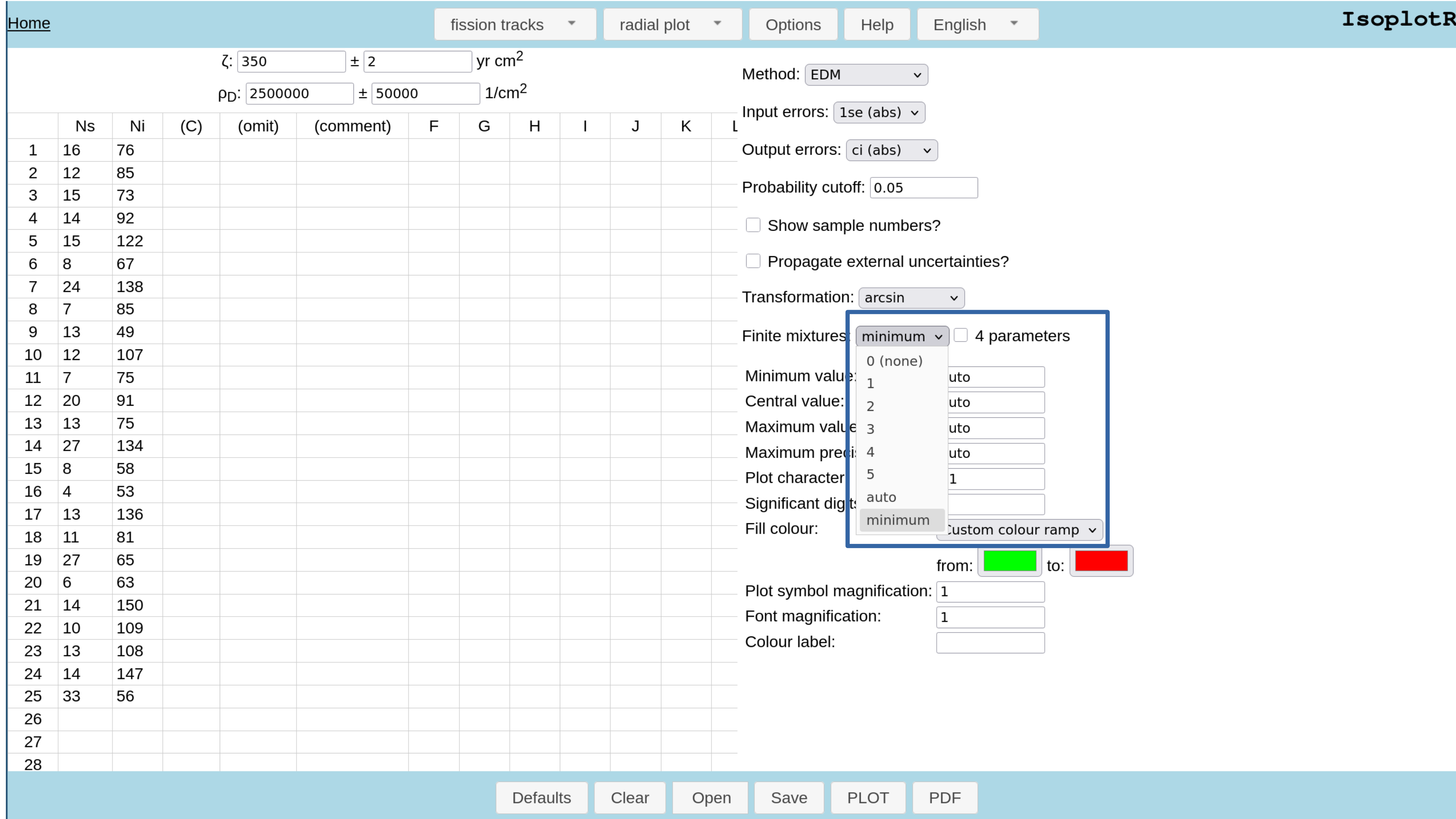
ρ_D: 2500000 ± 50000 1/cm^2

FT4.json

	Ns	Ni	(C)	(omit)	(comment)	F	G	H	I	J	K	L
1	16	76										
2	12	85										
3	15	73										
4	14	92										
5	15	122										
6	8	67										
7	24	138										
8	7	85										
9	13	49										
10	12	107										
11	7	75										
12	20	91										
13	13	75										
14	27	134										
15	8	58										
16	4	53										
17	13	136										
18	11	81										
19	27	65										
20	6	63										
21	14	150										
22	10	109										
23	13	108										
24	14	147										
25	33	56										
26												
27												
28												



Defaults | Clear | Open | Save | PLOT | PDF


Home
fission tracks
radial plot
Options
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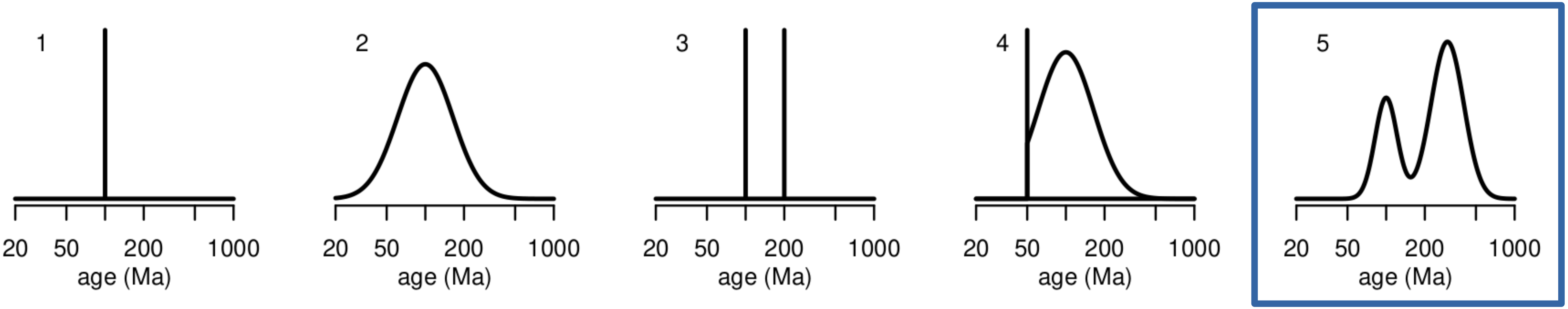
English
IsoplotR
ζ: 350 ± 2 yr cm2
ρD: 2500000 ± 50000 1/cm2
Ns Ni (C) (omit) (comment) F G H I J K L
1 16 76
2 12 85
3 15 73
4 14 92
5 15 122
6 8 67
7 24 138
8 7 85
9 13 49
10 12 107
11 7 75
12 20 91
13 13 75
14 27 134
15 8 58
16 4 53
17 13 136
18 11 81
19 27 65
20 6 63
21 14 150
22 10 109
23 13 108
24 14 147
25 33 56
26
27
28
Method: EDM
Input errors: 1se (abs)
Output errors: ci (abs)
Probability cutoff: 0.05
Show sample numbers?
Propagate external uncertainties?
Transformation: arcsin
Finite mixtures: minimum
4 parameters
0 (none)
1
2
3
4
5
auto
minimum
Minimum value
Central value:
Maximum value
Maximum preci
Plot character
Significant digits
Fill colour:
uto
uto
uto
uto
1
custom colour ramp
from:
to:
Plot symbol magnification: 1
Font magnification: 1
Colour label:
Defaults
Clear
Open
Save
PLOT
PDF

```
FT2a <- FTsamp(pop=2,ng=20)

radialplot(FT2a,k=2)

radialplot(FT2a,k="auto")


FT2b <- FTsamp(pop=2,ng=200)

radialplot(FT2b,k="auto")


FT3 <- FTsamp(pop=3,ng=20)

radialplot(FT3,k="min")
```

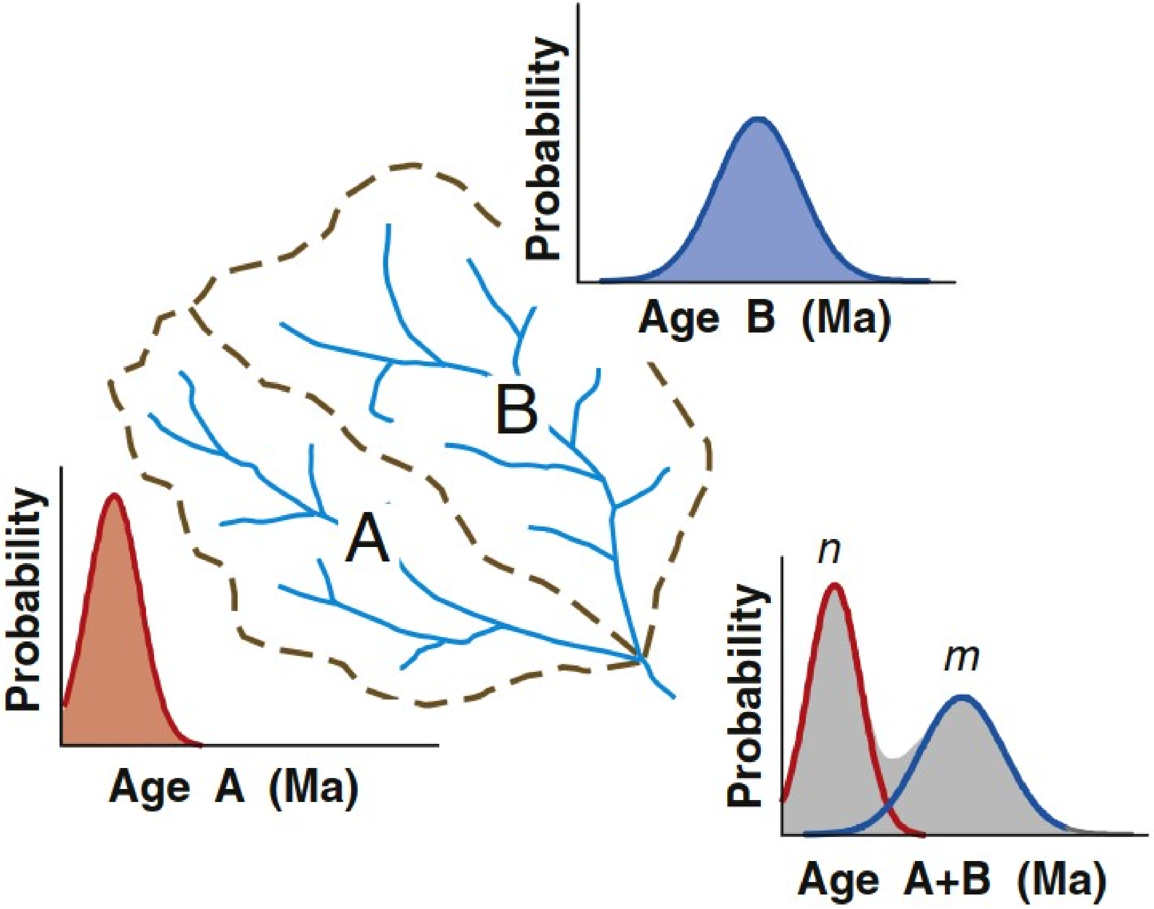
Ages:



Bayesian Mixture Modelling in Geochronology via Markov Chain Monte Carlo[1]

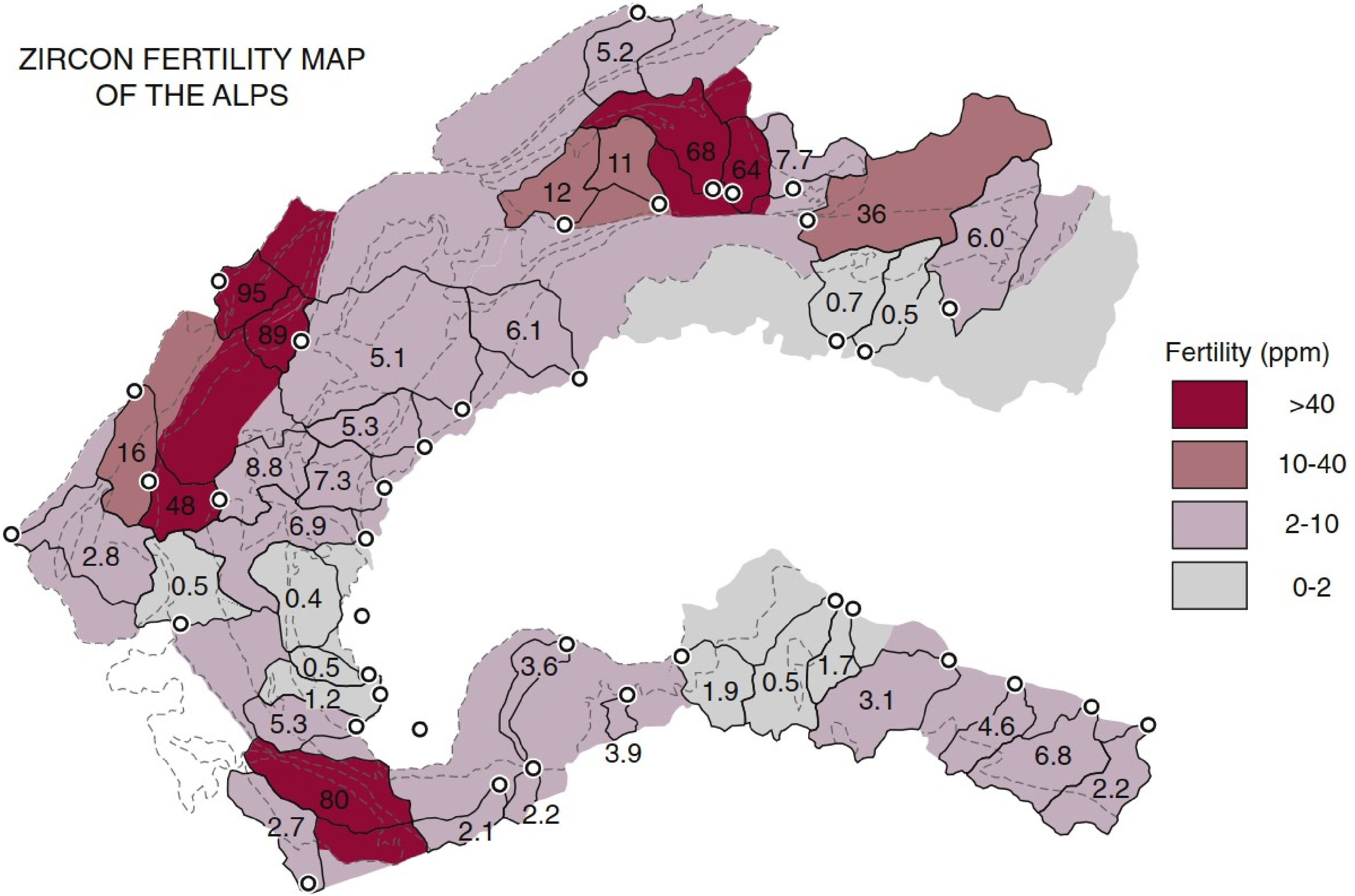
Ajay Jasra,[2] David A. Stephens,[2] Kerry Gallagher,[3] and Christopher C. Holmes[4,5]

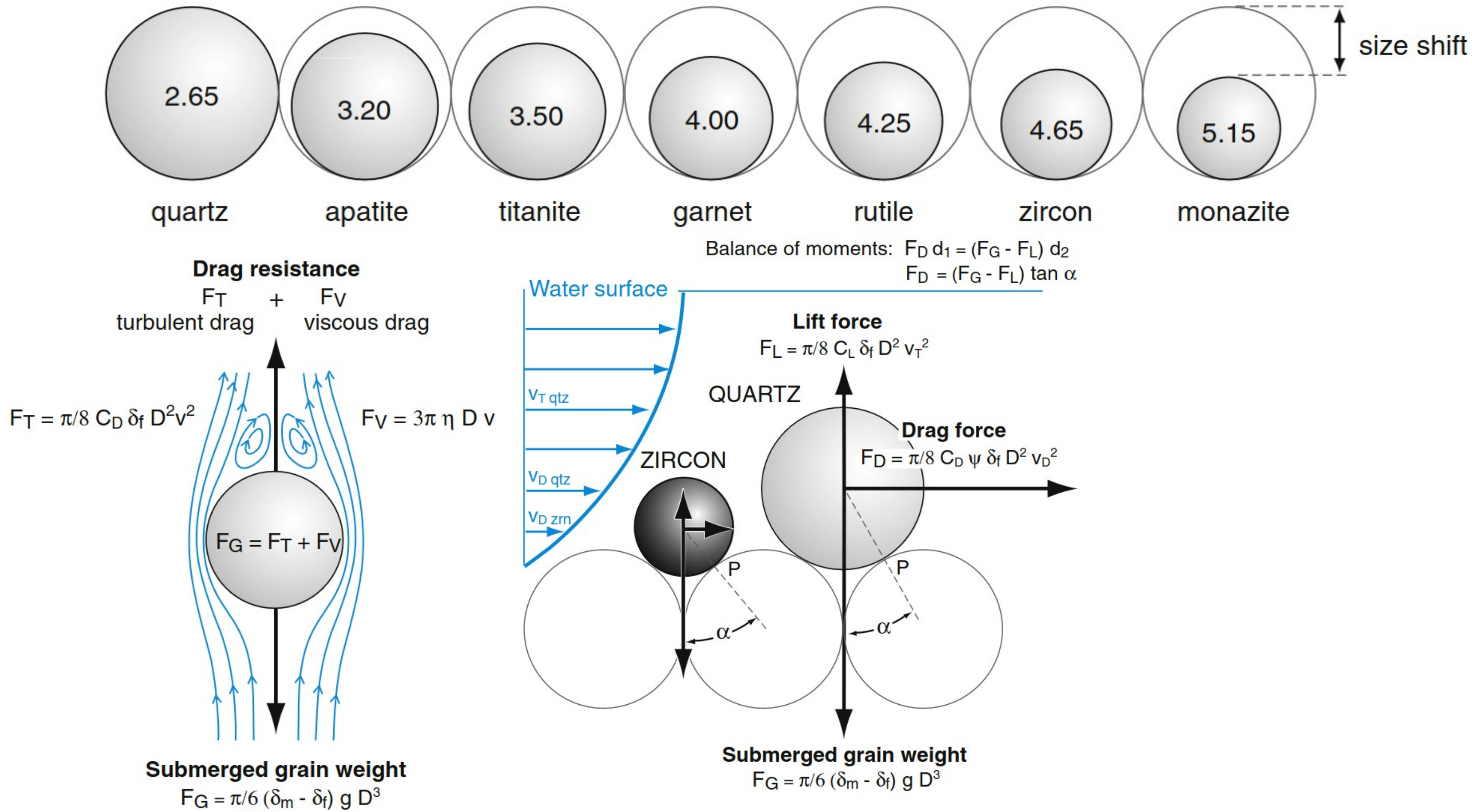
3p-1 parameters, including p σs→ requires very large datasets



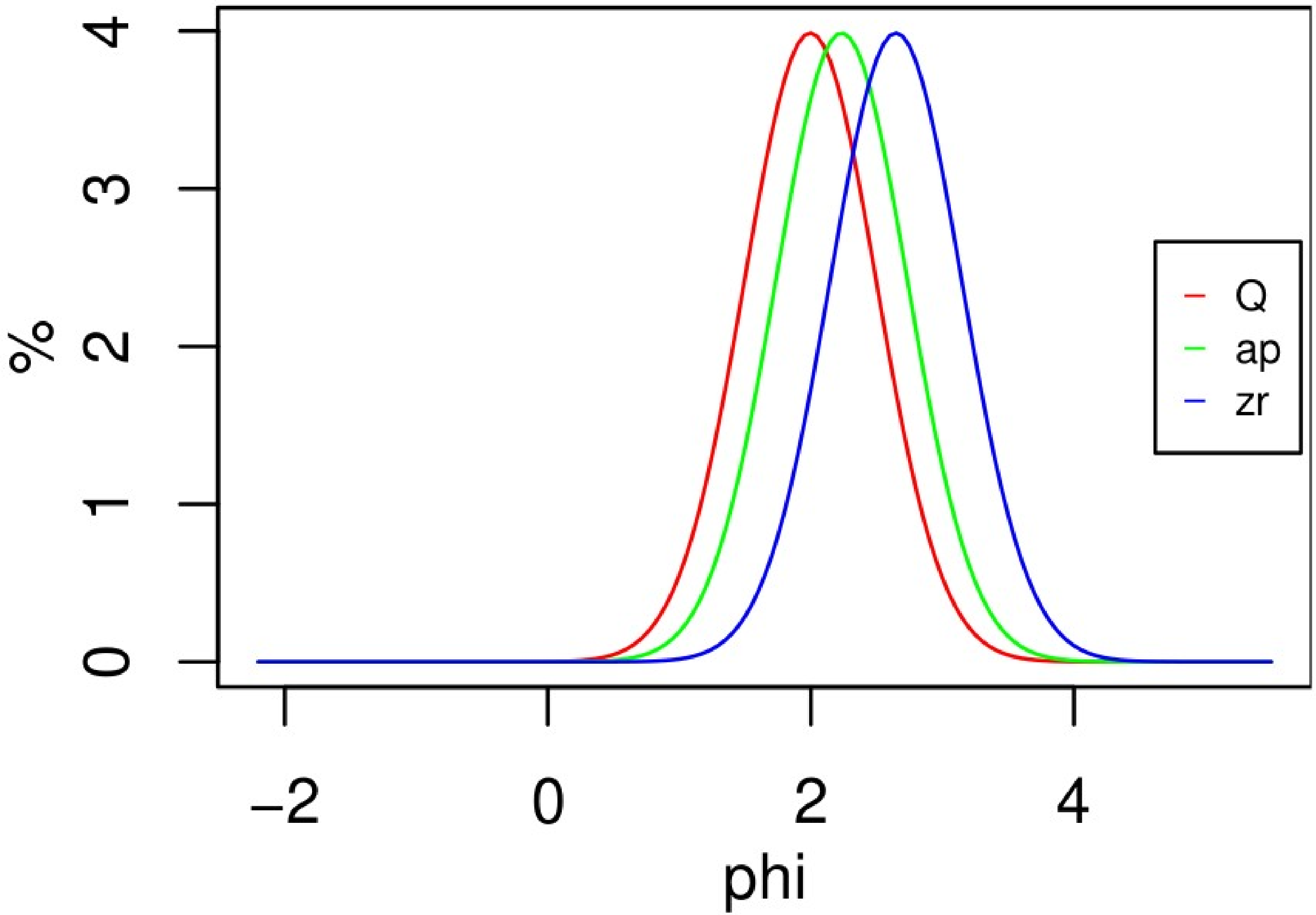
Hydraulic sorting and mineral fertility bias in detrital geochronology

Marco G. Malusà, Alberto Resentini *, Eduardo Garzanti

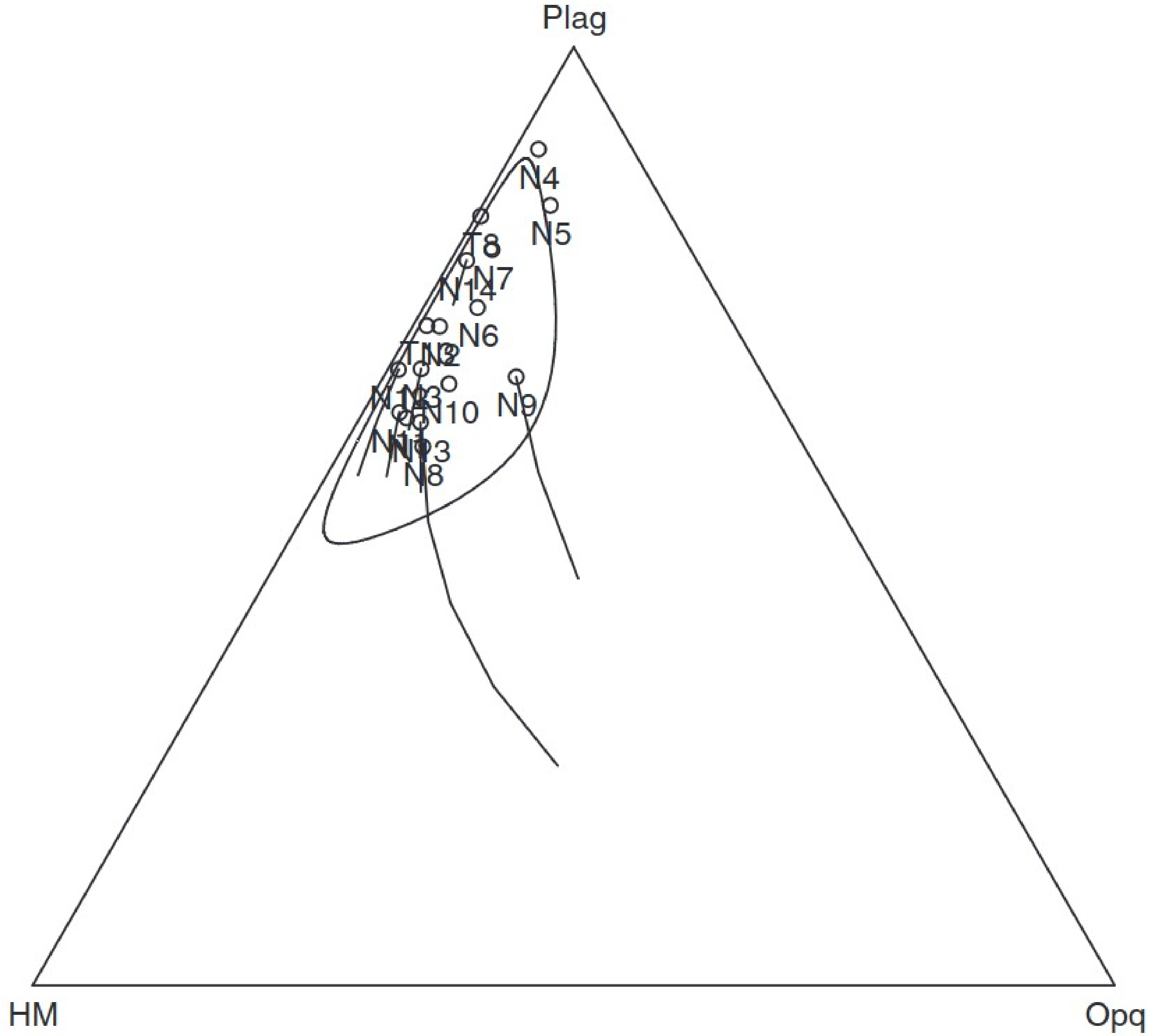

ZIRCON FERTILITY MAP
OF THE ALPS
5.2
11
68
64
7.7
12
36
6.0
95
89
0.7
0.5
6.1
5.1
5.3
16
8.8
7.3
48
6.9
2.8
0.5
0.4
0.5
1.2
5.3
3.6
1.9
0.5
1.7
3.1
3.9
4.6
6.8
80
2.2
2.2
2.7
2.1
Fertility (ppm)
>40
10-40
2-10
0-2

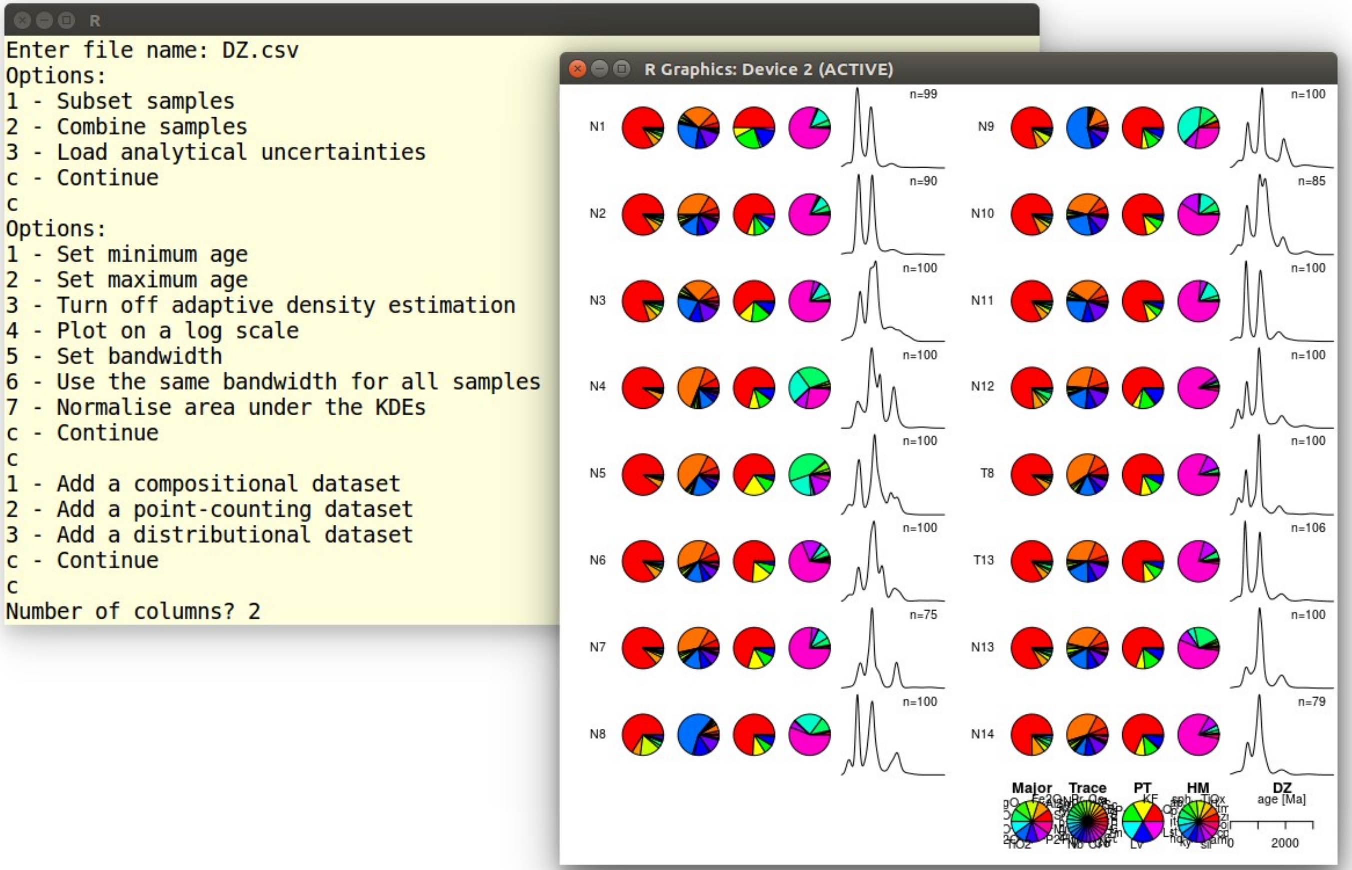


size shift
2.65
3.20
3.50
4.00
4.25
4.65
5.15
quartz
apatite
titanite
garnet
rutile
zircon
monazite
Balance of moments: $F_D\, d_1 = (F_G - F_L)\, d_2$
$F_D = (F_G - F_L) \tan\alpha$
Drag resistance
F_T + F_V
turbulent drag
viscous drag
$F_T = \pi/8\, C_D\, \delta_f\, D^2 v^2$
$F_V = 3\pi\, \eta\, D\, v$
$F_G = F_T + F_V$
Submerged grain weight
$F_G = \pi/6\, (\delta_m - \delta_f)\, g\, D^3$
Water surface
$v_{T\,qtz}$
$v_{D\,qtz}$
$v_{D\,zrn}$
Lift force
$F_L = \pi/8\, C_L\, \delta_f\, D^2\, v_T^2$
QUARTZ
ZIRCON
Drag force
$F_D = \pi/8\, C_D\, \psi\, \delta_f\, D^2\, v_D^2$
P
P
α
α
Submerged grain weight
$F_G = \pi/6\, (\delta_m - \delta_f)\, g\, D^3$


0
1
2
3
4
%
−2
0
2
4
phi
Q
ap
zr


ISS
STANDARD
TEST SIEVE
BSS
ASTM
MICS
INCHES
AS PER ISS : 460-1962


Plag
HM
Opq
N4
N5
N6
N7
N9
N10
N8


R
Enter file name: DZ.csv
Options:
1 - Subset samples
2 - Combine samples
3 - Load analytical uncertainties
c - Continue
c
Options:
1 - Set minimum age
2 - Set maximum age
3 - Turn off adaptive density estimation
4 - Plot on a log scale
5 - Set bandwidth
6 - Use the same bandwidth for all samples
7 - Normalise area under the KDEs
c - Continue
c
1 - Add a compositional dataset
2 - Add a point-counting dataset
3 - Add a distributional dataset
c - Continue
c
Number of columns? 2
R Graphics: Device 2 (ACTIVE)
Major
Trace
PT
HM
DZ
age [Ma]
2000